Supplementary Material

Sign compression for Muon: SignMuon, MuonSign, and the Limits of Error Feedback

A Appendix

A.1 Preliminaries: LMO, Muon, and generalized smoothness

This subsection states in full the definitions that Section 3 uses in abbreviated form.

Equip $\mathbb{R}^{m\times n}$ with the inner product $\langle \mathbf{M}, \mathbf{D}\rangle = \sum_{i,j} M_{ij}D_{ij}$, let $\|\cdot\|$ be a norm on it, let $\mathcal{B} = \{\mathbf{D}\in\mathbb{R}^{m\times n} : \|\mathbf{D}\|\le 1\}$ be its unit ball, and let $\|\mathbf{M}\|_{\mathrm{dual}} = \max_{\mathbf{D}\in\mathcal{B}}\langle \mathbf{M}, \mathbf{D}\rangle$ be the dual norm. The *Linear Minimization Oracle* (LMO) of $\|\cdot\|$ minimizes a linear form over the unit ball: it outputs

$$A(\mathbf{M}) \in \arg\min_{\mathbf{D}\in\mathcal{B}} \langle \mathbf{M}, \mathbf{D}\rangle = -\big\{\mathbf{D}\in\mathcal{B} : \langle \mathbf{M}, \mathbf{D}\rangle = \|\mathbf{M}\|_{\mathrm{dual}}\big\}, \tag{1}$$

Hence $\langle \mathbf{M}, A(\mathbf{M})\rangle = -\|\mathbf{M}\|_{\mathrm{dual}} \le 0$: the oracle returns a steepest-descent direction with respect to $\|\cdot\|$, normalized to the unit ball. The minimizer need not be unique, whence the inclusion; at rank-deficient $\mathbf{M}$ we use the rank-r selection fixed in the next paragraph. An LMO method minimizes a differentiable objective F by stepping along this direction: at the iterate $\mathbf{X}$ it forms an *effective update direction* $\mathbf{M}$, a stochastic gradient or a momentum estimate in every method below, and moves along $A(\mathbf{M})$, the minimizer over $\mathcal{B}$ of the first-order model $\mathbf{D}\mapsto F(\mathbf{X}) + \langle \mathbf{M}, \mathbf{D}\rangle$. Each layer of Section 3 carries its own norm and so its own oracle (Riabinin et al. 2025); for the role of the construction in optimizer design see (Bernstein and Newhouse 2024; Bernstein 2025; Pethick et al. 2025a; Kovalev 2025; Riabinin et al. 2025; Cesista 2025; Kravatskiy et al. 2025).

Muon is the instance in which each matrix layer carries the spectral norm $\|\cdot\|_{2\to2}$, so that the oracle uses the matrix structure of the gradient and the update direction is obtained by orthogonalizing the gradient matrix (Bernstein and Newhouse 2024). Let $\mathbf{M} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top$ be the singular value decomposition (SVD) of the matrix $\mathbf{M}\in\mathbb{R}^{m\times n}$, where $\mathbf{U}\in\mathbb{R}^{m\times r}$ and $\mathbf{V}\in\mathbb{R}^{n\times r}$ are the orthonormal matrices of singular vectors, $\boldsymbol{\Sigma}\in\mathbb{R}^{r\times r}$ is the diagonal matrix of singular values, and $r = \mathrm{rank}(\mathbf{M})$. Then the LMO direction is $A(\mathbf{M}) = -\mathbf{U}\mathbf{V}^\top$: Muon selects an orthonormal update direction corresponding to the solution of the linear minimization problem induced by the spectral norm geometry. Orthogonalization equalizes the singular values of the update, so that the directions in which the gradient is weak are not crowded out by the dominant ones (Jordan et al. 2024); large-scale pretraining studies report a corresponding efficiency gain over AdamW (Liu et al. 2025; Shah et al. 2025). The reference implementation obtains $\mathbf{U}_t\mathbf{V}_t^\top$ from a fixed number of Newton–Schulz iterations rather than from an SVD (Jordan et al. 2024); Algorithm A1 gives the procedure and the coefficients we use.

Choice of layer norm and the shape factor. Muon is also presented in the RMS→RMS operator norm, $\sqrt{n/m}\,\|\cdot\|_{2\to2}$ on $\mathbb{R}^{m\times n}$ (Bernstein and Newhouse 2024; Pethick et al. 2025a). The two readings give different oracles: the unit ball of $\sqrt{n/m}\,\|\cdot\|_{2\to2}$ is the spectral ball scaled by $\sqrt{m/n}$, so its LMO is $-\sqrt{m/n}\,\mathbf{U}\mathbf{V}^\top$. The unscaled $-\mathbf{U}\mathbf{V}^\top$ therefore belongs to the spectral norm and to no other, since a norm whose oracle it is has $\|\mathbf{M}\|_{\mathrm{dual}} = \langle \mathbf{M}, \mathbf{U}\mathbf{V}^\top\rangle = \|\mathbf{M}\|_*$ for every $\mathbf{M}$ and hence equals $\|\cdot\|_{2\to2}$ by biduality. We take the spectral norm, in which Assumption 2 and every statement below are stated, and treat the shape factor as a step-size question, for three reasons. It is a positive per-layer constant, so it changes neither $\mathrm{sign}(\cdot)$ nor the sign of any descent inner product, and no result of this paper is sensitive to it. The RMS→RMS value $\sqrt{m/n}$ is not in fact the factor Muon uses: the reference implementation applies $\sqrt{\max(1, m/n)}$ (Jordan et al. 2024), which agrees with it only for $m\ge n$ and is what our unit-gain rule returns (Appendix A.17). And that rule must also scale the two sign-terminated placements, for which no norm supplies a scale at all: by Theorems 1 and 3 they are oracles for none.

Finally, Corollary 2 requires Assumption 2 only in its weaker layer-wise (L^0, L^1) form, which replaces the constant L_i^g by $L_i^{0,g} + L_i^{1,g}\|\nabla_i g(\mathbf{X})\|_*$ (Riabinin et al. 2025; Pethick et al. 2025b): for every layer i and all parameter tuples $\mathbf{X}, \mathbf{Y}$,

$$\|\nabla_i g(\mathbf{X}) - \nabla_i g(\mathbf{Y})\|_* \le \big(L_i^{0,g} + L_i^{1,g}\|\nabla_i g(\mathbf{X})\|_*\big)\,\|\mathbf{X}_i - \mathbf{Y}_i\|_{2\to2}, \tag{2}$$

again with one pair of constants per layer for $g = f$ and per layer and client for $g = f_j$; at $L_i^{1,g} = 0$ the display is Assumption 2. It is Assumptions 8–9 of Gruntkowska et al. (2025), the hypotheses of the theorem Corollary 2 quotes, stated there as here for *arbitrary* pairs: the tuples may differ in every block although only block i enters the bound. That quantifier is strong. Fixing $\mathbf{X}_i = \mathbf{Y}_i$ while varying the remaining blocks forces $\nabla_i g(\mathbf{X}) = \nabla_i g(\mathbf{Y})$, so each $\nabla_i g$ depends on its own block alone, and a function satisfying the bound with finite constants is additively separable across layers. At $p = 1$, the setting of every counterexample and synthetic measurement in this paper, the restriction is empty and the display is ordinary (L^0, L^1)-smoothness. For a multilayer network it is the framework's idealization, and we inherit it unweakened: the descent lemma behind Theorem 5 applies the bound along a step in which every layer moves, which the restriction to pairs differing in one block would not license.

A.2 Width-one blocks: vector parameters

The setup of Section 3 asks each block to be a matrix, and $\min(m_i, n_i) = 1$ is permitted: a bias, a normalization gain, or any other one-dimensional parameter is the block $\mathbb{R}^{1\times n_i}$ or $\mathbb{R}^{m_i\times 1}$. Gluon states the same product space and leaves the norm on each block arbitrary, so it too admits them without comment (Riabinin et al. 2025); Scion is explicit, giving biases the RMS norm with oracle $\mathbf{b}/\|\mathbf{b}\|_{\mathrm{RMS}}$ (Pethick et al. 2025a). Fixing the spectral norm on every block recovers nearly that: on a width-one block the rank is one, the spectral, nuclear and Euclidean norms coincide, and $\mathrm{polar}(\mathbf{g}) = \mathbf{g}/\|\mathbf{g}\|_2$ for $\mathbf{g} \neq \mathbf{0}$. Assumptions 1 and 2 then read unchanged, and the norm-equivalence constant of Corollary 1 improves to $\bar{\rho}_i = \sqrt{r_i} = 1$.

Substituting $\mathrm{polar}(\mathbf{g}) = \mathbf{g}/\|\mathbf{g}\|_2$ into (6) gives, on $\mathbb{R}^{1\times n}$,

$$\mathrm{sign}(\mathrm{polar}(\mathbf{g})) = \mathrm{sign}(\mathbf{g}), \qquad \mathrm{polar}(\mathrm{sign}(\mathbf{g})) = \tfrac{1}{\sqrt{n}}\,\mathrm{sign}(\mathbf{g}),$$

and the two-sided placement returns $\mathrm{sign}(\mathbf{g})$ as well: all three are SignSGD up to a positive constant the step size absorbs. The ascent instances of Theorems 1–3 accordingly have $\min(m, n) \geq 2$.

The unit-gain multipliers of Appendix A.17 need no special case either: $\lambda = \sqrt{m}/\|\mathbf{P}\|_F$ returns $(1,\ 1/\sqrt{n})$ for the LMO and SIGN families on $\mathbb{R}^{1\times n}$, and $(\sqrt{m},\ 1)$ on $\mathbb{R}^{m\times 1}$.

Which parameters reach the methods. Little of this affects our experiments, since the sign methods are applied where Muon is. Centralized and federated runs give the rule to parameters of two or more dimensions other than the classifier head, and route biases, BatchNorm scales and that head to AdamW as the auxiliary group, whose bandwidth cost Appendix A.14 counts. On nanoGPT we keep record #40's grouping unchanged: one-dimensional scalars, the embeddings and the head go to its distributed Adam, the hidden matrices and the two kinds of gate weight to the method under test. One gate is width-one, the 1×12 `smear_gate`, and there polar is ℓ_2 normalization, so the three sign-terminated methods reduce exactly to SignSGD on it and the LMO five to normalized momentum. In the CIFAR runs the corresponding parameters are one-dimensional, and our implementation returns those from the oracle unchanged rather than ℓ_2-normalized, which differs from polar by a positive scale and so alters neither the sign nor the descent inner product.

A.3 Divergence on linear objectives: the ascent criterion and momentum

The criterion quoted in Section 4 collapses each of the three methods to a single scalar inequality and removes momentum from the discussion entirely.

Proposition 1 (Ascent criterion on linear objectives) *Run any of the three methods* (6) *on the linear objective* (8) *with* $\mathbf{G} \neq \mathbf{0}$*, from an arbitrary* $\mathbf{X}_0$*, with any momentum coefficient* $\mu \in [0, 1)$ *under either the Standard or the Nesterov rule. Then* $\tilde{\mathbf{M}}_t = \gamma_t\,\mathbf{G}$ *with* $\gamma_t > 0$*, the update direction is the constant matrix* $\mathbf{s}(\mathbf{G})$ *obtained by substituting* $\mathbf{G}$ *for* $\tilde{\mathbf{M}}_t$ *in* (6)*, and*

$$f(\mathbf{X}_t) - f(\mathbf{X}_{t-1}) = -\eta_t \left\langle \mathbf{G},\ \mathbf{s}(\mathbf{G}) \right\rangle.$$

In particular, if $\langle \mathbf{G}, \mathbf{s}(\mathbf{G})\rangle < 0$ *then* f *strictly increases at every iteration for any* $\eta_t > 0$*, and* $f(\mathbf{X}_t) \to +\infty$ *whenever* $\sum_t \eta_t = \infty$ *(for instance, for any constant step size).*

Proof of Proposition 1. On the linear objective (8) the gradient is globally constant, $\mathbf{G}_t = \mathbf{G}$, so the momentum buffer of (5) is $\mathbf{M}_t = (1-\mu)\sum_{i=0}^{t-1} \mu^i\,\mathbf{G} = (1-\mu^t)\,\mathbf{G}$. Both momentum rules then return a positive multiple of $\mathbf{G}$,

$$\tilde{\mathbf{M}}_t = \gamma_t\,\mathbf{G}, \qquad \gamma_t = \begin{cases} 1-\mu^t, & \text{(Standard)}, \\ 1-\mu^{t+1}, & \text{(Nesterov)}, \end{cases} \tag{3}$$

both positive for $t \geq 1$ because $\mu \in [0, 1)$; the Nesterov case is $(1-\mu)\mathbf{G} + \mu(1-\mu^t)\mathbf{G}$. (Under the heavy-ball convention every γ_t is multiplied by $1/(1-\mu)$, which changes nothing below.) The elementwise $\mathrm{sign}(\cdot)$ and the Muon LMO $\mathrm{polar}(\cdot)$ are each invariant under multiplication by a positive scalar, so evaluating (6) at $\tilde{\mathbf{M}}_t = \gamma_t\mathbf{G}$ returns the same matrix as evaluating it at $\mathbf{G}$; that is, $\mathbf{s}_t = \mathbf{s}(\mathbf{G})$ for every t, independently of μ and of the momentum variant. Hence $f(\mathbf{X}_t) - f(\mathbf{X}_{t-1}) = \langle \mathbf{G}, \mathbf{X}_t - \mathbf{X}_{t-1}\rangle = -\eta_t\langle \mathbf{G}, \mathbf{s}(\mathbf{G})\rangle$, which is (9). If $\langle \mathbf{G}, \mathbf{s}(\mathbf{G})\rangle < 0$, then $f(\mathbf{X}_t) = f(\mathbf{X}_0) - \langle \mathbf{G}, \mathbf{s}(\mathbf{G})\rangle \sum_{i=1}^{t} \eta_i$ increases strictly at every step and diverges to $+\infty$ whenever $\sum_t \eta_t = \infty$. ■

By Proposition 1, each divergence theorem reduces to exhibiting a single gradient $\mathbf{G}$ with $\langle \mathbf{G}, \mathbf{s}(\mathbf{G})\rangle < 0$; the three proofs below accomplish exactly this, and momentum requires no further comment.

A.4 Proof of Theorem 1 (Divergence of SignMuon)

Theorem 1 (Divergence of SignMuon) *There is a matrix* $\mathbf{G} \in \mathbb{R}^{4\times 4}$ *with* $\left\langle \mathbf{G}, \mathrm{sign}(\mathrm{polar}(\mathbf{G}))\right\rangle = -\frac{42468}{103} < 0$*. Hence SignMuon ascends on* (8)*:* f *strictly increases at every iteration for all* $\eta_t > 0$*, all* $\mu \in [0, 1)$*, and both momentum variants.*

For SignMuon $\mathbf{s}(\mathbf{G}) = \operatorname{sign}(\operatorname{polar}(\mathbf{G}))$, so by (9) it suffices to construct $\mathbf{G} \in \mathbb{R}^{4\times 4}$ with $\langle \mathbf{G}, \operatorname{sign}(\operatorname{polar}(\mathbf{G}))\rangle < 0$. Group the SVD $\mathbf{G} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ of an invertible $\mathbf{G}$ as the polar decomposition $\mathbf{G} = \mathbf{Q}\mathbf{H}$, with $\mathbf{Q} = \mathbf{U}\mathbf{V}^\top = \operatorname{polar}(\mathbf{G})$ orthogonal and $\mathbf{H} = \mathbf{V}\mathbf{\Sigma}\mathbf{V}^\top \succ 0$ symmetric. Only the symmetric part of $\mathbf{Q}^\top \operatorname{sign}(\mathbf{Q})$ then contributes to the trace against $\mathbf{H}$:

$$\langle \mathbf{G}, \operatorname{sign}(\operatorname{polar}(\mathbf{G}))\rangle = \operatorname{tr}\big(\mathbf{H}\mathbf{Q}^\top \operatorname{sign}(\mathbf{Q})\big) = \big\langle \mathbf{H}, \operatorname{sym}(\mathbf{Q}^\top \operatorname{sign}(\mathbf{Q}))\big\rangle, \tag{4}$$

so a counterexample with polar factor $\mathbf{Q}$ exists precisely when $\lambda_{\min}\big(\operatorname{sym}(\mathbf{Q}^\top \operatorname{sign}(\mathbf{Q}))\big) < 0$: necessity because $\mathbf{H} \succ 0$, sufficiency by taking $\mathbf{H} = \mathbf{w}\mathbf{w}^\top + \delta\mathbf{I}$ at a minimizing eigenvector $\mathbf{w}$ and $\delta > 0$ small, which keeps $\mathbf{G}$ invertible and its polar factor unique. The question thus concerns orthogonal matrices alone.

We construct $\mathbf{G} \in \mathbb{R}^{4\times 4}$ by defining a specific orthogonal matrix $\mathbf{O}$ and a specific rank-1 principal component $\mathbf{u}_1\mathbf{v}_1^\top$. Let the orthogonal matrix $\mathbf{O}$ be given by the following exact rational numbers:

$$\mathbf{O} = \frac{1}{103}\begin{pmatrix} 101 & 20 & 2 & -2 \\ -20 & 97 & 20 & -20 \\ -2 & 20 & 2 & 101 \\ -2 & 20 & -101 & -2 \end{pmatrix}. \tag{5}$$

Because no entry is zero, its element-wise sign matrix $\mathbf{S} = \operatorname{sign}(\mathbf{O})$ is uniquely defined. Now, let $\mathbf{u}_1$ and $\mathbf{v}_1$ be the following exact unit vectors:

$$\mathbf{u}_1 = \frac{1}{\sqrt{309}}\begin{pmatrix} 10 \\ -3 \\ 10 \\ 10 \end{pmatrix}, \quad \mathbf{v}_1 = \frac{1}{\sqrt{309}}\begin{pmatrix} 10 \\ 3 \\ -10 \\ 10 \end{pmatrix}. \tag{6}$$

One can easily verify that $\mathbf{O}\mathbf{v}_1 = \mathbf{u}_1$, meaning $\mathbf{u}_1$ and $\mathbf{v}_1$ act perfectly as left and right singular vectors for this orthogonal space. The crucial feature of this geometry is that the Frobenius inner product between this rank-1 component and the sign matrix $\mathbf{S}$ yields an exact, strictly negative fraction:

$$\langle \mathbf{u}_1\mathbf{v}_1^\top, \mathbf{S}\rangle = \mathbf{u}_1^\top \mathbf{S}\mathbf{v}_1 = -\frac{43}{103}. \tag{7}$$

We construct the gradient matrix $\mathbf{G}$ by assigning a large singular value ($\sigma_1 = 1001$) to this pathological component and a singular value of 1 to the rest of the orthogonal space. Exactly,

$$\mathbf{G} := 1000\,\mathbf{u}_1\mathbf{v}_1^\top + \mathbf{O} = \frac{1}{309}\begin{pmatrix} 100303 & 30060 & -99994 & 99994 \\ -30060 & -8709 & 30060 & -30060 \\ 99994 & 30060 & -99994 & 100303 \\ 99994 & 30060 & -100303 & 99994 \end{pmatrix}, \tag{8}$$

or, to one decimal,

$$\mathbf{G} \approx \begin{pmatrix} 324.6 & 97.3 & -323.6 & 323.6 \\ -97.3 & -28.2 & 97.3 & -97.3 \\ 323.6 & 97.3 & -323.6 & 324.6 \\ 323.6 & 97.3 & -324.6 & 323.6 \end{pmatrix}. \tag{9}$$

Since $\mathbf{O}\mathbf{v}_1 = \mathbf{u}_1$ and $\mathbf{v}_1^\top\mathbf{v}_1 = 1$,

$$\mathbf{G}\mathbf{v}_1 = 1000\,\mathbf{u}_1(\mathbf{v}_1^\top\mathbf{v}_1) + \mathbf{O}\mathbf{v}_1 = 1001\,\mathbf{u}_1,$$

so $(\mathbf{u}_1, \mathbf{v}_1)$ is a singular pair of $\mathbf{G}$ with $\sigma_1 = 1001$. For any $\mathbf{w} \perp \mathbf{v}_1$ we have $\mathbf{G}\mathbf{w} = \mathbf{O}\mathbf{w}$; as $\mathbf{O}$ is orthogonal and $\mathbf{O}\mathbf{v}_1 = \mathbf{u}_1$, the restriction $\mathbf{O}|_{\mathbf{v}_1^\perp}$ is an isometry onto $\mathbf{u}_1^\perp$, hence $\sigma_2 = \sigma_3 = \sigma_4 = 1$. Moreover, with $\mathbf{u}_k = \mathbf{O}\mathbf{v}_k$ for $k \geq 2$,

$$\textstyle\sum_{k\geq 2} \mathbf{u}_k\mathbf{v}_k^\top = \mathbf{O}\big(\mathbf{I} - \mathbf{v}_1\mathbf{v}_1^\top\big) = \mathbf{O} - \mathbf{u}_1\mathbf{v}_1^\top,$$

so $\mathbf{U}_G\mathbf{V}_G^\top = \mathbf{u}_1\mathbf{v}_1^\top + (\mathbf{O} - \mathbf{u}_1\mathbf{v}_1^\top) = \mathbf{O}$. All four singular values are positive, so $\mathbf{G}$ is invertible and $\operatorname{polar}(\mathbf{G}) = \mathbf{O}$ is its unique polar factor. The sign matrix (13) of Theorems 2–3 is full rank as well, so in all three proofs the argument of polar is invertible and its polar factor is unique: no statement depends on the selection rule fixed above for rank-deficient arguments.

Substituting the resulting matrix $\mathbf{G}$ into the descent condition yields:

$$\langle \mathbf{G}, \mathbf{S}\rangle = \langle 1000\,\mathbf{u}_1\mathbf{v}_1^\top + \mathbf{O}, \mathbf{S}\rangle = 1000\langle \mathbf{u}_1\mathbf{v}_1^\top, \mathbf{S}\rangle + \langle \mathbf{O}, \mathbf{S}\rangle. \tag{10}$$

The inner product $\langle \mathbf{O}, \mathbf{S}\rangle$ is equivalent to the L_1 norm (sum of absolute values) of $\mathbf{O}$, which equals exactly $\frac{532}{103}$. Therefore:

$$\langle \mathbf{G}, \mathbf{S}\rangle = 1000\left(-\frac{43}{103}\right) + \frac{532}{103} = -\frac{42468}{103} \approx -412.31. \tag{11}$$

Here $\mathbf{S} = \operatorname{sign}(\mathbf{O}) = \operatorname{sign}(\operatorname{polar}(\mathbf{G})) = \mathbf{s}(\mathbf{G})$, so $\langle \mathbf{G}, \mathbf{s}(\mathbf{G})\rangle = -\frac{42468}{103} < 0$. By Proposition 1, SignMuon strictly ascends, $f(\mathbf{X}_t) - f(\mathbf{X}_{t-1}) = \frac{42468}{103}\,\eta_t > 0$ at every iteration, for every $\eta_t > 0$, every $\mu \in [0, 1)$, and both momentum variants; $f(\mathbf{X}_t) \to +\infty$ under any non-summable step size. ■

A.5 Proof of Theorem 2 (Divergence of MuonUSign)

Theorem 2 (Divergence of MuonUSign) *There is a matrix* $\mathbf{G} \in \mathbb{R}^{5\times 5}$ *with* $\big\langle \mathbf{G}, \mathrm{polar}(\mathrm{sign}(\mathbf{G}))\big\rangle \approx -13.89 < 0$. *Hence MuonUSign ascends on* (8) *for all* $\eta_t > 0$, *all* $\mu \in [0,1)$, *and both momentum variants.*

MuonUSign (Algorithm A4) applies the sign *before* the LMO, so $\mathbf{s}(\mathbf{G}) = \mathrm{polar}(\mathrm{sign}(\mathbf{G}))$. By (9) it suffices to construct $\mathbf{G} \in \mathbb{R}^{5\times 5}$ with $\langle \mathbf{G}, \mathrm{polar}(\mathrm{sign}(\mathbf{G}))\rangle < 0$.

Here the step depends on $\mathbf{G}$ only through the magnitudes $|G_{ij}|$ and the sign pattern $\mathbf{S} = \mathrm{sign}(\mathbf{G})$; write $\mathbf{D} = \mathrm{polar}(\mathbf{S})$ for the direction it produces. Under the randomized convention $\mathbf{S} \in \{\pm 1\}^{m\times n}$ throughout, and the entrywise identity $G_{ij} = |G_{ij}|\, S_{ij}$ holds without exception, both sides vanishing wherever $G_{ij} = 0$. Consequently

$$\big\langle \mathbf{G}, \mathrm{polar}(\mathrm{sign}(\mathbf{G}))\big\rangle = \textstyle\sum_{i,j} |G_{ij}|\, S_{ij} D_{ij}, \tag{12}$$

and likewise $\langle \mathbf{G}, \mathrm{sign}(\mathbf{D})\rangle = \sum_{i,j} |G_{ij}|\, S_{ij}\, \mathrm{sign}(D_{ij})$ for the MuonSign step of Theorem 3. Every summand in (12) is nonnegative unless some entry is *mismatched*, $S_{ij} D_{ij} < 0$. It therefore suffices to exhibit one sign matrix carrying a mismatched entry: inflating $|G_{ij}|$ there, with the other magnitudes held fixed, drives the sum below zero.

Fix the full-rank sign matrix $\mathbf{S} \in \{-1, 1\}^{5\times 5}$,

$$\mathbf{S} = \begin{pmatrix} -1 & -1 & 1 & 1 & 1 \\ -1 & -1 & 1 & -1 & -1 \\ 1 & -1 & 1 & 1 & -1 \\ 1 & 1 & -1 & -1 & 1 \\ 1 & 1 & 1 & -1 & 1 \end{pmatrix}, \tag{13}$$

and let $\mathbf{D} = \mathrm{polar}(\mathbf{S})$ be its (unique, since $\mathbf{S}$ is full rank) polar factor. A direct computation gives $D_{4,2} = -1/\sqrt{17} \approx -0.2425 < 0$ while $\mathrm{sign}(D_{i,j}) = S_{i,j}$ at all 24 other entries; equivalently, $\mathrm{sign}(\mathbf{D})$ and $\mathbf{S}$ disagree at exactly the single entry $(4,2)$, where $S_{4,2} = +1$. We exploit this lone mismatch. Define

$$\mathbf{G} = \epsilon\, \mathbf{S} + (M - \epsilon)\, \mathbf{e}_4 \mathbf{e}_2^\top, \qquad \epsilon > 0,\ M > \epsilon, \tag{14}$$

so that $G_{4,2} = M > 0$ and $G_{i,j} = \epsilon S_{i,j}$ otherwise; hence $\mathrm{sign}(\mathbf{G}) = \mathbf{S}$ and $\mathrm{polar}(\mathrm{sign}(\mathbf{G})) = \mathbf{D}$ for every $M > 0$. The descent inner product splits as

$$\langle \mathbf{G}, \mathbf{D}\rangle = \epsilon \sum_{(i,j)\neq(4,2)} S_{i,j} D_{i,j} \;+\; M\, D_{4,2} = \epsilon\, C + M\, D_{4,2}, \tag{15}$$

where $C := \sum_{(i,j)\neq(4,2)} |D_{i,j}| = 10.366 > 0$ is fixed (every such entry agrees in sign with $\mathbf{S}$), while $M\, D_{4,2} = -M/\sqrt{17} \to -\infty$. Thus $\langle \mathbf{G}, \mathbf{D}\rangle < 0$ for any $M > \sqrt{17}\, C\epsilon \approx 42.7\,\epsilon$; with $\epsilon = 1,\ M = 100$ one obtains $\langle \mathbf{G}, \mathbf{D}\rangle = -13.89$, for the exact polar factor that the theorem is stated over. By Proposition 1, MuonUSign strictly ascends on $f(\mathbf{X}) = \langle \mathbf{G}, \mathbf{X}\rangle$ for every $\eta_t > 0$, every $\mu \in [0,1)$, and both momentum variants; $f(\mathbf{X}_t) \to +\infty$ under any non-summable step size. ■

A.6 Proof of Theorem 3 (Divergence of MuonSign)

Theorem 3 (Divergence of MuonSign) *For the* same *matrix* $\mathbf{G} \in \mathbb{R}^{5\times 5}$ *as in Theorem 2,* $\big\langle \mathbf{G}, \mathrm{sign}(\mathrm{polar}(\mathrm{sign}(\mathbf{G})))\big\rangle = -76 < 0$. *Hence MuonSign ascends on* (8) *for all* $\eta_t > 0$, *all* $\mu \in [0,1)$, *and both momentum variants.*

MuonSign (Algorithm A5) signs the polar factor as well, so $\mathbf{s}(\mathbf{G}) = \mathrm{sign}(\mathrm{polar}(\mathrm{sign}(\mathbf{G}))) = \mathrm{sign}(\mathbf{D})$. We reuse the *same* $\mathbf{S}$ and $\mathbf{G}$ of (13)–(14): since $\mathrm{sign}(\mathbf{G}) = \mathbf{S}$, the bidirectional step is the constant matrix $\mathrm{sign}(\mathbf{D})$, which agrees with $\mathbf{S}$ at all 24 entries except $(4,2)$, where $\mathrm{sign}(D_{4,2}) = -1 = -S_{4,2}$. Using $S_{i,j}^2 = 1$ everywhere,

$$\begin{aligned} \langle \mathbf{G}, \mathrm{sign}(\mathbf{D})\rangle &= \epsilon \sum_{(i,j)\neq(4,2)} S_{i,j}\, \mathrm{sign}(D_{i,j}) + M\ \mathrm{sign}(D_{4,2}) \\ &= 24\,\epsilon - M. \end{aligned} \tag{16}$$

This is negative for every $M > 24\epsilon$; with $\epsilon = 1,\ M = 100$ it equals exactly -76. By Proposition 1, MuonSign strictly ascends on $f(\mathbf{X}) = \langle \mathbf{G}, \mathbf{X}\rangle$ for every $\eta_t > 0$, every $\mu \in [0,1)$, and both momentum variants; $f(\mathbf{X}_t) \to +\infty$ under any non-summable step size. In particular, the *same* 5×5 linear instance is an ascent instance for the uplink-only placement (MuonUSign) and for the bidirectional one (MuonSign) alike. ■

A.7 Comparison with the convergence claim for SignMuon

Concurrently, Mishra, Trivedi, and Kumar (2026) introduced the sign-after-LMO method under the same name *SignMuon*: their algorithm forms the momentum $\mathbf{M}_t$, computes its polar factor, and steps along $\mathbf{S}_t = \operatorname{sign}(\operatorname{polar}(\mathbf{M}_t))$, normalized to $\mathbf{D}_t = \mathbf{S}_t/\sqrt{mn}$; after absorbing $1/\sqrt{mn}$ into η this is the SignMuon step of Section 4. Their abstract and contributions attribute to this method an $\mathcal{O}(1/\sqrt{T})$ stationarity guarantee; the theorem that establishes the rate is stated, accurately, for a *gradient-sign instantiation*. The distance between the two statements is the subject of this subsection. Two facts resolve it, and neither contradicts Theorem 1: the finalized rate is proved for an update that computes no polar factor and carries no momentum, and the one bound of theirs that does apply to the SignMuon update is an inequality whose right-hand side exceeds its left-hand side on the instance of Theorem 1, at every iteration and for every step size, so that it is satisfied there while the method ascends.

The two components of their analysis. Their stationarity measure is $\mathcal{G}_T = \frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\big[\|\nabla f(\mathbf{X}_t)\|_1/\sqrt{mn}\big]$, controlled through the per-entry sign-error probabilities $q_{ij,t} = \Pr\big(\mathbf{S}_{t,ij} \neq \operatorname{sign}([\nabla f(\mathbf{X}_t)]_{ij}) \mid \mathbf{X}_t\big)$ of the transmitted sign matrix $\mathbf{S}_t$. The first component is generic. For an *arbitrary* sign oracle, the conditional identity

$$\mathbb{E}\big[\langle \nabla f(\mathbf{X}_t),\, \mathbf{D}_t\rangle \,\big|\, \mathbf{X}_t\big] = \frac{1}{\sqrt{mn}}\sum_{i,j}\big|[\nabla f(\mathbf{X}_t)]_{ij}\big|\,\big(1-2q_{ij,t}\big) \tag{17}$$

and the descent lemma of spectral smoothness telescope into

$$\mathcal{G}_T \;\le\; \frac{f(\mathbf{X}_0)-f^*}{\eta T} + \frac{L_*}{2}\,\eta + \underbrace{\frac{2}{T}\sum_{t=0}^{T-1}\mathbb{E}\left[\frac{1}{\sqrt{mn}}\sum_{i,j}\big|[\nabla f(\mathbf{X}_t)]_{ij}\big|\,q_{ij,t}\right]}_{=:\,R_T}. \tag{18}$$

The second component estimates the residual R_T, and it is here that the oracle is fixed: for $\mathbf{S}_t = \operatorname{sign}(\widetilde{\mathbf{G}}_t)$, where $\widetilde{\mathbf{G}}_t$ is an unbiased stochastic gradient with coordinatewise variance at most σ_{ij}^2/n_b at mini-batch size n_b, a Markov–Jensen argument yields $\big|[\nabla f(\mathbf{X}_t)]_{ij}\big|\,q_{ij,t} \le \sigma_{ij}/\sqrt{n_b}$, hence $R_T \le 2\|\sigma\|_1/\sqrt{mn\,n_b}$ with $\|\sigma\|_1 = \sum_{i,j}\sigma_{ij}$. The residual vanishes as $n_b \to \infty$, and the choice $n_b = T$ produces the $\mathcal{O}(1/\sqrt{T})$ rate.

The finalized rate is not a result about SignMuon. The oracle of the second component transmits the sign of the stochastic gradient itself. That update invokes neither the momentum buffer nor the polar factor; as an algorithm it is SignSGD at batch size n_b, in single-worker and majority-vote form, normalized by $1/\sqrt{mn}$ and analysed under spectral rather than coordinatewise smoothness, which their own comparison identifies as the sole improvement over Bernstein et al. (2018). Nor is the restriction incidental. The Markov–Jensen estimate bounds the probability of a sign error by the ratio of noise to signal, $\sigma_{ij}/(\sqrt{n_b}\,|[\nabla f(\mathbf{X}_t)]_{ij}|)$, and is therefore available exactly when sampling noise is the only mechanism by which a transmitted sign can disagree with the gradient's. For the SignMuon oracle the disagreement is structural rather than stochastic: even with exact gradients ($\sigma \equiv 0$), $q_{ij,t}$ is the indicator that $\operatorname{sign}(\operatorname{polar}(\mathbf{M}_t))$ and $\operatorname{sign}(\nabla f(\mathbf{X}_t))$ differ at (i,j), a quantity that no batch size reduces. The $\mathcal{O}(1/\sqrt{T})$ rate accordingly attaches to the gradient-sign update, and to SignMuon their analysis offers only (18) with R_T unestimated.

Scope of the generic bound. Inequality (18) is valid for every sign oracle, and for that reason asserts nothing until R_T is estimated. By (17), the summand of R_T at time t exceeds the corresponding summand of $\mathcal{G}_T$ precisely when $\mathbb{E}[\langle \nabla f(\mathbf{X}_t), \mathbf{D}_t\rangle \mid \mathbf{X}_t] \le 0$, that is, precisely when the expected step fails to be a descent direction. Whenever this occurs at every t, the right-hand side of (18) exceeds the left-hand side termwise and the inequality holds irrespective of how the iterates behave. The bound therefore has content only where the transmitted sign is already positively aligned with the gradient in expectation; that alignment is the property a convergence proof for SignMuon would have to establish, and it is the property Theorem 1 refutes.

On the instance of Theorem 1. Consider the linear objective (8) with the 4×4 gradient $\mathbf{G}$ of Theorem 1. By Proposition 1 the gradient equals $\mathbf{G}$ and the update direction equals the constant matrix $\operatorname{sign}(\operatorname{polar}(\mathbf{G}))$ at every iteration, whatever the momentum, so the oracle is deterministic and q_{ij} is the indicator that $\operatorname{sign}(\operatorname{polar}(\mathbf{G}))_{ij} \neq \operatorname{sign}(\mathbf{G}_{ij})$. Identity (17) then evaluates exactly:

$$\sum_{i,j}|\mathbf{G}_{ij}|\,(1-2q_{ij}) = \big\langle \mathbf{G},\, \operatorname{sign}(\operatorname{polar}(\mathbf{G}))\big\rangle = -\tfrac{42468}{103} \;<\; 0. \tag{19}$$

Dividing by $\sqrt{mn} = 4$, every term of R_T exceeds the corresponding term of $\mathcal{G}_T$ by the same amount, so that

$$R_T = \mathcal{G}_T + \tfrac{10617}{103} \qquad \text{for every } T. \tag{20}$$

The right-hand side of (18) therefore exceeds the left-hand side by at least $10617/103$ for every $\eta > 0$, every L_* and every T: the inequality is satisfied and constrains nothing. What the trajectory actually does is read off the same identity,

$f(\mathbf{X}_t) - f(\mathbf{X}_{t-1}) = -\eta\langle \mathbf{G}, \mathbf{D}_t\rangle = \eta \cdot \frac{10617}{103} > 0$, the divergence of Theorem 1: the excess $R_T - \mathcal{G}_T$ of (20) and the per-iteration ascent rate $\eta^{-1}\big(f(\mathbf{X}_t) - f(\mathbf{X}_{t-1})\big)$ are the same number. The one assumption of theirs the linear instance lacks is lower boundedness, and the modification of Remark 1 applies unchanged: f is unbounded below only on a half-space the iterates never enter, where a smooth bounded replacement restores the assumption without moving the trajectory or any quantity above.

In their terms, Theorem 1 exhibits a smooth instance on which the sign-error probabilities $q_{ij,t}$ of the SignMuon oracle, averaged over the entries with weights $\big|[\nabla f(\mathbf{X}_t)]_{ij}\big|$, exceed $\frac{1}{2}$ at every iteration. Any rate extracted from (18) requires that weighted average to stay below $\frac{1}{2}$ by a uniform margin, and no assumption of theirs implies such a bound for $\mathrm{sign}(\mathrm{polar}(\mathbf{M}_t))$. Their theorems stand as guarantees for majority-vote SignSGD under spectral smoothness. A convergence guarantee for SignMuon they are not, and Theorem 1 shows that none is available at this level of generality.

A.8 Extended comparison with S-Muon

Taking the norms dual to the Ky Fan k-norms, Kravatskiy et al. (2025) obtain the *Fanion* family, whose updates $\sum_{i\leq k} \mathbf{u}_i\mathbf{v}_i^\top$ interpolate between the rank-one step of the nuclear norm and Muon's full-rank $\mathbf{U}\mathbf{V}^\top$ at $k = \min(m,n)$; a conic combination of LMO algorithms is again an LMO algorithm, for the norm dual to the corresponding combination of dual norms. Their S-Muon is one such combination, $\tau\,\mathbf{U}\mathbf{V}^\top + (1-\tau)\,c\,\mathrm{sign}(\mathbf{M}_t)$ with fixed $\tau \in [0,1]$, $c > 0$ (their notation differs; we reserve α for compressor contraction and η for the learning rate). There the sign enters *inside* the oracle, so the step is still an LMO for an explicit norm and inherits the convergence theory of one; our three placements act *around* the oracle. A caution from the same work applies to us directly: they exhibit an LMO method (rank-one Neon) markedly worse than Muon in practice despite sharing its convergence asymptotics in the bounds of Kovalev (2025) and Riabinin et al. (2025), from which our own guarantee descends. A rate of the form $\mathcal{O}(T^{-1/2})$ is not a prediction of the performance a method will attain.

A.9 Proof of Theorem 4 (Divergence of EF21-SignMuon)

Theorem 4 (Divergence of EF21-SignMuon) *For every $L > 0$, step size $\eta > 0$, momentum coefficient $\mu \in [0,1)$, and either momentum variant, there is an L-smooth (Assumption 2), bounded-below (Assumption 1) function $f : \mathbb{R}^{2\times 2} \to \mathbb{R}$ on which EF21-SignMuon started at $\mathbf{X}_0 = \mathbf{0}$ diverges: for an explicit constant $c = c(f) > 0$,*

$$f(\mathbf{X}_{t+2}) - f(\mathbf{X}_t) = c\,L\eta^2 > 0 \qquad (t \geq 3), \tag{21}$$

so $f(\mathbf{X}_t) \to +\infty$. In particular, no step-size rule $\eta = \eta(L,\mu)$ using only the smoothness and momentum constants can make the method convergent.

Recall from the main text that EF21-SignMuon (Algorithm A3) does not step along the polar factor $\mathbf{D}_t = \mathrm{polar}(\tilde{\mathbf{M}}_t)$ itself, but along the error-feedback estimate $\mathbf{d}_t^{\mathrm{est}}$ of (10), followed by $\mathbf{X}_t = \mathbf{X}_{t-1} - \eta\,\mathbf{d}_t^{\mathrm{est}}$. A *single* magnitude α_t rescales the signs of *all* entries at once, and it is this coupling that the counterexample exploits. We now prove Theorem 4.

Proof idea *The mechanism.* The error-feedback update (10) moves *every* entry of $\mathbf{d}_t^{\mathrm{est}}$ by the same magnitude α_t; only the signs are individual. Suppose then that one entry must track a target alternating between $+1$ and -1 while another must track a small constant $-\varepsilon$. The alternating entry keeps its residual, and with it α_t, at $\Theta(1)$; the constant entry is therefore displaced by $\pm\Theta(1)$ at every step and can only oscillate about its target, never settle on it. Which side of the target the oscillation favours is decided by its *phase*, and a one-bit sign carries no information by which a phase could be corrected. In the unfavourable phase the estimate of the constant entry has a time average of the sign *opposite* to $-\varepsilon$, and the iterate driven by that estimate moves the wrong way forever.

From the sketch to an instance. The sketch is not yet a counterexample: in EF21-SignMuon the quantity tracked is the polar factor $\mathbf{D}_t = \mathrm{polar}(\tilde{\mathbf{M}}_t)$, of unit spectral norm, and the gradients behind it must all come from one smooth function. Both constraints are met at size 2×2 by the reflections $\bar{\mathbf{D}}^{\pm} = \left(\begin{smallmatrix} a & \pm b \\ \pm b & -a \end{smallmatrix}\right)$ with $a^2 + b^2 = 1$: one matrix the oracle can emit carries both roles of the sketch at once, the large alternating entries on the off-diagonal ($b = \frac{24}{25}$) and the small constant ones on the diagonal ($a = \frac{7}{25}$), coupled by the shared α_t. On these targets the estimate enters a period-two cycle in which the time average of the $(2,2)$-entry is positive although every target value is $-\frac{7}{25}$, so $(\mathbf{X}_t)_{22}$ travels to $-\infty$, the direction in which the objective increases (Figure 1, right). Nor does the mechanism rest on degeneracy: the momentum matrices of the divergent tail have condition number $\frac{16}{9}$ throughout.

The role of the preamble. The unfavourable phase must be arranged. From $\mathbf{d}_0^{\mathrm{est}} = \mathbf{0}$, the alternating targets alone lock the estimate into a period-two cycle whose diagonal average has the *correct* sign. The recursion (10) is piecewise affine, the pieces indexed by the sign pattern of the residual, and the harmless cycle and the wrong-sign one lie in different pieces, which the dynamics cannot join. The target sequence of Part 2 therefore opens with two preamble steps, whose sole purpose is to place the estimate in the piece containing the wrong-sign cycle.

Eliminating the parameters. The step size and the smoothness constant only rescale the trajectory, so $\eta = 1$ and one value of L suffice (Part 1). Momentum determines only which gradients produce a given target sequence, not how the recursion (10) responds to it; solving the momentum recursion for the gradients therefore settles every $(\mu, \text{variant})$ at once (Part 3).

Accordingly the proof has three parts, and they are independent. Part 1 removes L and η by rescaling. Part 2 carries the dynamics, and is a finite computation in exact rational arithmetic: on one fixed sequence of LMO targets, the estimate enters a limit cycle whose diagonal has the wrong sign. Part 3 is an existence argument only: it exhibits a smooth function whose gradients generate that target sequence for every μ and either momentum variant. A reader willing to grant that some smooth objective produces the targets can stop after Part 2.

Proof

Part 1: rescaling.

Lemma 1 (Scale reduction) *If $\tilde f$ is $\tilde L$-smooth with EF21-SignMuon iterates $\tilde{\mathbf{X}}_t$ at step size 1, then $f(\mathbf{X}) := \frac{L\eta^2}{\tilde L}\,\tilde f(\mathbf{X}/\eta)$ is L-smooth and its run at step size η (same μ, same variant, from $\mathbf{0}$) satisfies $\mathbf{X}_t = \eta\,\tilde{\mathbf{X}}_t$ and $f(\mathbf{X}_t) = \frac{L\eta^2}{\tilde L}\,\tilde f(\tilde{\mathbf{X}}_t)$.*

Proof. $\nabla f(\mathbf{X}) = \frac{L\eta}{\tilde L}\nabla \tilde f(\mathbf{X}/\eta)$, so f is L-smooth. Suppose $\mathbf{X}_s = \eta\tilde{\mathbf{X}}_s$ for $s < t$: the two gradients then differ by the positive factor $\frac{L\eta}{\tilde L}$, which $\mathbf{M}_t$ and $\tilde{\mathbf{M}}_t$ (positive combinations of past gradients) inherit and polar discards. Hence $\mathbf{D}_t$ and $\mathbf{d}_t^{\mathrm{est}}$ match the normalized run, and $\mathbf{X}_t = \eta\tilde{\mathbf{X}}_t$. ■

It therefore suffices to exhibit, for each (μ, variant), a C^∞ $\tilde L$-smooth $\tilde f$ on which the *normalized* run ($\eta = 1$) obeys (21); that $\tilde f$ may be taken bounded below (Assumption 1) is shown afterwards (Remark 1). We fix $\eta = 1$ from now on.

Part 2: the limit cycle of the estimate. The dynamical core of the proof is the behavior of the recursion (10) on the fixed target sequence

$$\mathbf{D}_1 = \mathbf{S}_1, \quad \mathbf{D}_2 = \mathbf{S}_2, \quad \mathbf{D}_t = \bar{\mathbf{D}}^{(-1)^t}\ (t \ge 3), \tag{22}$$

$$\begin{gathered}\bar{\mathbf{D}}^{\pm} = \begin{pmatrix} 7/25 & \pm 24/25 \\ \pm 24/25 & -7/25 \end{pmatrix}, \\ \mathbf{S}_1 = \begin{pmatrix} -4/5 & 3/5 \\ -3/5 & -4/5 \end{pmatrix}, \quad \mathbf{S}_2 = \begin{pmatrix} 3/5 & -4/5 \\ 0 & 0 \end{pmatrix}.\end{gathered} \tag{23}$$

The divergence originates in the tail ($t \ge 3$): the reflections $\bar{\mathbf{D}}^{\pm}$ share the diagonal $(\frac{7}{25}, -\frac{7}{25})$, while their off-diagonal $\pm\frac{24}{25}$ reverses sign at every step. The rotation $\mathbf{S}_1$ and the rank-one $\mathbf{S}_2$ form a two-step preamble. Only $\mathbf{S}_2$ requires comment: at a rank-deficient argument the spectral-ball LMO is not unique, and Section 3 resolves it through the thin SVD that retains only the nonzero singular directions, $\mathrm{polar}(\mathbf{M}) = \mathbf{U}\mathbf{V}^\top$ with one column of $\mathbf{U}, \mathbf{V}$ per nonzero singular value of $\mathbf{M}$; under that convention a rank-one matrix of unit spectral norm, such as $\mathbf{S}_2$, is its own polar factor, and this is also what the implementation computes. The next lemma says what the preamble is for, and that something like it is unavoidable.

Lemma 2 (The cycle reached without the preamble) *Write $\bar{\mathbf{D}}^{\pm} = \begin{pmatrix} a & \pm b \\ \pm b & -a \end{pmatrix}$ with $a^2 + b^2 = 1$ and $0 < a < b$. On the purely alternating targets $\mathbf{D}_t = \bar{\mathbf{D}}^{(-1)^t}$ ($t \ge 1$), the recursion* (10) *started from $\mathbf{d}_0^{\mathrm{est}} = \mathbf{0}$ enters a period-two cycle immediately, namely $\mathbf{d}_t^{\mathrm{est}} = \frac{a+b}{2}\begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}$ for odd t and $\frac{b-a}{2}\begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}$ for even t. Over a period its $(2,2)$-entry averages $-\frac{a}{2}$: the* same *sign as every target value $-a$, at half the magnitude.*

Proof. Put $m = \frac{a+b}{2}$. The residual $\bar{\mathbf{D}}^{-} - \mathbf{0}$ has signs $\begin{pmatrix} + & - \\ - & - \end{pmatrix}$ and mean modulus m, so $\mathbf{d}_1^{\mathrm{est}} = m\begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}$. Next, $\bar{\mathbf{D}}^{+} - \mathbf{d}_1^{\mathrm{est}} = \begin{pmatrix} a-m & b+m \\ b+m & m-a \end{pmatrix}$ has signs $\begin{pmatrix} - & + \\ + & + \end{pmatrix}$ (as $a < m$) and mean modulus b, giving $\mathbf{d}_2^{\mathrm{est}} = \frac{b-a}{2}\begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}$. Repeating once returns $\mathbf{d}_1^{\mathrm{est}}$. The average of the two $(2,2)$-entries is $\frac{1}{2}\left(-m + \frac{b-a}{2}\right) = -\frac{a}{2}$. ■

That cycle is harmless: the alternation by itself does not diverge, and no choice of (a, b) makes it do so. The divergence comes from a *second* period-two cycle of the same recursion, whose residuals carry a *uniform* sign pattern ($\begin{pmatrix} + & + \\ + & + \end{pmatrix}$ and $\begin{pmatrix} - & - \\ - & - \end{pmatrix}$) where those of the harmless cycle are mixed. Since (10) is affine on each sign-pattern cell, the dynamics cannot pass from one cycle to the other; the preamble exists solely to place $\mathbf{d}_2^{\mathrm{est}}$ in the cell of the wrong-sign cycle, which is what Lemma 3 verifies.

Lemma 3 (Wrong-sign limit cycle) *On the targets* (22)*, the recursion* (10) *from $\mathbf{d}_0^{\mathrm{est}} = \mathbf{0}$ enters at $t = 3$ the exact period-two cycle $\mathbf{d}_t^{\mathrm{est}} = \mathbf{d}_B$ (odd t), $\mathbf{d}_A$ (even t), where*

$$\mathbf{d}_B = \tfrac{1}{200}\begin{pmatrix} -61 & -201 \\ -61 & -61 \end{pmatrix}, \qquad \mathbf{d}_A = \tfrac{1}{200}\begin{pmatrix} 131 & -9 \\ 131 & 131 \end{pmatrix}. \tag{24}$$

Over a period its $(2,2)$-entry averages $\frac{1}{2}\big((\mathbf{d}_A)_{22} + (\mathbf{d}_B)_{22}\big) = +\frac{7}{40}$, opposite in sign to every target value $(\mathbf{D}_t)_{22} = -\frac{7}{25}$.

Proof. Substituting (22) into (10) gives the values

t	1	2	3	4 (then 2-periodic)
α_t	$\frac{7}{10}$	$\frac{21}{20}$	$\frac{131}{200}$	$\frac{24}{25}$
$\mathbf{d}_t^{\mathrm{est}}$	$\frac{1}{10}\begin{pmatrix} -7 & 7 \\ -7 & -7 \end{pmatrix}$	$\frac{1}{20}\begin{pmatrix} 7 & -7 \\ 7 & 7 \end{pmatrix}$	$\mathbf{d}_B$	$\mathbf{d}_A$

(25)

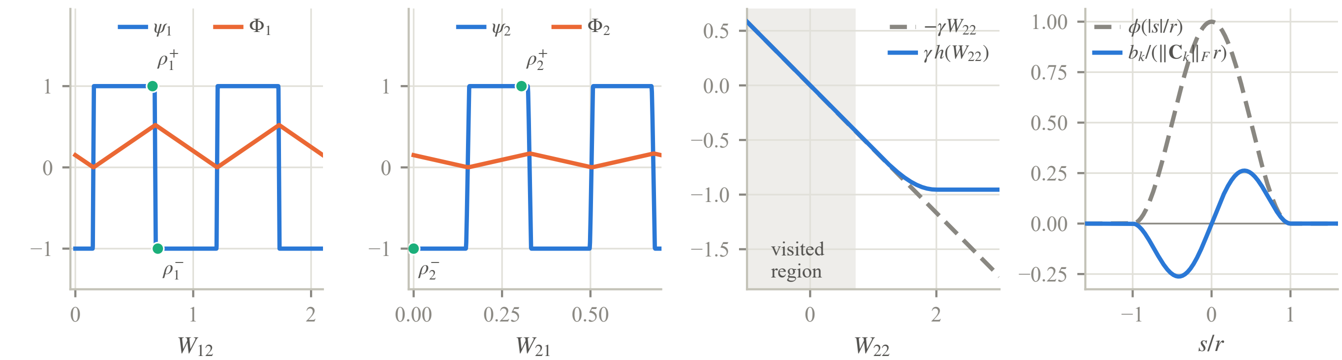

Figure A1: The components of the objective $\tilde{f} = g + \sum_k b_k$ of Part 3, as implemented. *First two panels:* the periodic ramps ψ_1, ψ_2 and their bounded antiderivatives Φ_1, Φ_2 over two periods; the marked residues ρ_i^+ (kept at odd iterates) and ρ_i^- (even iterates) lie on the plateaus where $\psi_i = \pm 1$ exactly, so from $t \geq 4$ the off-diagonal gradient entries alternate between $+A$ and $-A$. The drawn ψ_i is the implementation's ramp; the trajectory samples only the plateaus, on which any C^∞ choice agrees with it. *Third panel:* the divergence term $-\gamma W_{22}$ and its bounded replacement $\gamma\, h(W_{22})$ of Remark 1; the two agree on the visited region $\{W_{22} \leq \frac{7}{10}\}$ (shaded), so the floor changes no iterate while restoring Assumption 1. *Fourth panel:* the correction b_k of (28) along the ray $\mathbf{Z}_k + s\, \mathbf{C}_k / \|\mathbf{C}_k\|_F$, normalized by $\|\mathbf{C}_k\|_F\, r$, with the cutoff ϕ: b_k vanishes at $\mathbf{Z}_k$ while $\nabla b_k(\mathbf{Z}_k) = \mathbf{C}_k$, and its support, the ball $\{\|\mathbf{W} - \mathbf{Z}_k\|_F \leq r\}$, contains no iterate other than its own center.

Each residual $\mathbf{D}_t - \mathbf{d}_{t-1}^{\text{est}}$ has strictly nonzero entries, so the signs are unambiguous; e.g. at $t = 3$ it is $\frac{1}{100}\begin{pmatrix} -7 & -61 \\ -131 & -63 \end{pmatrix}$, all negative. From $t = 3$ the targets are 2-periodic and the pair $(\mathbf{d}_B, \mathbf{d}_A)$ reproduces itself: $\bar{\mathbf{D}}^+ - \mathbf{d}_B$ and $-(\bar{\mathbf{D}}^- - \mathbf{d}_A)$ are both entrywise positive with mean $\frac{24}{25}$, so $\mathbf{d}_B + \frac{24}{25}\mathbf{J} = \mathbf{d}_A$ and $\mathbf{d}_A - \frac{24}{25}\mathbf{J} = \mathbf{d}_B$ ($\mathbf{J}$ all-ones). ■

Since $\mathbf{X}_t = \mathbf{X}_{t-1} - \mathbf{d}_t^{\text{est}}$ (recall $\eta = 1$), over one period the $(2,2)$-coordinate changes by $-(\mathbf{d}_A + \mathbf{d}_B)_{22} = -\frac{7}{20}$. Consequently a term $-\gamma W_{22}$ in $\tilde{f}$, with $\gamma := \frac{7}{12}$, increases by $\gamma \cdot \frac{7}{20} = \frac{49}{240}$ per period. This is the divergence, provided a genuine smooth function produces the targets (22); Part 3 constructs one.

Part 3: realization by a smooth function. Fix (μ, variant) and set $\gamma = \frac{7}{12}$, $\nu = \frac{1}{1+2\mu}$, and

$$A = \frac{1+\mu}{1-\mu} \text{ (standard)}, \qquad A = \frac{1+\mu}{(1-\mu)(1+2\mu)} \text{ (Nesterov)}. \tag{26}$$

We build $\tilde{f} = g + \sum_{k=1}^{3} b_k$ from a periodic-plus-linear *field* g and three *localized corrections* b_k, all explicit.

The field is

$$g(\mathbf{W}) = -\gamma\, W_{22} + A\, \Phi_1(W_{12}) + A\, \Phi_2(W_{21}), \tag{27}$$

where $\Phi_i(w) = \int_0^w \psi_i$ and $\psi_i \colon \mathbb{R} \to [-1, 1]$ is a fixed C^∞, p_i-periodic function with $\int_0^{p_i} \psi_i = 0$,

$$p_1 = \tfrac{21}{20}, \quad p_2 = \tfrac{7}{20}, \qquad \delta = \tfrac{1}{100},$$

equal to $+1$ on $[\rho_i^+ - \delta, \rho_i^+ + \delta]$ and -1 on $[\rho_i^- - \delta, \rho_i^- + \delta]$ (mod p_i), where

$$\rho_1^+ = \tfrac{131}{200},\ \rho_1^- = \tfrac{140}{200}; \qquad \rho_2^+ = \tfrac{61}{200},\ \rho_2^- = 0.$$

The two intervals are disjoint mod p_i (their centers are $\frac{9}{200} > 2\delta$ apart), so such a ψ_i exists; the zero-mean condition, met by balancing the rest of the period, makes Φ_i periodic (hence bounded). On each of the two intervals $\Phi_i' = \psi_i = \pm 1$ exactly. Figure A1 draws both ramps, together with the remaining components of $\tilde{f}$.

The corrections pin the first three gradients. Fix once a C^∞ cutoff $\phi \colon \mathbb{R} \to [0, 1]$ with $\phi \equiv 1$ on $[0, \frac{1}{2}]$ and $\phi \equiv 0$ on $[1, \infty)$, put $r = \frac{1}{50}$, and set

$$b_k(\mathbf{W}) = \big\langle \mathbf{C}_k,\ \mathbf{W} - \mathbf{Z}_k \big\rangle\, \phi\big(\|\mathbf{W} - \mathbf{Z}_k\|_F / r\big), \tag{28}$$

centered at the first three iterates (from Lemma 3)

$$\mathbf{Z}_1 = \mathbf{0}, \quad \mathbf{Z}_2 = \tfrac{1}{10}\begin{pmatrix} 7 & -7 \\ 7 & 7 \end{pmatrix}, \quad \mathbf{Z}_3 = \tfrac{1}{20}\begin{pmatrix} 7 & -7 \\ 7 & 7 \end{pmatrix}.$$

Each b_k is C^∞, supported in $\{\|\mathbf{W}-\mathbf{Z}_k\|_F \le r\}$; since its linear factor vanishes at $\mathbf{Z}_k$ while $\phi(0)=1$, $\nabla b_k(\mathbf{Z}_k)=\mathbf{C}_k$. We choose $\mathbf{C}_k := \hat{\mathbf{G}}_k - \nabla g(\mathbf{Z}_k)$ with the explicit $\hat{\mathbf{G}}_k$ of (29) below, so that $\nabla \tilde f(\mathbf{Z}_k)=\hat{\mathbf{G}}_k$. Every term of $\tilde f$ has a bounded Hessian, so $\tilde f$ is C^∞ and $\tilde L$-smooth for a finite $\tilde L(\mu, \text{variant})$.

Lemma 4 (Realization) *For every $\mu \in [0,1)$ and either variant, EF21-SignMuon on this $\tilde f$ ($\eta=1$, from $\mathbf{0}$) generates exactly the targets* (22).

Proof. *The required gradients.* The buffer recursion $\mathbf{M}_t = \mu\mathbf{M}_{t-1} + (1-\mu)\mathbf{G}_t$ of (5) can be solved for the gradients: prescribing the buffers $(\mathbf{M}_t)_{t\ge1}$ forces $\mathbf{G}_t = (\mathbf{M}_t - \mu\mathbf{M}_{t-1})/(1-\mu)$, and these are the gradients the function must deliver. Define accordingly the *transient gradients*

$$\hat{\mathbf{G}}_t = \frac{\mathbf{M}_t - \mu\mathbf{M}_{t-1}}{1-\mu} \quad (t=1,2,3;\ \mathbf{M}_0=\mathbf{0}) \tag{29}$$

and the *field gradient* $\mathbf{G}^\pm = \left(\begin{smallmatrix} 0 & \pm A \\ \pm A & -\gamma \end{smallmatrix}\right)$, where the prescribed buffer values $\mathbf{M}_{1,2,3}$ are given below. With $\bar{\mathbf{M}}^\pm = \left(\begin{smallmatrix} 0 & \pm 1 \\ \pm 1 & -\gamma \end{smallmatrix}\right)$, the factorization

$$\begin{gathered}\bar{\mathbf{M}}^\pm = \bar{\mathbf{D}}^\pm \left(\begin{smallmatrix} 24/25 & \mp 7/25 \\ \mp 7/25 & 337/300 \end{smallmatrix}\right) \\ (\text{second factor } \succ 0,\ \det = 1)\end{gathered} \tag{30}$$

shows $\operatorname{polar}(\bar{\mathbf{M}}^\pm)=\bar{\mathbf{D}}^\pm$, with singular values $\frac34$ and $\frac43$; this is the condition number $\frac{16}{9}$ cited in the proof idea.

Standard momentum ($\tilde{\mathbf{M}}_t = \mathbf{M}_t$). Take $\mathbf{M}_1=\mathbf{S}_1, \mathbf{M}_2=\mathbf{S}_2, \mathbf{M}_3=\bar{\mathbf{M}}^-$; then $\hat{\mathbf{G}}_{1,2,3}$ are the explicit matrices (29). Feeding $\mathbf{G}^{(-1)^t}$ for $t\ge4$ keeps $\mathbf{M}_t=\bar{\mathbf{M}}^{(-1)^t}$, because $(1-\mu)\mathbf{G}^\pm = \bar{\mathbf{M}}^\pm - \mu\bar{\mathbf{M}}^\mp$ with A as in (26). As $\operatorname{polar}(\mathbf{S}_1)=\mathbf{S}_1$ (orthogonal) and $\operatorname{polar}(\mathbf{S}_2)=\mathbf{S}_2$ (rank one), (30) yields the targets (22).

Nesterov momentum ($\tilde{\mathbf{M}}_t = (1+\mu)\mathbf{M}_t - \mu\mathbf{M}_{t-1}$). It is here that the orthogonality of $\mathbf{S}_1$ is needed. The Nesterov direction involves two consecutive buffers, so prescribing the tail $t\ge4$ already fixes $\mathbf{M}_3$, and only the preamble remains to absorb the mismatch at $t=3$; and the set of matrices with a given polar factor $\mathbf{D}$ is $\{\mathbf{DH} : \mathbf{H}\succ 0\}$, which is three-dimensional when $\mathbf{D}$ is orthogonal but only one-dimensional (a positive scalar) when $\mathbf{D}$ has rank one. A preamble of two rank-one targets leaves too little freedom: a symbolic check shows that it admits no realization once $\mu \gtrsim 0.3$. One orthogonal target supplies enough freedom, of which the construction uses a single scalar, the τ below. Take $\mathbf{M}_1 = \frac{1}{1+\mu}\mathbf{S}_1\mathbf{H}_1$, $\mathbf{M}_2=\frac{1}{1+\mu}(\mathbf{S}_2+\mu\mathbf{M}_1)$, $\mathbf{M}_3=\bar{\mathbf{M}}'^-$ and $\mathbf{M}_t=\bar{\mathbf{M}}'^{(-1)^t}$ ($t\ge4$), where $\bar{\mathbf{M}}'^\pm = \left(\begin{smallmatrix} 0 & \pm\nu \\ \pm\nu & -\gamma \end{smallmatrix}\right)$, $\mathbf{H}_1=\operatorname{diag}(1,1+\tau)$, and, for $\mu>0$,

$$\tau = \left(\tfrac{140(1+\mu)^2}{1+2\mu} - 44\right)\tfrac{1+\mu}{117\mu} > 0.$$

(At $\mu=0$ the Nesterov rule reads $\tilde{\mathbf{M}}_t=\mathbf{M}_t$ and is the standard case already treated, so nothing is left to prove there.) Then $\tilde{\mathbf{M}}_1=\mathbf{S}_1\mathbf{H}_1$ and $\tilde{\mathbf{M}}_2=\mathbf{S}_2$ have polar factors $\mathbf{S}_1,\mathbf{S}_2$, and $\tilde{\mathbf{M}}_t=\bar{\mathbf{M}}^{(-1)^t}$ for $t\ge4$ (since $\nu(1+2\mu)=1$). The one nontrivial step is $t=3$. Set $\mathbf{H}_3 := \bar{\mathbf{D}}^-\tilde{\mathbf{M}}_3$; since $(\bar{\mathbf{D}}^-)^2=\mathbf{I}$, this is the same as $\tilde{\mathbf{M}}_3=\bar{\mathbf{D}}^-\mathbf{H}_3$. The stated τ is exactly the value making $\mathbf{H}_3$ symmetric, and $\mathbf{H}_3\succ0$ for all $\mu\in(0,1)$: under $\mu=\frac{s}{1+s}$ ($s>0$) the numerators of its two leading minors are polynomials in s with nonnegative coefficients. Hence $\operatorname{polar}(\tilde{\mathbf{M}}_3)=\bar{\mathbf{D}}^-$, completing (22).

The function delivers these gradients. It remains to verify $\nabla\tilde f(\tilde{\mathbf{X}}_{t-1})=\hat{\mathbf{G}}_t$ for $t\le3$ and $=\mathbf{G}^{(-1)^t}$ for $t\ge4$, by induction along the run: as long as the gradients match this prescription, the iterates are those computed in Part 2, and the prescription need only be checked at those points. By Lemma 3 the iterates $\tilde{\mathbf{X}}_0,\tilde{\mathbf{X}}_1,\tilde{\mathbf{X}}_2$ are exactly the centers $\mathbf{Z}_1,\mathbf{Z}_2,\mathbf{Z}_3$, where $\nabla\tilde f=\hat{\mathbf{G}}_{1,2,3}$ by construction. The three balls are disjoint ($\|\mathbf{Z}_j-\mathbf{Z}_k\|_F\ge\frac{7}{10}>2r$), and every later iterate has $(1,2)$-entry $\ge\frac{131}{200}$ while the centers have it ≤0, so no ball is re-entered. For $t\ge4$ the query lies in the field, where

$$\nabla g(\mathbf{W}) = \begin{pmatrix} 0 & A\psi_1(W_{12}) \\ A\psi_2(W_{21}) & -\gamma \end{pmatrix}.$$

The period shift $\tilde{\mathbf{X}}_{t+2}-\tilde{\mathbf{X}}_t = -(\mathbf{d}_A+\mathbf{d}_B)$ advances W_{12} by exactly $+p_1$ and W_{21} by exactly $-p_2$ per period, so the two coordinates travel to $+\infty$ and $-\infty$ respectively while, mod p_i, they hold the residues ρ_i^+ at odd indices and ρ_i^- at even ones; there $\psi_i=\pm1$, giving $\nabla g=\mathbf{G}^{(-1)^t}$. ■

Proof of Theorem 4. By Lemma 4 the normalized run produces the targets (22), so by Lemma 3 its estimate locks onto the wrong-sign cycle and $\tilde{\mathbf{X}}_{t+2}-\tilde{\mathbf{X}}_t$ is a constant shift. Along it Φ_1,Φ_2 return to their values and the corrections b_k vanish, so only the linear term acts: $\tilde f(\tilde{\mathbf{X}}_{t+2})-\tilde f(\tilde{\mathbf{X}}_t) = -\gamma(-\frac{7}{20}) = \frac{49}{240}>0$. By Lemma 1, f then obeys (21) with $c=\frac{49}{240\tilde L}$. As $(L,\eta,\mu,\text{variant})$ were arbitrary, no rule $\eta=\eta(L,\mu)$ can prevent divergence. ■

Remark 1 (Boundedness below) *Only the linear term $-\gamma W_{22}$ makes $\tilde f$ unbounded below, and only as $W_{22}\to+\infty$, a region the iterates never enter, since $(\tilde{\mathbf{X}}_t)_{22}\le\frac{7}{10}$ throughout (it decreases after the transient). Replacing $-\gamma W_{22}$ by any C^∞ function that agrees with it on $\{W_{22}\le1\}$ and is constant on $\{W_{22}\ge2\}$ therefore leaves the whole trajectory, and* (21) *with it, unchanged while rendering $\tilde f$ bounded below (Assumption 1); the theorem is stated with this modification in force.*

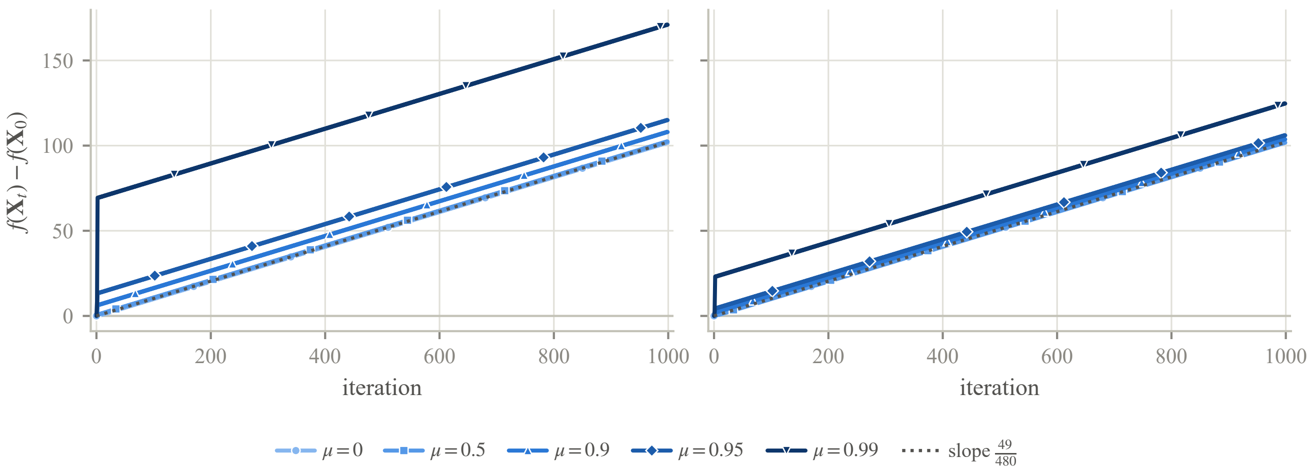

Figure A2: Momentum does not prevent the divergence of EF21-SignMuon. For each momentum coefficient μ and variant, EF21-SignMuon is run on the corresponding instance of Theorem 4 and $f(\mathbf{X}_t) - f(\mathbf{X}_0)$ is plotted; *left:* standard momentum, *right:* Nesterov. Every setting diverges at the common rate $\frac{49}{480}$ (dotted). Subtracting $f(\mathbf{X}_0)$ removes the only genuinely μ-dependent offset (the field constant $A(\mu)$ scales a bounded periodic term); what remains is a bounded transient, largest as $\mu \to 1$, on top of the shared linear divergence.

Remark 2 (The construction is not convex) *Theorems 1–3 run on a linear, hence convex, objective; $\tilde{f}$ is nonconvex, through the periodic terms $A\Phi_i$ and the corrections b_k. The nonconvexity is forced by the run rather than chosen by the realization: along the divergent trajectory,*

$$\langle \nabla \tilde{f}(\tilde{\mathbf{X}}_6) - \nabla \tilde{f}(\tilde{\mathbf{X}}_3),\ \tilde{\mathbf{X}}_6 - \tilde{\mathbf{X}}_3 \rangle = -\tfrac{9A}{50} < 0,$$

violating the gradient monotonicity that every convex function obeys, so no *convex function generates these iterates and gradients. The theorem is stated under Assumptions 1–2 because that is where it is needed: EF21-MuonUSign and EF21-MuonSign converge under exactly these hypotheses (Theorem 5), so the divergence and the guarantees concern one problem class. Whether some convex instance, necessarily through a different trajectory, also defeats EF21-SignMuon we leave open.*

Remark 3 (Verification) *The construction is checked in two independent ways: symbolically, in exact rational arithmetic, and numerically, by running the float64 reference implementation of Algorithm A3 on the assembled $\tilde{f}$. The right panel of Figure 1 confirms that at $\mu = 0$ EF21-SignMuon is the only one of the eight methods that diverges, the others (SignMuon, MuonUSign, MuonSign, EF21-MuonUSign, EF21-MuonSign, SignSGD, Muon) staying bounded; Figure A2 confirms that EF21-SignMuon diverges at the exact rate $\frac{49}{480}$ for every $\mu \in \{0, \frac{1}{2}, 0.9, 0.95, 0.99\}$ under both standard and Nesterov momentum, as the reduction predicts.*

A.10 Convergence of EF21-MuonUSign and EF21-MuonSign

We do not analyse the two error-feedback methods from scratch. EF21-MuonUSign and EF21-MuonSign are *exact instances* of EF21-Muon (Gruntkowska et al. 2025, Algorithm 3), already analysed in the layer-wise, stochastic, federated setting. Three conditions must be verified before its guarantees transfer: our step is their LMO step, our loop is their loop (Proposition 2), and our messages come from contractive compressors (Lemma 5). Only the last is non-trivial. Table A1 is the change of variables.

Notation and constants. For the layer tuple $\mathbf{X} = [\mathbf{X}_1, \dots, \mathbf{X}_p]$, $\mathbf{X}_i \in \mathbb{R}^{m_i \times n_i}$, of the Problem Statement write $d_i := m_i n_i$, $r_i := \min(m_i, n_i)$, $d_{\max} := \max_i d_i$; a second subscript selects a layer ($\mathbf{X}_{t,i}$, $\mathbf{g}_{t,i}$). We use $\|\mathbf{Y}\|_{2\to 2} \le \|\mathbf{Y}\|_F \le \|\mathbf{Y}\|_* \le \sqrt{\operatorname{rank} \mathbf{Y}}\, \|\mathbf{Y}\|_F$. Smoothness constants: L_i for f and $L_{i,j}$ for f_j in Assumption 2, with $\tilde{L}_i^2 := \frac{1}{N}\sum_j L_{i,j}^2$ and $L := \max_i L_i$; in the (L^0, L^1) form the pairs are again per layer for f and per layer and client for f_j, with $L^1_{i,\max} := \max_j L^1_{i,j}$.

Assumption 3 (Stochastic gradient) *Each client's stochastic gradient is unbiased, $\mathbb{E}_\xi[\nabla f_j(\mathbf{X}; \xi)] = \nabla f_j(\mathbf{X})$, with bounded variance $\mathbb{E}_\xi \|\nabla f_j(\mathbf{X}; \xi) - \nabla f_j(\mathbf{X})\|_*^2 \le \sigma^2$.*

Assumptions 1–3 are Assumptions 1–2 and 6–10 of Gruntkowska et al. (2025) with the layer norms taken spectral; our variance bound is stated in the nuclear norm and implies theirs via $\|\cdot\|_F \le \|\cdot\|_*$. The clause $f_j \ge f_j^*$ of Assumption 1 is needed only for Corollary 2.

Main result

Theorem 5 (Convergence of the EF21 methods) *Run the federated Algorithm A9 with the EF21 uplink and EMA momentum $\mu \in [0,1)$. Then:*

(i) (smooth; EF21-MuonUSign and EF21-MuonSign) *under Assumptions 1–3, with the "sharp" learning rate $\eta_{t,i} = \gamma_i \|\mathbf{g}_{t,i}\|_*$ and tuned (γ_i, μ), both methods reach $\frac{1}{T}\sum_{t<T} \mathbb{E}\|\nabla f(\mathbf{X}_t)\|_*^2 = \mathcal{O}(T^{-1/2})$ (Corollary 1);*

(ii) (generalized smooth; EF21-MuonUSign only) *under (L^0, L^1)-smoothness, EF21-MuonUSign with a plain* constant *learning rate reaches $\min_{t\le T} \sum_i \mathbb{E}\|\nabla_i f(\mathbf{X}_t)\|_* = \mathcal{O}(T^{-1/4})$ (Corollary 2).*

Both statements are written for a per-layer constant common to all layers ($\gamma_i \equiv \gamma$, $\eta_i \equiv \eta$); for unequal constants the left-hand sides carry the step-size weights of Corollaries 1–2.

Part (i) implies the rate announced in the main text, since the minimum over $t \le T$ is at most the average. The centralized Algorithms A6–A7 are the federated method at $N = 1$, so both parts cover the centralized runs as well.

The reduction

The step is their LMO step. For $\mathbf{G} = \mathbf{U\Sigma V}^\top$ the framework's oracle over the spectral ball of radius τ is $\mathrm{lmo}(\mathbf{G}) = \mathbf{X} - \tau \mathbf{U}\mathbf{V}^\top$, which is our server step with $\tau = \eta_{t,i}$, since $\mathbf{D}_{t,i} = -A(\mathbf{g}_{t,i}) = \mathbf{U}_{t,i}\mathbf{V}_{t,i}^\top$:

$$\mathbf{X}_{t,i} = \mathbf{X}_{t-1,i} - \eta_{t,i}\,\mathbf{D}_{t,i} = \mathrm{lmo}_{\mathcal{B}(\mathbf{X}_{t-1,i},\eta_{t,i})}(\mathbf{g}_{t,i}). \tag{31}$$

Their "sharp" step $\mathbf{X} - \gamma\,\mathbf{G}^\sharp$ uses $\mathbf{G}^\sharp = \|\mathbf{G}\|_* \mathbf{U}\mathbf{V}^\top$, so a constant γ_i amounts to the schedule $\eta_{t,i} = \gamma_i\|\mathbf{g}_{t,i}\|_*$ and a constant $\eta_{t,i}$ to the plain Muon rate: the two differ as learning-rate choices, not as algorithms, and either run is an instance of the framework. The guarantees, however, are attached to specific choices. Part (i) of Theorem 5 assumes the nuclear-norm schedule, which our implementation does not use; what the corollaries then cover of the constant-rate runs we actually perform is recorded in the Scope paragraph below. The spectral norm-equivalence constants are $\underline{\rho}_i = 1$, $\bar{\rho}_i = \sqrt{r_i}$, defined by $\underline{\rho}_i\|\mathbf{Y}\|_{2\to2} \le \|\mathbf{Y}\|_F \le \bar{\rho}_i\|\mathbf{Y}\|_{2\to2}$. At rank-deficient $\mathbf{G}$ the LMO is non-unique, but any selection serves: writing $\mathbf{\Delta} := \mathrm{lmo}_{\mathcal{B}(\mathbf{X},\tau)}(\mathbf{G}) - \mathbf{X}$ for the displacement it produces, the analysis uses only $\langle \mathbf{G}, \mathbf{\Delta}\rangle = -\tau\|\mathbf{G}\|_*$ and $\|\mathbf{\Delta}\|_{2\to2} \le \tau$, with $\mathbf{G} = \mathbf{0}$ read as the zero step ($\mathbf{0}^\sharp = \mathbf{0}$).

The momentum is their momentum. Algorithm A9 uses $\mathbf{M}_t = \mu\mathbf{M}_{t-1} + (1-\mu)\mathbf{G}_t$, which is (5) and is the framework's momentum with $\beta = 1-\mu$. Rescaling the momentum stream by a constant, as the heavy-ball convention does by the factor $1/(1-\mu)$, alters nothing: the EF21 recursion is positively homogeneous and the Muon LMO scale-invariant, so the factor leaves the iterates unchanged under a constant $\eta_{t,i}$ and is absorbed into γ_i under the "sharp" schedule. (The Nesterov branch is a different filter; see the end of this appendix.)

Proposition 2 (Exact instance) *Fix μ, a learning-rate schedule, and the LMO selection above. Started from $\mathbf{M}_0^{(j)} = \mathbf{g}_0^{(j)} = \mathbf{g}_0 = \mathbf{0}$, $\mathbf{W}_0 = \mathbf{X}_0$, Algorithm A9 with the EF21 uplink and either downlink mode ($\mathcal{C}^\downarrow \in \{$exact, EF21-P$\}$) produces the same trajectory as Algorithm 3 of Gruntkowska et al. (2025) with spectral norms, scaled-sign worker compressors, identity/scaled-sign server compressor, $\beta = 1-\mu$, and radii $t_i^k = \eta_{k,i}$, up to a one-step index shift $X^{t+1} = \mathbf{X}_t$.*

Proof The loops differ only in where the round is cut: they order it *step → downlink → gradient → uplink*, we order it *downlink → gradient → uplink → step*. Their iteration $k = 0$ is vacuous under our initialization: $\mathbf{g}_0 = \mathbf{0}$ gives $X^1 = X^0 = \mathbf{X}_0$ and a zero downlink residual, so $W^1 = \mathbf{X}_0$. Thereafter their iteration $k = t$ performs our round t verbatim, with the same momentum, the same compressed residual and the same LMO step (31), giving $X^{t+1} = \mathbf{X}_t$, $W^{t+1} = \mathbf{W}_t$ by induction. Running their method for $K = T+1$ iterations therefore yields $\{\mathbf{X}_0, \mathbf{X}_0, \mathbf{X}_1, \dots, \mathbf{X}_{T-1}\}$, and any average or minimum over these equals ours up to one duplicated nonnegative term. ■

Zero initialization also makes their initial-error constant Ψ^0 explicit: the $\|\mathbf{M}_0 - \mathbf{g}_0\|$ term vanishes and the gradient-deviation terms become $\|\nabla_i f(\mathbf{X}_0)\|$. The β^{-1} surviving in Ψ^0 enters only through the Ψ^0/T term, which $T^{-1/2}$ dominates.

The scaled sign is a contractive compressor The framework requires every transmitted message to originate in a *contractive compressor*: a map with $\mathbb{E}\|\mathcal{C}(\mathbf{Y}) - \mathbf{Y}\|^2 \le (1-\alpha)\|\mathbf{Y}\|^2$ for some $\alpha \in (0,1]$ (Gruntkowska et al. 2025, Def. 1). This is precisely the property that fails for a bare sign and holds once it is scaled.

Lemma 5 (Contractivity of the scaled sign) *For every $\mathbf{Y} \in \mathbb{R}^{m\times n}$ ($d = mn$), the scaled sign $\mathcal{C}(\mathbf{Y}) = \mathrm{mean}(|\mathbf{Y}|)\,\mathrm{sign}(\mathbf{Y})$, with exact zeros resolved to ±1 as in Section 4, satisfies*

$$\|\mathcal{C}(\mathbf{Y}) - \mathbf{Y}\|_F^2 = \|\mathbf{Y}\|_F^2 - \tfrac{1}{d}\|\mathbf{Y}\|_1^2 \le \left(1 - \tfrac{1}{d}\right)\|\mathbf{Y}\|_F^2, \tag{32}$$

so $\mathcal{C}$ is Euclidean-contractive with $\alpha = 1/d$. The identity holds for every draw of the random signs, not merely in expectation, and gives the exact contraction $\alpha(\mathbf{Y}) = \|\mathbf{Y}\|_1^2/(d\|\mathbf{Y}\|_F^2)$, which is $\Theta(1)$ for dense $\mathbf{Y}$ (e.g. $\to 2/\pi$ for i.i.d. Gaussian entries) and equals $1/d$ exactly at any 1-sparse $\mathbf{Y}$.

	This paper	EF21-Muon
clients / rounds	N, T	$n, K = T+1$
iterate	$\mathbf{X}_t$	X^{t+1}
momentum	μ	$1-\beta$
learning rate	$\eta_{t,i}$	radius t_i^t
norm equivalence	$1,\ \sqrt{r_i}$	$\underline{\rho}_i,\ \bar{\rho}_i$
uplink compr.	scaled sign	$\mathbb{B}_2(\alpha_D)$
downlink compr.	exact / sign	$\mathcal{I}$ / $\mathbb{B}_2(\alpha_P)^\dagger$
compression α	$1/d_{\max}$	α_D

Table A1: Change of variables from our notation to that of Gruntkowska et al. (2025). Layer norms $\|\cdot\|_{(i)}, \|\cdot\|_{(i)\star}$ are read as $\|\cdot\|_{2\to 2}, \|\cdot\|_*$ and smoothness constants $L_i^0, \tilde{L}_i^0$ as $L_i, \tilde{L}_i$; the index shift is Proposition 2. †Their Theorem 19 as printed asks for a server compressor in $\mathbb{B}(\alpha_P)$, contractive in the *layer* norm; the Euclidean class $\mathbb{B}_2(\alpha_P)$, to which the scaled sign does belong, is admitted by their Remark 23, which introduces the factor $\bar{\rho}_i^2$ appearing in Corollary 1 and Remark 5.

Proof Write $c := \|\mathbf{Y}\|_1/d$ and $s_{kl} = \pm 1$ for the transmitted signs. A nonzero entry contributes $(|Y_{kl}| - c)^2$ and a zero entry contributes $(c\, s_{kl})^2 = c^2$ whichever sign was drawn, so $\|\mathcal{C}(\mathbf{Y}) - \mathbf{Y}\|_F^2 = \|\mathbf{Y}\|_F^2 - 2c\|\mathbf{Y}\|_1 + c^2\|\mathbf{Y}\|_0 + c^2(d - \|\mathbf{Y}\|_0) = \|\mathbf{Y}\|_F^2 - 2c\|\mathbf{Y}\|_1 + dc^2$, which is (32) on substituting c. The bound then follows from $\|\mathbf{Y}\|_1 \ge \|\mathbf{Y}\|_F$, with equality exactly at the 1-sparse $\mathbf{Y}$; the Gaussian limit uses $\mathbb{E}|Y_{kl}| = \sqrt{2/\pi}\,\sigma$. ■

The randomized $\operatorname{sign}(0)$ is what makes (32) hold with equality for every $\mathbf{Y}$: under the ternary convention the $\|\mathbf{Y}\|_0$-terms in the proof do not cancel, the error depends on the sparsity of $\mathbf{Y}$, and $\alpha = 1/d$ becomes an infimum rather than an attained value.

The scaling is essential to Lemma 5: a bare sign contracts for no α at all, since as $\|\mathbf{Y}\|_F \to 0$ on a fixed support, $\|\operatorname{sign}(\mathbf{Y}) - \mathbf{Y}\|_F \to \sqrt{\|\mathbf{Y}\|_0}$. That is the dividing line between our divergent and convergent methods: the majority-vote methods transmit unscaled signs of full quantities, the EF21 variants the scaled sign of a *residual*, at the cost of one extra scalar per layer per round. Per layer the uplink scaled sign lies in $\mathbb{B}_2(1/d_i) \subseteq \mathbb{B}_2(\alpha)$ with $\alpha := 1/d_{\max}$; the downlink is exact for EF21-MuonUSign ($\alpha_P = 1$) and the same scaled sign for EF21-MuonSign.

Transferred guarantees All requirements hold, so the framework's theorems apply through Table A1. We state the rates and stepsize rules; the explicit non-asymptotic bounds are those of Theorems 19 (smooth) and 24 ((L^0, L^1)-smooth) of Gruntkowska et al. (2025), evaluated at the constants of Table A1.

Corollary 1 (Smooth case; EF21-MuonUSign and EF21-MuonSign) *Let Assumptions 1–3 hold. Run EF21-MuonUSign or EF21-MuonSign (Algorithm A9 with the EF21 uplink and the exact or the scaled-sign downlink, respectively) with EMA momentum and $\eta_{t,i} = \gamma_i\|\mathbf{g}_{t,i}\|_*$, for any γ_i below the per-layer threshold of Gruntkowska et al. (2025, Thm. 19) under Table A1. Then, applying the momentum tuning of Gruntkowska et al. (2025, Cor. 2) per layer (their Corollary 1 supplies the layer-wise initialization, their Corollary 2 the tuning at p = 1),*

$$\tfrac{1}{T}\textstyle\sum_{t<T}\sum_{i=1}^p w_i\,\mathbb{E}\|\nabla_i f(\mathbf{X}_t)\|_*^2 = \mathcal{O}(T^{-1/2}),$$

with $w_i := \gamma_i/(\frac{1}{p}\sum_l \gamma_l)$; for a common γ_i all $w_i = 1$ and the left side is $\frac{1}{T}\sum_{t<T}\mathbb{E}\|\nabla f(\mathbf{X}_t)\|_*^2$.

Proof By Proposition 2 the run is an instance of Algorithm 3 with $K = T+1$, and by Lemma 5 its compressors satisfy $\alpha_D = \alpha$ and $\alpha_P \in \{1, \alpha\}$ (EF21-MuonSign's Euclidean downlink is admissible by their Remark 23, which multiplies the α_P-terms of the threshold constant ζ_i by $\bar{\rho}_i^2 = r_i$). Substituting $\underline{\rho}_i = 1, \bar{\rho}_i = \sqrt{r_i}$ into Gruntkowska et al. (2025, Thm. 19) gives the threshold and the rate; the duplicated $\mathbf{X}_0$-term changes the constant by at most the factor $\frac{T+1}{T}$. ■

The threshold in question is $\gamma_i \le (2L_i + 2\sqrt{\zeta_i})^{-1}$, with ζ_i the constant of Gruntkowska et al. (2025, Thm. 19). Under Table A1, ζ_i grows polynomially in the layer dimension, through $\bar{\rho}_i^2 = r_i$ and the uplink $\alpha = 1/d_{\max}$; for EF21-MuonSign its α_P-terms carry the additional factor r_i of Remark 23, shrinking the admissible γ_i by a further $\sqrt{r_i}$. That additional factor is structural: the scaled sign is spectrally contractive for no parameter at all (Remark 5).

Corollary 2 ((L^0, L^1) case; EF21-MuonUSign) *Let Assumption 2 hold in its (L^0, L^1) form. Run EF21-MuonUSign (exact downlink) with $S := T+2$, momentum $\mu = 1 - S^{-1/2}$, and the* constant *per-layer rate $\eta_{t,i} \equiv \eta_i/S^{3/4}$ for any $\eta_i \le 1$ with $\eta_i^2\sqrt{r_i}\,(L_{i,\max}^1)^2 = \mathcal{O}(1)$.*[1] *Then*

$$\min_{0\le t\le T}\textstyle\sum_{i=1}^p v_i\,\mathbb{E}\|\nabla_i f(\mathbf{X}_t)\|_* = \mathcal{O}(T^{-1/4}), \quad v_i := \frac{\eta_i}{\frac{1}{p}\sum_l \eta_l}.$$

[1] Gruntkowska et al. (2025, Thm. 24) imposes four upper bounds on η_i^2; at $\beta = (K+1)^{-1/2}$ the two quoted here are the binding ones, the first bound relaxing as T grows. Their second bound, as printed, carries a factor $K+1$ in the denominator that the proof does not use: the display it is chosen to ensure requires only $\eta_i^2 \le (1 - \sqrt{1-\alpha_D})\,\underline{\rho}_i\,(K+1)^{1/2}/(24\sqrt{1-\alpha_D}\,\bar{\rho}_i(L_{i,\max}^1)^2)$, which also relaxes as T grows, and this is the requirement we work from.

Proof Their Theorem 24 requires the identity server compressor, i.e. the exact downlink of EF21-MuonUSign (so EF21-MuonSign is excluded). Applying it with $K = T + 1$, $\beta = S^{-1/2}$ and Table A1 gives the schedule and the rate. ■

Under plain smoothness ($L^1 = 0$) every constraint above collapses to $\eta_i \le 1$. The *constant*-learning-rate EF21-MuonUSign we actually run is therefore covered as it stands, and at no loss of order: $\mathcal{O}(T^{-1/4})$ for the norm corresponds to $\mathcal{O}(T^{-1/2})$ for its square, the quantity of part (i). What the "sharp" schedule changes is the bounded quantity, an average of the squared norm in place of a minimum of the norm, not the effective speed.

Scope The reduction covers the two error-feedback methods and no others, which matches our negative results. The majority-vote methods send unscaled signs and fall outside the framework: their server aggregates by a vote, $\mathrm{sign}(\sum_j \mathbf{s}_t^{(j)})$, where Algorithm 3 averages, and it is the average that Proposition 2 matches. EF21-SignMuon compresses the LMO *output* rather than the gradient, so it tracks the non-Lipschitz polar factor and the momentum-tracking step breaks (Theorem 4). EF21-MuonSign gets the smooth guarantee but not the (L^0, L^1) one, whose theory assumes an uncompressed downlink.

One further gap is ours and not the framework's. Corollary 1 reaches EF21-MuonSign only under the "sharp" schedule $\eta_{t,i} = \gamma_i \|\mathbf{g}_{t,i}\|_*$, while Corollary 2, the constant-rate statement, excludes it for want of an identity server compressor. Our experiments run a cosine-annealed constant η_0, so the EF21-MuonSign runs are covered by neither, whereas constant-rate EF21-MuonUSign is covered by Corollary 2 at $L^1 = 0$. The trajectories remain exact instances of Algorithm 3 either way; what the constant rate costs is the step-size hypothesis of the rate, not the reduction.

The remaining gap is the Nesterov branch, which our language-model runs use (Appendix A.16): with $\beta := 1 - \mu$ it steers by $\tilde{\mathbf{M}}_t = (1-\beta)\mathbf{M}_t + \beta \mathbf{G}_t$ rather than by the buffer $\mathbf{M}_t$, and the framework's analysis is written for a direction that itself satisfies the recursion $\mathbf{M}_t = (1-\beta)\mathbf{M}_{t-1} + \beta \mathbf{G}_t$, which $\tilde{\mathbf{M}}_t$ does not. The discrepancy is small and explicit: writing $\mathbf{n}_t := \mathbf{G}_{t,i} - \nabla_i f(\mathbf{X}_t)$, we have $\nabla_i f - \tilde{\mathbf{M}}_t = (1-\beta)(\nabla_i f - \mathbf{M}_t) - \beta\, \mathbf{n}_t$, and since $\mathbb{E}\langle \nabla_i f - \mathbf{M}_t, \mathbf{n}_t \rangle = -\beta\, \mathbb{E}\|\mathbf{n}_t\|_2^2$,

$$\mathbb{E}\|\nabla_i f - \tilde{\mathbf{M}}_t\|_2^2 \le (1-\beta)^2\, \mathbb{E}\|\nabla_i f - \mathbf{M}_t\|_2^2 + 3\beta^2 \sigma_i^2.$$

The first term is exactly the deviation Gruntkowska et al. (2025, Thm. 19) already tracks, contracted rather than enlarged; the second is dominated by the $\beta\sigma_i^2$ term already in its bound. Nesterov should therefore degrade the constants rather than the rate. We state this as an expectation, not as a corollary, since the tracking recursion for $\|\tilde{\mathbf{M}}_{t+1} - \tilde{\mathbf{M}}_t\|$ would have to be redone as well.

Remark 4 (Worst-case and realized contraction) *The uplink parameter $\alpha = 1/d_{\max}$ enters the threshold of Corollary 1 through the $1/\alpha^2$-terms of ζ_i, so the admissible γ_i shrinks linearly in $d_{\max}$, the dimension of the largest layer; the same dimension factor arises for Top-1 compressors in Euclidean EF21 (Richtárik, Sokolov, and Fatkhullin 2021). On the uplink this is a worst case only: by Lemma 5, $\alpha = 1/d$ requires a residual concentrated on a single coordinate, whereas the momentum residual is dense, with $\alpha(\mathbf{\Delta}) = \Theta(1)$, and the framework is stated to extend to iteration-dependent α (Gruntkowska et al. 2025, Rem. 12), under which the dense value would enter in place of the worst case.*

The downlink residual of EF21-MuonSign does not stay dense. Unlike the uplink residual, which every round's gradient refreshes, it is produced by the compressor's own recursion

$$\mathbf{\Delta}_{t+1}^{\downarrow} = \mathbf{\Delta}_t^{\downarrow} - \eta_{t+1}\mathbf{D}_{t+1} - \mathrm{mean}\,|\mathbf{\Delta}_t^{\downarrow}|\,\mathrm{sign}(\mathbf{\Delta}_t^{\downarrow}),$$

which corrects every coordinate by the same scalar $\mathrm{mean}\,|\mathbf{\Delta}_t^{\downarrow}|$. *A coordinate whose per-step drive $\eta_{t+1}(\mathbf{D}_{t+1})_{kl}$ exceeds that scalar receives a correction smaller than its drive at every step, while the remaining coordinates keep the scalar small, so the residual concentrates on few coordinates; by the exact expression $\alpha(\mathbf{Y}) = \|\mathbf{Y}\|_1^2/(d\|\mathbf{Y}\|_F^2)$ of Lemma 5, concentration is precisely what lowers α. Section 5.3 measures the effect: on the layers built from a zero initialization, $\alpha(\mathbf{\Delta}^{\downarrow})$ falls to 1.2×10^{-4}, about four orders of magnitude below the uplink value on the same layers, yet still far above the floor $1/d = 4.2 \times 10^{-7}$. The dimension dependence of Corollary 1 is therefore not attained on the downlink either, but the downlink lacks the $\Theta(1)$ contraction that suppresses it on the uplink.*

Remark 5 (The scaled sign is not layer-norm contractive) *Corollary 1 reaches EF21-MuonSign only through Gruntkowska et al. (2025, Rem. 23) and its factor $\bar{\rho}_i^2$, for two reasons. First, the alternative, a server compressor contractive in the* layer *norm, is unavailable: the scaled sign is contractive there for no $\alpha_P > 0$; for $\mathbf{Y}_\delta = (1-\delta)\mathbf{I}_n + \delta \mathbf{J}_n$ ($\mathbf{J}_n$ all ones, $0 < \delta < 1$, $n \ge 2$), every entry is positive, so $\mathrm{sign}(\mathbf{Y}_\delta) = \mathbf{J}_n$, $\mathrm{mean}\,|\mathbf{Y}_\delta| = (1 + (n-1)\delta)/n$, and*

$$\mathcal{C}(\mathbf{Y}_\delta) - \mathbf{Y}_\delta = (1-\delta)\big(\tfrac{1}{n}\mathbf{J}_n - \mathbf{I}_n\big),$$

a matrix with eigenvalues 0 and $-(1-\delta)$; hence $\|\mathcal{C}(\mathbf{Y}_\delta) - \mathbf{Y}_\delta\|_{2\to2} = 1 - \delta$ while $\|\mathbf{Y}_\delta\|_{2\to2} = 1 + (n-1)\delta$, and the ratio tends to 1 as $\delta \downarrow 0$. Second, the sufficient condition of Gruntkowska et al. (2025, App. D), which certifies a compressor in $\mathbb{B}_2(\alpha)$ as layer-norm contractive when $\alpha > 1 - 1/r_i$, is out of reach: for a 768×3072 layer it demands $\alpha > 0.9987$, where the scaled sign attains $2/\pi$ on dense inputs (Lemma 5). The factor $\bar{\rho}_i^2 = r_i$ in ζ_i can therefore be removed only by replacing the compressor, for instance by random dropout ($\alpha_P = p$) or a Top-K SVD compressor ($\alpha_P = 1 - \sigma_{K+1}^2/\sigma_1^2$), both layer-norm contractive (Gruntkowska et al. 2025, App. D) and neither one-bit.

Remark 6 (From Muon to Gluon) *Only* $(\underline{\rho}_i, \bar{\rho}_i) = (1, \sqrt{r_i})$ *is spectral-norm-specific: Lemma 5 is Euclidean and the framework's theorems hold for arbitrary layer norms. A new geometry has to supply only its LMO and its norm-equivalence pair; both corollaries then hold with those constants in place of* $(1, \sqrt{r_i})$*, giving* EF21-GluonUSign *and* EF21-GluonSign *for the Gluon setting (Riabinin et al. 2025). Admissible geometries abound:* $\ell_1 \to \ell_\infty$ *for embeddings (Pethick et al. 2025a), the Schatten-*p *norms (Cesista 2025), the Ky Fan duals of the Fanion family (Kravatskiy et al. 2025), and, by the closure property of the last work, the norms whose LMO is a conic combination of these LMOs; their unit ball is the corresponding Minkowski sum, whence* $\bar{\rho}_i \leq \sum_j \alpha_j \bar{\rho}_i^{(j)}$*. Convergence is thus a property of error feedback together with scaled-sign compression, not of the spectral geometry.*

We assume the exact spectral LMO, as do all analyses of Muon-type methods (Li and Hong 2025; Kovalev 2025; Riabinin et al. 2025; Gruntkowska et al. 2025); in practice the polar factor is approximated by polynomial iterations (Amsel et al. 2025; Grishina, Smirnov, and Rakhuba 2025), in our case the five Newton–Schulz steps of Algorithm A1, with vector parameters and the last layer trained by AdamW as usual (Jordan et al. 2024). That substitution is not covered by Theorem 5 either; Shulgin et al. (2026) analyse the inexact Muon update directly and find the method tolerant of oracle error.

A.11 Reproducibility Details

Choice of benchmarks. Each benchmark answers a question the others cannot. On the convex quadratic of Appendix A.12 the smoothness constant and the minimizer are known in closed form, so the alignment the counterexamples attack can be measured there rather than inferred. CIFAR-10 with a ResNet-18 (Krizhevsky 2009) carries no such special property: we use it because it is the benchmark on which the sign-compression line of work is quoted (Bernstein et al. 2018, 2019; Karimireddy et al. 2019) and on which SignMuon's own concurrent proposal is evaluated (Mishra, Trivedi, and Kumar 2026); it is small enough to run every method at several step sizes and several seeds, which is what the claims about seed spread require. The federated split of the same data at $N = 11$ tests the methods in the setting they are designed for, a bandwidth-limited link between clients and a server. NanoGPT supplies what CIFAR cannot: a transformer language model at practical scale, with matrices wide enough for the layer-rank term of Corollary 1 to be visible, which is where the two models of EF21-MuonSign separate.

Computing infrastructure. The experiments were not all run on the same machine. The synthetic study ran on one NVIDIA RTX A4000 (128-core AMD EPYC 7543 host, 472 GB RAM; Linux 6.12, Python 3.12, PyTorch 2.7.0, CUDA 12.8, driver 575.51.03); the language-modelling runs used a rented 8×H100 SXM node, specified in full in Appendix A.16. The centralized and federated CIFAR-10 runs were executed on single-GPU workstations; each run records its machine, commit and wall time in its `metrics.json`. All 126 centralized runs and all 175 federated runs behind the tables and figures below were executed on the same machine and at one commit: an NVIDIA RTX A4500 (19.6 GB) in a 32-core AMD Ryzen 9 5950X host with 62.7 GB of memory, under Linux 5.15, Python 3.12.11, PyTorch 2.5.1+cu124, CUDA 12.4, driver 560.35.05.

Randomness and seeds. The network experiments are seeded through a single routine that seeds Python's `random` module, NumPy and PyTorch on all CUDA devices; the federated runs additionally pin cuDNN to deterministic kernels, while the centralized sweep leaves cuDNN autotuning enabled, since it reports the spread across seeds rather than a bitwise-reproducible trajectory. The synthetic study forks and re-seeds its own generator per configuration. The federated experiment of Table 2 uses five seeds (0–4) per method and the centralized experiment of Table 1 three (0–2), each reported as mean $\pm$ one sample standard deviation across seeds; the weight-decay ablations use seed 0. The nanoGPT runs of Table 3 are single runs at the speedrun's own unpinned initialization, and the table quotes the five-seed spread published upstream; our released script accepts an explicit seed but pins the generator only, since deterministic kernels would forfeit the wall-clock time the same table reports. The synthetic study of Appendix A.12 averages over three draws of the problem (seeds 1337–1339) at a fixed $\mathbf{X}_0$ (seed 42), its claims concerning random instances. Differences smaller than the seed spread are not claimed as results, and we report spreads rather than significance tests because at these seed counts no test could reject: a paired Wilcoxon signed-rank test over n seeds has smallest attainable two-sided exact p-value 2^{1-n}, which is 0.0625 at five seeds and 0.25 at three, above the 5% level in both cases. The comparisons we claim are separated by several standard deviations. Learning-rate selection is performed once, at seed 0, on a validation split disjoint from the test set, and the selected rate is reused unchanged for every seed.

Step-size schedules. Each experiment inherits the schedule conventional to its domain, and none of them is the constant rate our rates are proved for. The ResNet runs anneal cosinally to zero, the standard for this architecture and the premise of every accuracy we can be compared against; the nanoGPT runs keep record #40's stable-then-decay schedule, flat for the first 55% of steps and decaying linearly to a tenth of the base rate η_0 of (7) thereafter (Appendix A.16), because altering it would forfeit the reproduction that validates our port. The discrepancy is deliberate: matching the analysed step size would sacrifice comparability on both benchmarks and close only one of several gaps between the theory and the runs, the others being Newton–Schulz in place of an exact oracle, momentum, and normalization layers. One measurement is exempt: the growth-exponent diagnostic of Appendix A.17, which is run at a constant rate because under a decaying one the accumulated update saturates and the fit reports the schedule instead of the alignment.

Learning-rate selection. The only tuned hyperparameter is η_0. Every method with a norm-fixed step is tuned and reported under the unit-gain rule (7), so that η_0 is the per-step RMS gain for each of them; SGD and Adam have no norm-fixed step and run at one global rate. The rule is a heuristic, and Appendix A.17 bounds and measures what depends on it: the three SIGN methods keep their order when re-tuned from scratch under one global rate and under μP (Table A8). Momentum is fixed at 0.9 and weight decay at 0 in the primary tables, the setting the sweep of Mishra, Trivedi, and Kumar (2026) itself selects; the regularized case is an ablation (Appendix A.13 centralized, and below for the federated study). The auxiliary group, biases, normalization parameters and the classifier head, is trained by AdamW at 10^{-3} for every method, a convention rather than a verified common optimum: a sweep at matched budget places SignMuon's optimum at 10^{-3} and Muon's at 2×10^{-3}, a difference of 0.16 points that lies within the seed spread, so the auxiliary rate is method-dependent, to a degree the sweep does not quantify.

In the centralized study, selection uses a fixed 45k/5k train/validation partition and validation accuracy averaged over the last five epochs; the test set is never consulted during tuning. Each method starts from the same five-point 1–2–5 lattice (three points per order of magnitude); an optimum at a grid endpoint triggers a widening and a re-run, up to four times, extending Muon and EF21-MuonUSign to seven points and SignSGD to nine. Selection runs use the same 75-epoch cosine schedule as the reported ones, so the tuning and reporting horizons coincide. The selected η_0 is then retrained on the full 50k training set at three seeds, and we report the mean and standard deviation of the test accuracy over the last five epochs.

Tuning the federated study. So that no placement is handicapped by its step size, every method receives the same tuning budget: a five-point 1–2–5 lattice in η_0, ranked on a 5k validation split held out of the 50k *before* the client partition, at the full 2000-round horizon Table 2 reports, with the grid widened and the method re-tuned whenever an optimum occurred at an endpoint. SignMuon is the one method that required the widening, settling at $\eta_0 = 0.1$ on seven points; every selected rate is interior to its own grid. The reported runs then use the full 50k at the selected rate, so no test image is ever scored during selection.

One selection margin requires comment. Eight of the nine methods carrying a per-layer multiplier (the eight of Appendix A.17 and the server-side-LMO control) separate their selected rate from the runner-up by 0.15 to 1.07 validation points; SignMuon separates 0.1 from 0.05 by 0.02 at the single tuning seed, which is no separation at all, so its row in Table 2 is to be read as either of two adjacent lattice points. The ambiguity does not extend further, the next points out lying 0.5 and 1.3 points behind.

The selected rates span a factor of ten, from $\eta_0 = 0.01$ for SignSGD and EF21-MuonUSign to 0.1 for Muon and SignMuon, with both families covering that range. This spread is between *methods* and carries no verdict on the per-layer rule, whose claim concerns layer *shape*; the measurement that does bear on the rule, re-tuning under competing conventions, is Table A8. Weight decay is 0 in the reported table. Switching on a decoupled 5×10^{-4} at seed 0 moves Muon by -0.32 points, SignMuon by $+0.27$ and the server-side-LMO control by $+0.05$: Muon leads in either setting, and the other two exchange places by margins below the 0.24 standard deviation SignMuon carries over five seeds, so the ablation separates nothing that the primary table does not.

Accuracy and the threshold column. Alongside final accuracy we report the number of epochs (rounds, in the federated tables) to reach a fixed test accuracy. The two measure distinct quantities: with the methods spanning about a point and a half at 75 epochs (Table A5), final accuracy is close to the noise floor, whereas a threshold crossing on a monotonically rising curve separates the methods by factors rather than by tenths of a point, and is the analogue of the "steps to 3.35" column of Table 3.

Conventions with numerical consequences. Three implementation conventions can displace reported numbers and are recorded here. (i) The Muon LMO is computed in `bfloat16` (five Newton–Schulz steps) unless stated otherwise; for the methods that sign the LMO *output*, entries of $\mathrm{polar}(\cdot)$ near zero may flip at this precision, so that their trajectories carry a precision-dependent component (`-lmo-dtype float32` is available). (ii) In the federated runs, BatchNorm running statistics are never updated: local models are discarded each round and BN runs in inference mode during gradient accumulation, so the statistics stay at their initialization for the entire run, in training and evaluation alike. The result is a fixed normalization with learnable affine parameters, self-consistent between train and test, applied identically to every method. It is also one reason a channel may remain inactive across an entire local batch and so contribute an exactly zero row to the momentum. (iii) For EF21-MuonSign, training metrics are logged at the broadcast model $\mathbf{W}$ (where gradients must be evaluated) while validation and test metrics are evaluated at the server model $\mathbf{X}$ of (13), the iterate the guarantee bounds, except where a table states otherwise.

A.12 The smooth convex problem

On a deterministic L-smooth convex quadratic we measure the scalar the guarantees rest on: the alignment $\rho_t = \langle \nabla F(\mathbf{X}_t), \mathbf{D}_t \rangle / (\|\nabla F(\mathbf{X}_t)\|_F \|\mathbf{D}_t\|_F)$ between the gradient and the step. Theorems 1–3 construct instances driving it negative; on random instances the three methods they cover keep $\rho_t \geq 0.142$ throughout a tuned trajectory, so the construction is not one that random data reproduces. Only EF21-MuonSign becomes negative, on 0.95% of steps and to -0.026; it is also the one method here possessing a convergence guarantee (Appendix A.10), obtained without per-step descent.

The same experiment separates the two effects that a fixed-target iteration count confounds. Sign compression of an LMO step secures a lower accuracy floor, not a faster rate: SignMuon's floor lies a factor 1.74–1.80 below SignSGD's at every step size, and the two share $\|\mathbf{S}\|_F$ exactly, so the separation resides in the floor rather than in the step length.

Algorithm	**iters**	**best** F	$\min \|\nabla F\|_F$	**tuned** (η, μ)
SignMuon	364	$6.9 \cdot 10^{-4}$	$2.4 \cdot 10^{-2}$	$(1.0 \cdot 10^{-3}, 0)$
MuonUSign	556	$7.1 \cdot 10^{-4}$	$2.3 \cdot 10^{-2}$	$(1.0 \cdot 10^{-2}, 0.5)$
MuonSign	595	$4.8 \cdot 10^{-4}$	$1.9 \cdot 10^{-2}$	$(6.8 \cdot 10^{-4}, 0.5)$
EF21-SignMuon	401	$5.4 \cdot 10^{-4}$	$2.1 \cdot 10^{-2}$	$(6.8 \cdot 10^{-3}, 0)$
EF21-MuonUSign	521	$5.3 \cdot 10^{-4}$	$1.8 \cdot 10^{-2}$	$(6.8 \cdot 10^{-3}, 0)$
EF21-MuonSign	695	$3.8 \cdot 10^{-4}$	$1.7 \cdot 10^{-2}$	$(4.6 \cdot 10^{-3}, 0)$
Muon	267	$4.7 \cdot 10^{-4}$	$2.0 \cdot 10^{-2}$	$(1.0 \cdot 10^{-2}, 0.5)$
SignSGD	486	$4.3 \cdot 10^{-4}$	$2.2 \cdot 10^{-2}$	$(6.8 \cdot 10^{-4}, 0.5)$
SGD	66	$1.9 \cdot 10^{-8}$	$1.6 \cdot 10^{-6}$	$(2.15, 0.5)$
Adam	85	$9.2 \cdot 10^{-6}$	$6.5 \cdot 10^{-4}$	$(6.8 \cdot 10^{-2}, —)$

Table A2: Fixed-target criterion under the current protocol: fewest iterations to $F \leq 10^{-3}$ within $T_{\max} = 5000$ at $m = n = 100$, over three problem draws, $(\eta, \mu, \text{schedule})$ tuned on logarithmic grids spanning five orders of magnitude. A constant schedule is selected for every method. Best F and $\min \|\nabla F\|_F$ are minima over all 5000 iterations rather than values at the crossing, so they report the floor the method settles into.

Construction. To isolate the effect of matrix structure from stochastic noise and from the complexity of DNN architectures, we use a deterministic L-smooth convex quadratic:

$$F(\mathbf{X}) = \frac{1}{2}\|\mathbf{A}^{1/2}\mathbf{X}\mathbf{B}^{1/2}\|_F^2 = \frac{1}{2}\langle \mathbf{X}, \mathbf{A}\mathbf{X}\mathbf{B}\rangle \to \min_{\mathbf{X}\in\mathbb{R}^{m\times n}}, \tag{33}$$

with $\mathbf{A} \in \mathbb{S}^m_{++}, \mathbf{B} \in \mathbb{S}^n_{++}$ symmetric and $\mathbf{X}_0$ drawn entrywise from $\mathcal{N}(0, 0.01)$. Eigenvalues are sampled uniformly from $(0, 1)$ in a Haar-random eigenbasis, so the matrices are almost surely positive *definite* and the minimizer is $\mathbf{X}^\star = \mathbf{0}$ with $F^\star = 0$.

Two facts about this instance are exact rather than estimated, and both are used below. The Hessian of F is the Kronecker product $\mathbf{B} \otimes \mathbf{A}$, so its eigenvalues are the products $\lambda_i(\mathbf{A})\lambda_j(\mathbf{B})$: the Frobenius smoothness constant is $L = \max_{ij} \lambda_i(\mathbf{A})\lambda_j(\mathbf{B}) \leq 1$ and the strong-convexity constant is $\sigma = \min_{ij} \lambda_i(\mathbf{A})\lambda_j(\mathbf{B}) > 0$. The uniform draw leaves the resulting condition number L/σ uncontrolled (it is near 3.7×10^4 at the $m = n = 100$ every measurement in this subsection uses), so where conditioning is the variable we instead use log-spaced spectra with $L = 1$ and L/σ set exactly. And since $\nabla F(\mathbf{X}) = \mathbf{A}\mathbf{X}\mathbf{B}$ in closed form, the gradient can be evaluated at any point without an autograd graph, which permits the bidirectional method to be scored on its exact model $\mathbf{X}$ while its gradient is taken at the broadcast model $\mathbf{W}$, as its algorithm requires.

The fixed-target criterion. The natural criterion, fewest iterations to $F(\mathbf{X}) \leq 10^{-3}$ within $T_{\max} = 5000$ with learning rate and momentum tuned per method, does not measure a convergence rate. Eight of the ten methods take a *norm-fixed* step, $\|\operatorname{sign}(\cdot)\|_F = \sqrt{mn}$ and $\|\operatorname{polar}(\cdot)\|_F = \sqrt{r}$, so at a constant η the iterate settles into a ball of radius $\eta\|\mathbf{S}\|_F$ and F plateaus; Adam, bounded entrywise by $\approx\eta$, plateaus as well, and SGD, whose step vanishes with the gradient, is the only method that does not. Write F_∞ and g_∞ for the settled values of $F(\mathbf{X}_t)$ and $\|\nabla F(\mathbf{X}_t)\|_F$. Measured directly, $g_\infty \propto \eta$ for every method possessing a floor, $F_\infty \propto \eta^2$ for every such method but SignSGD, whose exponent is 1.33, and the iteration count is const/η. The tuner accordingly returns the largest η whose plateau falls under the target, and the resulting ranking is one of *accuracy floors*. Over the seven step sizes at which both methods were run, SignMuon's g_∞ lies a factor 1.74 to 1.80 below SignSGD's, flat in η as two floors of equal exponent must be, while its F_∞ lies 6.7 to 71 times below, the two F-exponents differing. Tuned per budget, SignMuon holds the smaller $\min_t \|\nabla F\|_F$ at every horizon but $T = 250$, where SignSGD prevails by 2%, a margin below what three draws resolve. We therefore report floor and rate separately, reading the descent lemma

$$F(\mathbf{X}_{t+1}) \leq F(\mathbf{X}_t) - \eta\langle \nabla F(\mathbf{X}_t), \mathbf{D}_t\rangle + \tfrac{\eta^2 L}{2}\|\mathbf{D}_t\|_F^2 \tag{34}$$

as the statement that separates them: the second term is the floor, the first is the rate.

Table A2 nonetheless reports the criterion. Read as a ranking of floors, it orders the six placements identically within both families: sign after the LMO is the least expensive, sign on both channels the most. Error feedback adds 10% on SignMuon and 17% on MuonSign, and removes 6% on MuonUSign, the one placement whose LMO already receives a compressed argument. Muon surpasses all six placements, and SGD (66) and Adam (85) surpass every normalized step by factors of three to ten, a quadratic with an exactly known Hessian being precisely the case for which a scaled gradient step is designed. Figure A3 plots the trajectories behind those counts, and shows the plateau that makes the criterion a ranking of floors.

Alignment. Equation (34) makes progress contingent on a single scalar, the normalized alignment between the gradient and the step actually taken,

$$\rho_t := \frac{\langle \nabla F(\mathbf{X}_t), \mathbf{D}_t\rangle}{\|\nabla F(\mathbf{X}_t)\|_F\,\|\mathbf{D}_t\|_F} \in [-1, 1]. \tag{35}$$

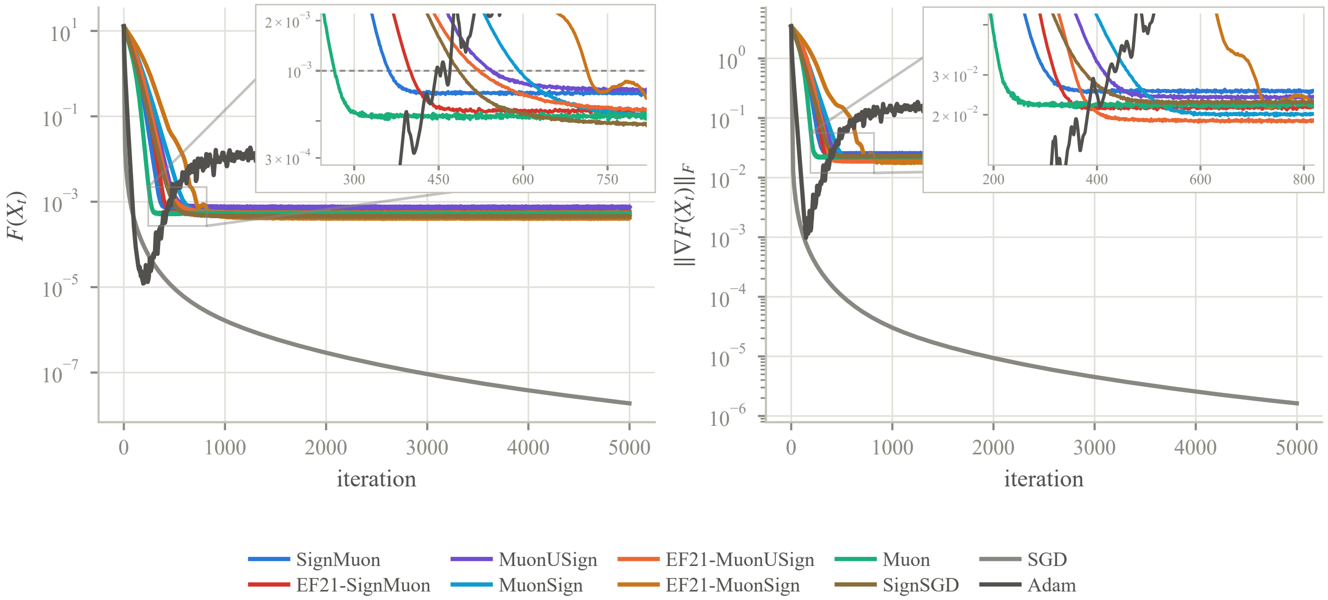

Figure A3: Trajectories at the optima of Table A2, geometric mean over three problem draws. Every normalized method plateaus within a few hundred iterations and stays there, which renders the crossing time a ranking of floors; SGD, whose step vanishes with the gradient, is the only curve still descending at $T_{\max}$, and Adam oscillates about its floor rather than settling onto it. The insets magnify the arrival window, over the full axis a fold of curves in the first sixth of the range, and are scaled to the eight norm-fixed methods: the dashed line is the target $F \leq 10^{-3}$, and the order in which the curves cut it is the iteration count of Table A2, Muon first and EF21-MuonSign last. Those counts are means of the per-draw crossing times rather than crossings of the plotted mean, and for the slowest method the two differ by enough to see.

Algorithm	$\min_t \rho_t$	**median** ρ_t	**mean** ρ_t	**% of steps with** $\rho_t < 0$	**closed form**	**tuned** (η, μ)
SignMuon	0.208	0.381	0.399	0.00%	—	$(2.2 \cdot 10^{-4}, 0)$
MuonUSign	0.161	0.357	0.378	0.00%	—	$(3.2 \cdot 10^{-3}, 0)$
MuonSign	0.142	0.360	0.364	0.00%	—	$(2.2 \cdot 10^{-4}, 0)$
EF21-SignMuon	0.333	0.659	0.573	0.00%	—	$(1.5 \cdot 10^{-3}, 0)$
EF21-MuonUSign	0.346	0.558	0.553	0.00%	—	$(1.5 \cdot 10^{-3}, 0)$
EF21-MuonSign	-0.026	0.489	0.419	0.95%	—	$(1.5 \cdot 10^{-3}, 0)$
Muon	0.454	0.666	0.633	0.00%	0.695	$(1.5 \cdot 10^{-3}, 0)$
SignSGD	0.174	0.503	0.499	0.00%	0.794	$(2.2 \cdot 10^{-4}, 0.5)$
SGD	—	—	—	—	1	$(3.2, 0.95)$

Table A3: Alignment ρ_t of Equation (35) along the tuned trajectory, over three problem draws at $m = n = 100$. The closed form is evaluated at $\mathbf{X}_0$ without momentum and is comparable only to rows whose tuned μ is 0; the six sign-around-the-LMO methods admit none. SGD's closed form is 1 but its step is not instrumented, so its distribution column is empty; Adam is omitted, its step lying outside the descent lemma. Every method of Theorems 1–3 stays bounded away from zero; the single negative excursion is EF21-MuonSign's, which Appendix A.10 proves convergent through the EF21 estimator rather than per-step descent.

Theorems 1–3 are constructions that drive ρ_t negative. Table A3 reports its distribution along the tuned trajectory on random instances, which is the empirical counterpart of those theorems and the one measurement here that is about the methods rather than about the tuning protocol. Three closed forms anchor it: $\rho = 1$ for SGD, $\rho = \|\mathbf{G}\|_1/(\|\mathbf{G}\|_F\sqrt{mn})$ for SignSGD, and $\rho = \|\mathbf{G}\|_*/(\|\mathbf{G}\|_F\sqrt{r})$ for Muon. The six sign-around-the-LMO methods admit none, which is the subject of this paper.

Closed-form checks. Three measurements test quantities the theory predicts in closed form. (i) The floor: balancing the two terms of (34) at $\langle \nabla F, \mathbf{D}\rangle = \rho\|\nabla F\|_F\|\mathbf{D}\|_F$ gives $g_\infty = \eta\, L\|\mathbf{S}\|_F/(2\rho)$, a slope of 1 in $\log \eta$ with that coefficient. SignMuon and SignSGD share $\|\mathbf{S}\|_F = \sqrt{mn}$ exactly, so any separation between their floors is attributable to ρ alone. (ii) The budget exponent: tuning $(\eta, \mu, \text{schedule})$ separately at each horizon T and fitting err $\propto T^{-p}$, $\eta^\star \propto T^{-q}$, with err $:= \min_{t\leq T} \|\nabla F(\mathbf{X}_t)\|_*^2$ the squared dual norm our theorems bound. The nonconvex bound gives $p = q = \frac{1}{2}$; a strongly convex problem gives $p = q = 1$, and this instance, having $\sigma > 0$, need not occupy the nonconvex regime. The fit reports both at once: the eight norm-fixed

	floor		budget				stability
Algorithm	$\mathrm{d}\log g_\infty/\mathrm{d}\log\eta$	R^2	p	R^2	q	R^2	$\eta_{\max}\lVert\mathbf{S}\rVert_F$
SignMuon	1.000	1.000	1.88	0.996	0.39	0.94	13.4
MuonUSign	1.000	1.000	1.86	0.995	0.39	0.94	14.9
MuonSign	1.000	1.000	1.93	0.993	0.44	0.94	11.8
EF21-SignMuon	1.000	1.000	2.07	1.000	0.55	1.00	18.9
EF21-MuonUSign	1.000	1.000	2.04	1.000	0.55	1.00	14.9
EF21-MuonSign	1.000	1.000	2.14	1.000	0.55	1.00	14.9
Muon	1.000	1.000	2.03	0.999	0.55	1.00	17.6
SignSGD	0.989	1.000	1.76	0.990	0.39	0.94	11.1
SGD	*no floor*		8.97	0.985	-0.11	0.50	$2.06^\dagger$
Adam	1.000	1.000	2.65	0.915	-0.66	0.62	$> 100^\ddagger$
predicted	1		1/2 or 1		1/2 or 1		$2/L$

Table A4: Floor exponent, budget exponents and stability edge against the predicted values, at $m = n = 100$ over three problem draws. The floor exponent attains its prediction exactly for eight of the nine methods possessing a floor and to within 1.1% for SignSGD, fitted on the seven step sizes at which the plateau is reached (four for Adam). SGD possesses no floor, its step vanishing with the gradient, so its floor columns are empty and its $\eta_{\max}$ is the control reproducing $2/L$; neither baseline admits a power law in $\eta^\star$. † SGD has no $\lVert\mathbf{S}\rVert_F$; the entry is $\eta_{\max}$, 1.02 times $2/L$. ‡ The search reached its ceiling without encountering an edge: Adam's step is bounded by $\approx\eta$ irrespective of the gradient, so it oscillates rather than diverges.

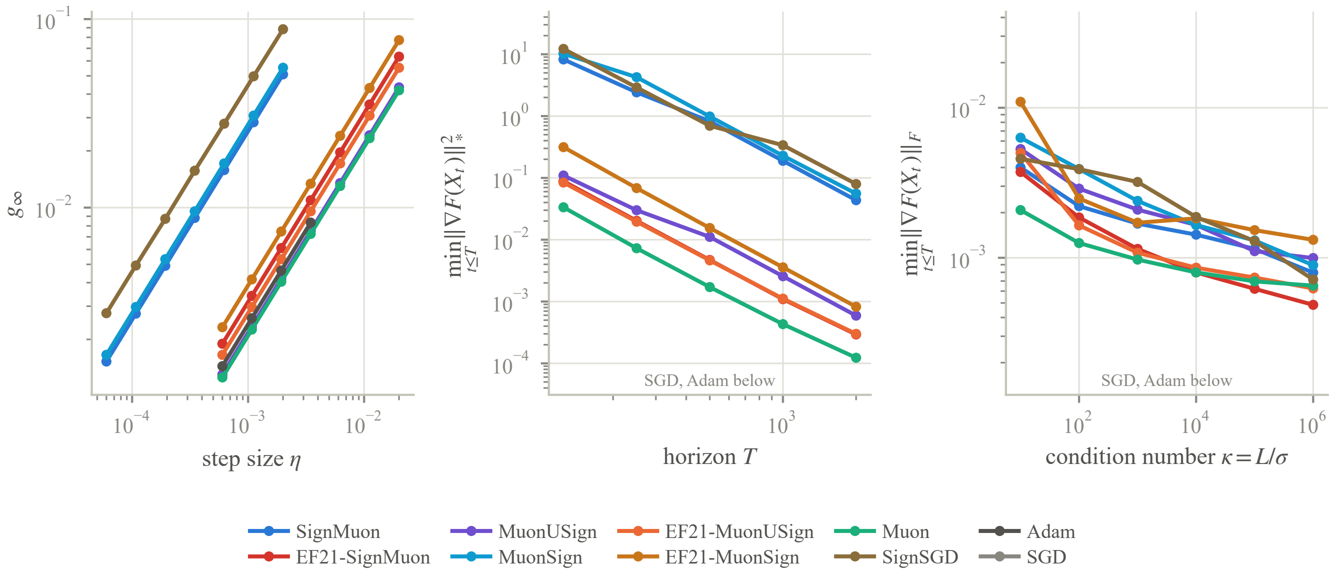

Figure A4: Log–log fits behind Table A4, at $m = n = 100$ over three problem draws. *Left*, the accuracy floor g_∞ of a constant step, the settled $\lVert\nabla F\rVert_F$: the lines are parallel because every exponent is 1, and the two clusters are the two step lengths, the three methods with $\lVert\mathbf{S}\rVert_F = \sqrt{mn}$ on the left and the five with $\sqrt{r}$ on the right, each swept over the window its length calls for. *Centre*, error at each retuned horizon, in the norm dual to each method's own LMO ball, whence the two bands: ℓ_1 above and nuclear below, comparable within a family but not across families. *Right*, tuned at each κ; over the five orders of magnitude swept, the eight norm-fixed methods vary by a factor of 3.2 to 8.3, against the sixteen orders of magnitude spanned by SGD and Adam. Both of the last two panels are scaled to those eight, leaving SGD and Adam below the frame: on a quadratic with an exactly known Hessian a scaled gradient step converges to machine precision. The gap reaches nineteen orders of magnitude at $\kappa = 10$, where SGD attains $2 \cdot 10^{-22}$ against the $2 \cdot 10^{-3}$ of the closest norm-fixed method.

methods tune $\eta^\star$ as the nonconvex bound prescribes, q scattering about $\frac{1}{2}$, while the attained error falls at p between 1.76 and 2.14, a step size chosen for the worst case of the smoothness class applied to an instance far easier than that worst case. (iii) The stability edge: the largest stable η, with SGD as a control required to reproduce the textbook $2/L$. Reported as the step length $\eta_{\max}\lVert\mathbf{S}\rVert_F$, this would be family-independent were the operative bound the Frobenius ball; the spread measures how far that bound stands from the geometry in which each step actually resides.

Dataset	Optimizer	Epochs	η_0	Train Acc	Test Acc	Ep. to 90%	s/epoch
CIFAR-10	SignMuon	75	0.02	99.99	94.60 ± 0.15	7.7	16.6
	Muon		0.1	99.99	94.35 ± 0.27	7.7	14.4
	EF21-SignMuon		0.02	100.00	94.31 ± 0.11	9.0	16.6
	EF21-MuonUSign		0.05	99.99	94.14 ± 0.07	10.3	16.7
	EF21-MuonSign		0.005	99.99	94.04 ± 0.10	11.0	18.1
	MuonUSign		0.02	99.99	93.98 ± 0.12	10.3	16.2
	Adam		0.001	99.97	93.37 ± 0.27	20.0	12.7
	SignSGD		0.002	99.97	93.37 ± 0.26	19.0	12.5
	MuonSign		0.1	99.98	93.31 ± 0.21	17.7	17.2
	SGD		0.02	99.96	93.04 ± 0.14	21.3	12.2

Table A5: CIFAR-10: centralized learning on ResNet-18, three seeds. Test accuracy is the mean $\pm$ standard deviation of the last five epochs; train accuracy, "Ep. to 90%" and "s/epoch" are means over the same three seeds. η_0 was selected on a held-out 5k validation split at the same 75-epoch horizon the table reports, and the selected rate then retrained on the full 50k set, as described in Appendix A.11. Every method fits the training set to within 0.05 points of the others, so the column separates nothing on its own; the differences in the test column are not differences in how far training got.

Conditioning. Conditioning governs the dynamics of a quadratic, and the construction above leaves it to chance, so the right panel of Figure A4 sweeps κ over five orders of magnitude, from 10 to 10^6, at fixed $L = 1$. The eight norm-fixed methods are insensitive to it: over the entire sweep the attained $\|\nabla F\|$ moves by a factor of 3.2 (Muon) to 8.3 (EF21-MuonSign), and the fitted $\mathrm{d}\log\|\nabla F\|/\mathrm{d}\log\kappa$ lies between -0.096 (Muon) and -0.17 (EF21-SignMuon). The factor is the measurement and the slope a summary of it, which is the order to read them in: six of the eight fits have $R^2 \geq 0.90$, but EF21-MuonUSign's is 0.83 and EF21-MuonSign's 0.67, so for those two rows the slope is not itself a quantity we would quote. The sign is mildly negative because at fixed L a larger κ entails a smaller σ, hence a flatter landscape to occupy rather than a harder one to descend; their floors are fixed by $\eta\|\mathbf{S}\|_F$, which is independent of the spectrum. SGD and Adam are sensitive to it, jointly spanning sixteen orders of magnitude across the sweep, which is why the panel is scaled to the eight and leaves those two below it. SGD's fitted slope of 3.6 is not an exponent: its first three points lie at or below 10^{-20}, which on this problem constitutes exact convergence rather than a measurement.

Protocol and reproduction. Every measurement above is at $m = n = 100$ over three problem draws (seeds 1337–1339, $\mathbf{X}_0$ from seed 42), with the LMO taken in `bfloat16` as in the network experiments, and is reported as the geometric mean over the draws for the error metrics and the arithmetic mean for iteration counts.

Learning rates are searched on logarithmic grids spanning five orders of magnitude, one per step-norm family, each ending past the largest stability edge measured for that family so that no stable step size falls outside the search. A linear grid spanning a single order of magnitude, which an earlier version of this experiment used, is narrow enough to censor an optimum at a grid boundary, giving an upper bound rather than a tuned value; the search flags any optimum at an edge. No learning rate reported above is flagged. The windows are derived from $\|\mathbf{S}\|_F$ rather than from the method name, which matters for MuonUSign and EF21-SignMuon: both take a step of length $\sqrt{r}$ despite the SIGN in their names, and under a window assigned by name both were searched an order of magnitude below the range their step length calls for.

The remaining flags are on momentum, at the top of its grid (0.99), and belong to SGD and SignSGD at the largest condition numbers: an ill-conditioned quadratic asks for heavy momentum, so those two rows of the κ sweep bound the dependence from one side rather than measuring it. The other eight methods select 0.5 or less at every κ, and the three carrying an error-feedback estimator select 0 throughout.

Every number in this subsection comes from a single scripted run of the released benchmark, which records each stage together with the commit, GPU and wall time it ran under.

A.13 Centralized training results

Table A5 reports the centralized CIFAR-10 results on ResNet-18 in full: the selected η_0, the final train and test accuracies, the epoch count to the 90% threshold, and the per-epoch cost. Figure 2 starts at epoch 25, where the methods are a few points apart rather than twenty. Every run uses batch 128, momentum 0.9, auxiliary rate 10^{-3}, and zero weight decay, so that the matrix rule and η_0 are the only quantities that vary. Figure A5 gives the accuracy curves from epoch 3, Figure A6 the training loss, and Figure A7 the learning-rate sweep behind the selection.

Selection horizon. Selection and reporting share the 75-epoch horizon, so each η_0 above is the argmax of a validation sweep conducted at the length the table reports. An earlier protocol that selected at 15 epochs and reported at 75 was abandoned because the two horizons selected different rates for two of the methods. The selection is not sharply peaked for the sign-after

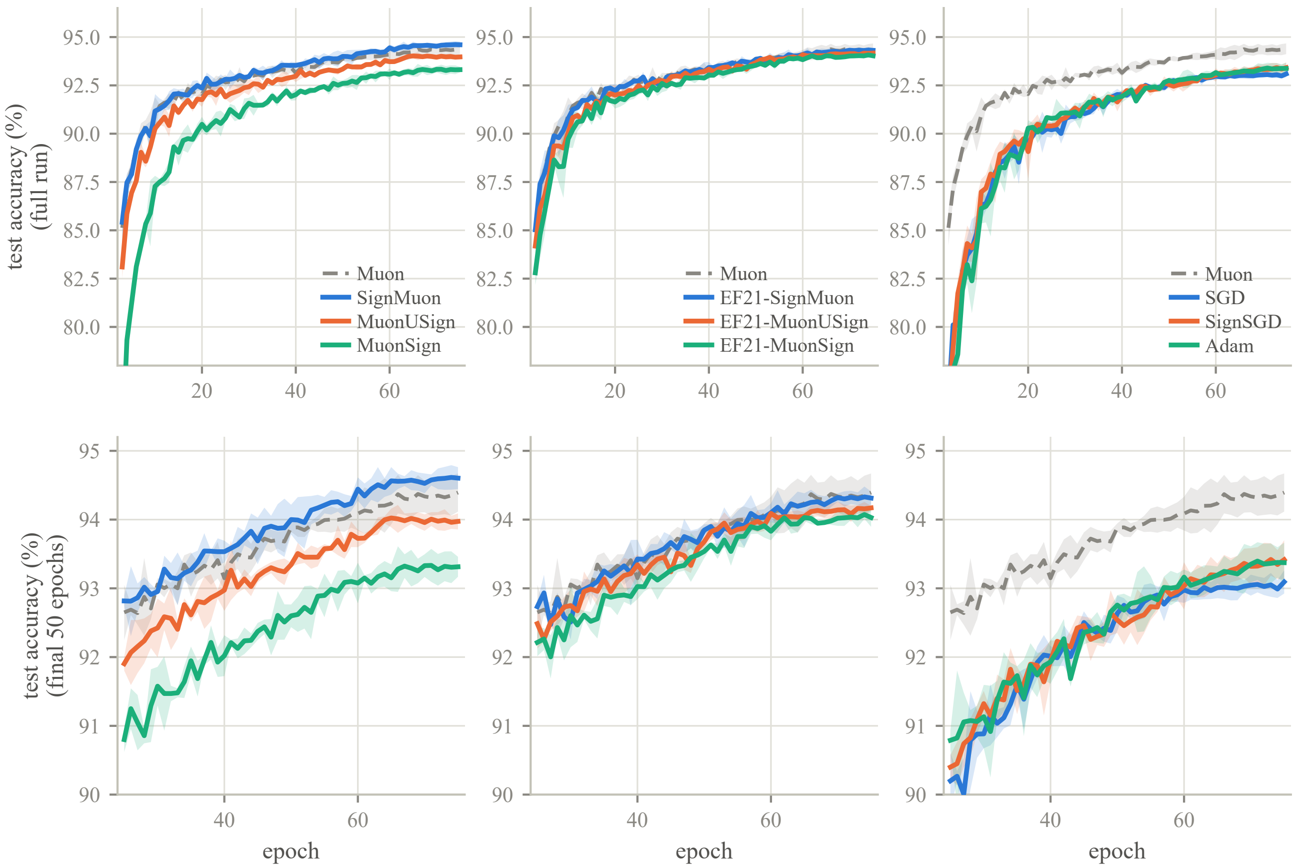

Figure A5: Test accuracy over the whole run (top row) and from epoch 25 (bottom row), the same data at two scales, grouped as in Figure 2. Bands are ± 1 standard deviation over three seeds; Muon is the gray dashed reference in every panel. Series are named in each panel's legend.

methods: over the rates within a factor of five of its own optimum, SignMuon's validation accuracy varies by 0.13 points, against 0.65 for Muon and 3.00 for SignSGD (Figure A7).

Weight decay. The primary table is unregularized, the setting the theorems analyse and the one Mishra, Trivedi, and Kumar (2026)'s sweep selects. Repeating the top three at the same η_0 with decoupled decay 5×10^{-4} (seed 0, decay not re-tuned) displaces each by at most 0.34 points, of the order of the three-seed standard deviation of the undecayed runs, and leaves the three within 0.23 of one another: Muon 94.46% against 94.12% undecayed at that seed, EF21-SignMuon 94.45% against 94.42%, SignMuon 94.23% against 94.43%. The ordering within that interval does change, but at a single seed and over so narrow a range it is not a measured effect; what the ablation establishes is that decay introduces no separation where the primary table shows none. Decay is applied decoupled, $\mathbf{X} \mathrel{*}= 1 - \eta\lambda$, so the LMO sees the true gradient; the coupled convention would only rotate the direction, since every step here is scale-invariant.

A.14 Communication accounting

Table 2 quotes mean bits per parameter per round. This section states precisely what is counted, since the headline "32×" of the sign-compression literature is an idealization eroded by three separate effects, only one of which is customarily acknowledged.

Write P_{mat} for the number of matrix parameters, P_{aux} for the auxiliary group (biases, BatchNorm affine parameters, the classifier head), L for the number of matrix layers, and $P = P_{\mathrm{mat}} + P_{\mathrm{aux}}$. On CNN2, $P_{\mathrm{mat}} = 762{,}560$, $P_{\mathrm{aux}} = 2{,}146$ and $L = 3$. All figures below are *per client per round*: the uplink is what one client sends, the downlink what the server sends to one client.

(i) The uplink alphabet. The randomized sign of Section 4 renders every transmitted symbol a genuine bit, so the uplink costs one bit per parameter with no entropy coding required to realize it.

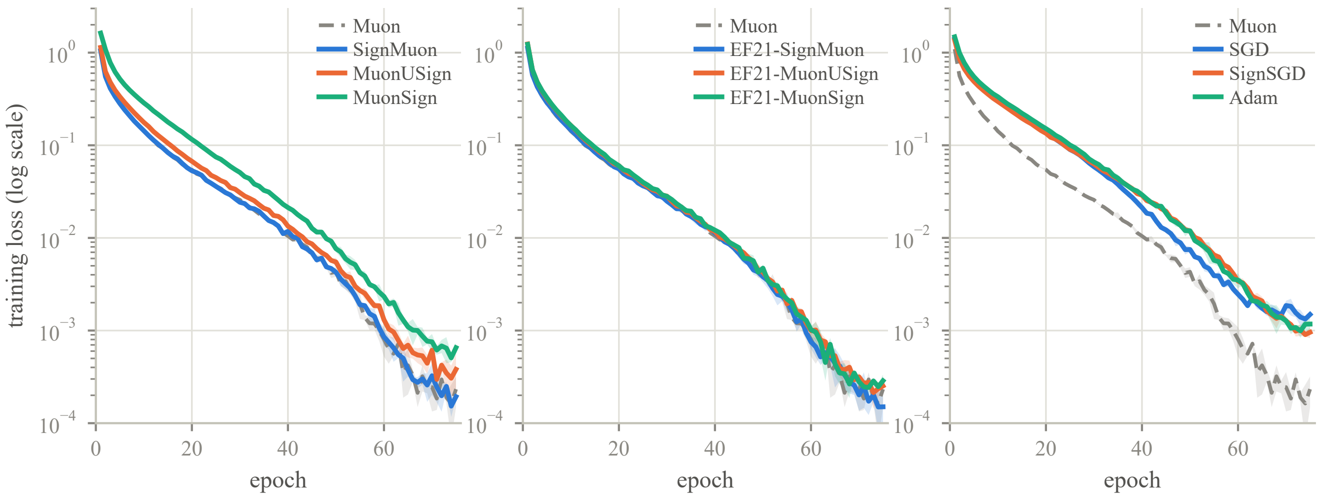

Figure A6: Training loss (log scale), grouped as in Figure 2 and over the same runs. Within a panel the methods are hard to separate (eight of the ten fall below 10^{-3}, differing mainly in how early), which is why the body figure carries the accuracy alone. What the loss does show is the separation between families: in the right panel Muon reaches 10^{-2} six to eleven epochs before SGD, SignSGD and Adam and terminates four to six times lower, while in the left panel SignMuon tracks Muon and the sign-before and both-sides orderings lie above it, in the same order as their accuracies. The three EF21 variants (centre) are indistinguishable in loss, whereas their accuracies span 0.27 points.

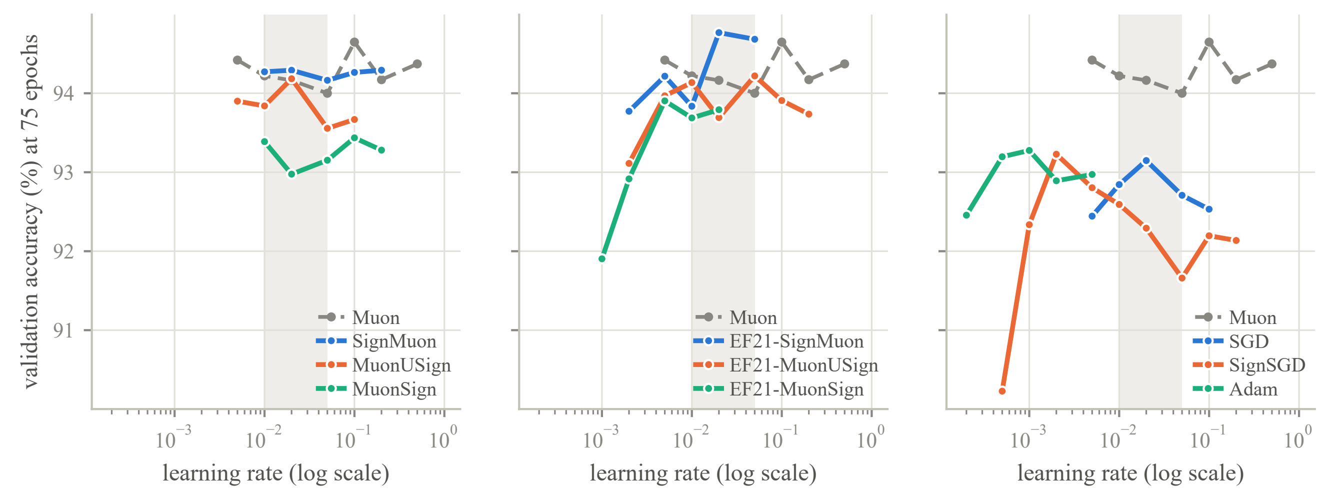

Figure A7: Learning-rate sweep at the 75-epoch selection horizon, on the tuning split and on the validation accuracy that selection ranked, grouped as in Figure 2, with Muon repeated as the gray dashed reference. The shaded band marks $\eta_0 \in [0.01, 0.05]$, the only interval every method but Adam was swept over; the nine vary by 0.10 to 0.93 points within it, SGD (0.44) among them, so an *absolute* window no longer discriminates between the families, their optima lying two orders of magnitude apart. The matched comparison is the relative one. Within a factor of five of its own optimum SignMuon varies by 0.13 points and SignSGD by 3.00; the two emit steps of identical Frobenius length, so the difference is attributable to the LMO. The remaining eight lie between these, from 0.46 (MuonSign) to 2.00 (EF21-MuonSign). Taking the sign of the oracle's output does not reduce this insensitivity: SignMuon's sweep is the flattest of the ten.

(ii) The auxiliary group is never compressed. It travels at full precision in both directions for every method, so a "one-bit" channel actually costs

$$\frac{1 \cdot P_{\mathrm{mat}} + 32\, P_{\mathrm{aux}}}{P} = 1.087 \text{ bits},$$

Method	Up (bits)	Down (bits)	Up	Down
Muon, MuonServer	32	32	$1.0\times$	$1.0\times$
SGD, Adam	32	32	$1.0\times$	$1.0\times$
SignMuon	1.0870	1.0870	$29.4\times$	$29.4\times$
SignSGD	1.0870	1.0870	$29.4\times$	$29.4\times$
MuonSign	1.0870	1.0870	$29.4\times$	$29.4\times$
EF21-MuonSign	1.0871	1.0871	$29.4\times$	$29.4\times$
MuonUSign	1.0870	32	$29.4\times$	$1.0\times$
EF21-SignMuon	1.0871	32	$29.4\times$	$1.0\times$
EF21-MuonUSign	1.0871	32	$29.4\times$	$1.0\times$

Table A6: Communication per client per round on CNN2, under the randomized-zero convention. Bits are per model parameter; reductions are against a 32-bit baseline. The fourth decimal separates the error-feedback methods from the rest: it is the per-layer scale of point (iii). Groups: uncompressed references, one bit in both directions, one bit on the uplink only.

a $29.4\times$ reduction rather than $32\times$. On CNN2 the group is 0.28% of the parameters; on an architecture with a large embedding or head it would dominate this table, which is why the quantity is computed per model rather than quoted.

(iii) Error feedback carries one scale per layer. Both EF21 channels transmit the pair $(\mathbf{s}_t, \alpha_t)$ with one full-precision α_t per matrix layer: on the uplink for every EF21 method, and again on the downlink for EF21-MuonSign. That is $32L$ bits, adding $32L/P \approx 1.3 \times 10^{-4}$ bits per parameter here. That lies four decimal places in, and it is reported for completeness rather than because it alters a conclusion: it is the difference between "one bit per parameter" and "one bit per parameter, plus a constant".

(iv) Which methods compress the downlink. The criterion is not whether the method applies a compressor but whether the object the server must distribute is already ± 1-valued. Three cases qualify: the majority vote $\mathbf{s}_t^{\mathrm{agg}}$ itself (SignMuon, SignSGD), a signed LMO output (MuonSign), and a primal error-feedback residual (EF21-MuonSign). In the first of these the server broadcasts the vote rather than the model and each client applies the step to its local copy; the copies start from a common $\mathbf{X}_0$ and receive identical updates, so they never drift. The remaining three methods must distribute a dense server-side quantity, namely $\mathrm{polar}(\cdot)$ of the aggregate for MuonUSign and EF21-MuonUSign and a scaled average of signs for EF21-SignMuon, and therefore transmit it at full precision.

Table A6 collects the four effects into the per-round cost of each method, and is the source of the Up and Down columns of Table 2. It is computed by `federated.algorithms.communication_bits` from the alphabet and the measured zero rate of the run it describes, so a run made under the legacy ternary convention reports its own higher figure rather than the idealized one. Every run behind Table 2 was made under the randomized convention, which makes these figures realized rather than idealized. Two diagnostics record what the convention had to absorb, both counted *before* the randomized mapping and therefore feeding no accounting. Exact zeros do occur on the uplink, at up to 3.7% of coordinates for MuonSign and 0.5% for SignMuon, and at none at all for the three error-feedback methods, whose compressed quantity is a residual rather than a direction. And the majority vote tied in no coordinate of any evaluated round of any reported run, as at $N = 11$ and ± 1 client messages it cannot.

A.15 Federated training results

Figure A8 (together with Table 2 in the main text) reports the comparison of optimizers at $N = 11$ clients. The figure additionally shows Muon with a server-side LMO, which is not a communication-efficient method but isolates whether moving the oracle off the clients carries a penalty on its own; it does not ($85.74 \pm 0.11\%$ against Muon's $85.98 \pm 0.26\%$), so the gaps in Table 2 are attributable to compression and sign placement rather than to where the oracle runs.

On the sign-after placement the cost of error feedback depends on the setting, and federation is where it is largest: EF21-SignMuon lies 1.00 points below SignMuon here, against 0.29 points centrally (Table 1), while on nanoGPT the two are indistinguishable (Table 3).

Two features of the figure do not appear in the table. The threshold column does not order the methods as final accuracy does: MuonUSign crosses 80% in 500 rounds, ahead of SignMuon's 540 and EF21-SignMuon's 640, and finishes below both, so which method leads depends on where the round budget is cut. And the left panel orders them differently again: the lowest test cross-entropy of the eleven is EF21-SignMuon's, 0.482 ± 0.005, and the highest bar Adam's is Muon's, 0.586 ± 0.016, the reverse of how those two stand on accuracy. Cross-entropy and accuracy are not obliged to agree, and it is accuracy the comparison is about; the loss panel is drawn so that the disagreement is on the record rather than suppressed by the choice of metric.

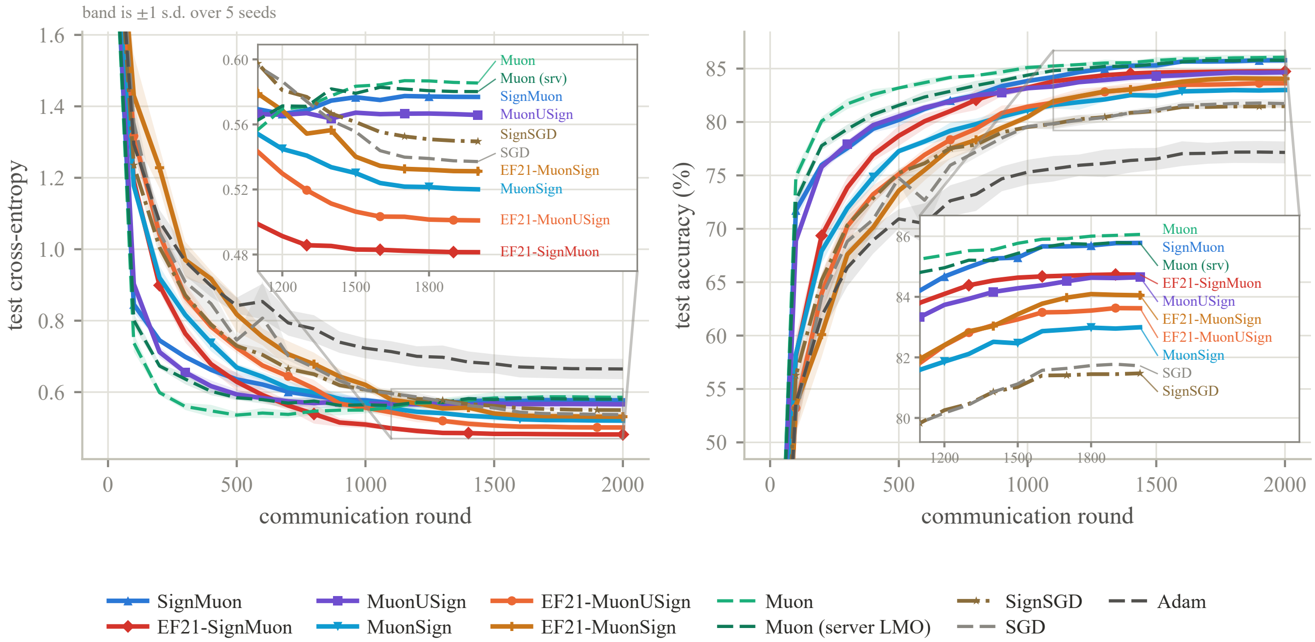

Figure A8: CIFAR-10 federated learning on CNN2 at $N = 11$ clients and batch 192 per client, every method evaluated on the exact server model $\mathbf{X}_t$. Both panels are clipped to exclude round 0, the untrained model, which is identical for every method. Solid lines are the one-bit methods, dashed the uncompressed references and dash-dot SignSGD, so that hue is not the only channel separating eleven curves. The insets magnify the final 45% of rounds, where every number in Table 2 is read, and name each curve at its own end; Adam finishes clear of the other ten and so falls outside the magnified window. Eleven methods are drawn, one more than Table 2, the extra being the server-side-LMO Muon control.

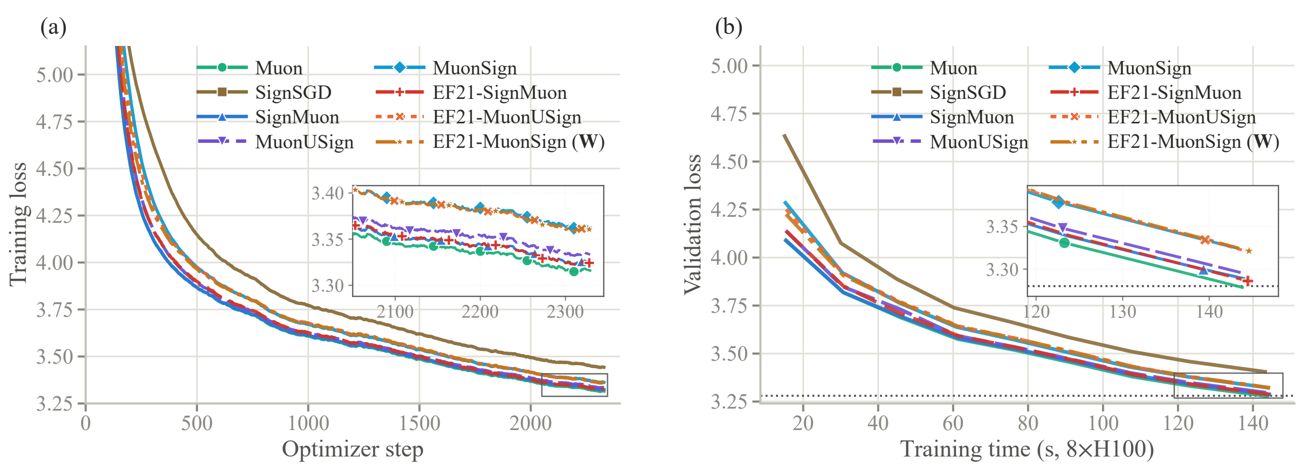

Figure A9: NanoGPT, supporting curves. (a) Training loss against optimizer step (EMA-smoothed; the logged quantity is a per-rank sum over 32,768 tokens, divided out here). (b) Validation loss against the speedrun clock, which excludes validation and compilation. Insets magnify the boxed tails, as in Figure 3. Marker positions are staggered in both figures so that curves lying within a line width of one another can still be followed individually. The ordering is the same on both axes: no method incurs a measurable wall-clock premium for its compressor or its error-feedback buffers. Note in (a) that EF21-MuonSign's training loss, taken at $\mathbf{W}$, exhibits no degradation throughout, which is what localizes its validation gap to the tracking of $\mathbf{X}$ rather than to training.

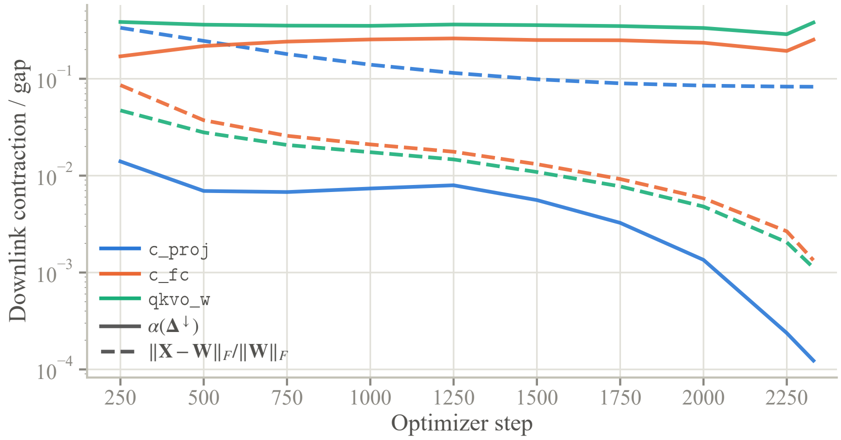

Figure A10: The downlink measurement for EF21-MuonSign, per layer type: the contraction $\alpha(\mathbf{\Delta}^{\downarrow})$ the scaled sign actually achieves (solid) and the resulting server/broadcast gap $\|\mathbf{X}_t - \mathbf{W}_t\|_F / \|\mathbf{W}_t\|_F$ (dashed). On the zero-initialized `c_proj` the contraction sits one to two orders of magnitude below the other two from step 250 on, then collapses over the last 500 steps to 1.2×10^{-4}, three orders below them, while the gap stops decreasing near 10^{-1}; the other two contract at $\Theta(10^{-1})$ throughout and close their gaps by 42 and $64\times$ over the run, ending near 10^{-3}. This is Remark 4 rendered as a measurement.

A.16 Language-modelling details

Setup. Upstream modded-nanoGPT record #40 (2025-10-04), the last record before NorMuon and hence the last whose hidden-matrix optimizer is a clean, separable momentum → LMO → step Muon, so our variants inject exactly at the LMO. Model: 12 layers, model dimension 768, 6 heads of dimension 128, vocabulary 50,257; hidden matrices are 768×3072 (the merged $\mathbf{Q}/\mathbf{K}/\mathbf{V}/\mathbf{O}$ weight `qkvo_w` is used as four 768×768 blocks, and both the LMO and the compressor scale are applied per block; the two MLP matrices are the up-projection `c_fc` and the zero-initialized output projection `c_proj`, which maps the 3072-dimensional hidden activation back to the model dimension). Data: FineWeb10B, the 10B-token sample of FineWeb (Penedo et al. 2024) that the speedrun repository prepares and tokenizes; 262,144 tokens per step, 2330 steps (= 611M tokens), validation on the fixed 10,485,760-token split. Hardware: one rented 8×H100 SXM node (80 GB per GPU; dual Xeon Platinum host, 224 vCPU, 2 TB RAM, PCIe 5.0 ×16, NVMe scratch), driver 595.71.05, running PyTorch 2.10.0+cu128 under Python 3.12.3 in a virtual environment of its own rather than the container's torch, since the prebuilt Flash-Attention-3 kernel the record fetches exists for no CUDA-13 build; one process per GPU. Gradients are averaged by `reduce_scatter` so the owning rank runs the centralized update and `all_gather` returns the parameter, i.e. the compression is a property of the update rule, as in the centralized algorithms we analyze.

Hyperparameters. Matrix/gate optimizer: $\eta_0 = 0.06$ (LMO family) or 0.03 (SIGN family), per-layer scaled by the unit-gain rule (Appendix A.17); Nesterov momentum $\mu = 0.95$, warmed up linearly from 0.85 over the first 300 steps and cooled back to 0.85 over the last 50; weight decay 0 (the record's own value); η constant then linearly cooled to $0.1\eta_0$ over the final 45% of the 2290 scheduled iterations, with the 40-step extension held at that floor; LMO by 5 Polar-Express iterations. Auxiliary parameters (embeddings, scalars, head) use the record's distributed Adam unchanged: $\eta = 0.008$, $\beta = (0.65, 0.95)$, $\varepsilon = 10^{-8}$, no weight decay, per-parameter multipliers 75 on embeddings and 5 on scalars, stepped every other iteration. Nothing above was tuned by us: the LMO family runs at the record's own η_0, and every value outside the matrix optimizer is the record's. Wall-clock varies by at most 1.1% across all eight methods (61.4–62.1 ms/step), all of them some 2% above the record's own 60.4 ms/step, which every method pays equally: our port replaces its Triton kernels and batched sharded transport with a pure-torch per-parameter equivalent.

Supporting curves. Figure A9 gives the two views Figure 3 omits: training loss against optimizer step, and validation loss against the speedrun clock. The second is what supports the claim of equal wall-clock in Section 5.3; the first shows that EF21-MuonSign trains normally at $\mathbf{W}$, which is what places its validation gap in the tracking of $\mathbf{X}$ rather than in training. Figure A10 then measures that tracking directly, per layer type.

Compressor diagnostics. Table A7 reports, per layer type at the final step, the contraction each scaled sign achieves, $\alpha(\mathbf{\Delta}) = \|\mathbf{\Delta}\|_1^2/(d\|\mathbf{\Delta}\|_F^2)$, and the relative estimator lag. Three observations merit separate comment. *(i)* Every uplink is well-contractive, $\alpha \in [0.32, 0.64]$ against the isotropic $2/\pi = 0.637$, and EF21-SignMuon's is uniformly the best, the entries of its orthogonal target being the most evenly spread; yet the uplink *lag* is large (0.61–0.82), so that the estimator remains far from its target at every step and the methods nonetheless train well, the LMO being scale-invariant and requiring only the

Layer type	uplink α	uplink lag	downlink α	downlink gap
EF21-SignMuon				
`qkvo_w` ($\times 10$)	0.640	0.66	–	–
`c_fc` ($\times 11$)	0.633	0.62	–	–
`c_proj` ($\times 11$)	0.634	0.62	–	–
EF21-MuonUSign				
`qkvo_w`	0.374	0.81	–	–
`c_fc`	0.323	0.82	–	–
`c_proj`	0.597	0.62	–	–
EF21-MuonSign				
`qkvo_w`	0.406	0.79	0.380	0.0011
`c_fc`	0.355	0.78	0.253	0.0013
`c_proj`	0.596	0.61	$\mathbf{1.2{\times}10^{-4}}$	$\mathbf{0.083}$

Table A7: Compressor diagnostics at step 2330, medians over the identical layers of each type. α is the contraction the scaled sign achieves on that round's residual ($2/\pi$ for an isotropic residual, $1/d$ in the worst case); "lag" is $\|\text{target} - \text{estimator}\|_F/\|\text{target}\|_F$; "gap" is $\|\mathbf{X}_t - \mathbf{W}_t\|_F/\|\mathbf{W}_t\|_F$. Downlink columns are empty for the two methods that broadcast exactly. Gates are omitted (they are 6×12 and 1×12).

direction. *(ii)* The single anomaly in the table is the `c_proj` downlink, $\alpha = 1.2 \times 10^{-4}$, a factor 4.8×10^3 below its uplink on the very same layer. Since both compressors are the same operator, the difference is a property of the residual, not of the compressor: the uplink residual is refreshed by an exogenous stochastic gradient each round, the downlink residual is generated by the compressor's own recursion (Remark 4). *(iii)* The resulting validation loss at the server model $\mathbf{X}$ falls to 4.20 by step 750, rises to 5.52 by step 1500, and then holds there (5.51–5.58 to the end of the run): a persistent offset above the broadcast model, not a divergence.

Where the two models part. The `c_proj` anomaly of item *(ii)* is what separates EF21-MuonSign's two models: on that layer the gap $\|\mathbf{X}_t - \mathbf{W}_t\|_F/\|\mathbf{W}_t\|_F$ stops decreasing at 8% against $<1\%$ for every other layer and both gates, and it is the only layer whose gap fails to close, which is what localizes the ≈ 2.2-nat offset of item *(iii)* to it. The reason is the initialization. A layer built from zero receives maximally correlated updates, so its downlink residual concentrates, and a compressor that moves every coordinate by $\operatorname{mean}|\boldsymbol{\Delta}^{\downarrow}|$ cannot catch the coordinates driven hardest. This is the mechanism of Remark 4 rather than a tuning failure. Lowering η_0 does not repair it, because the admissible step size would have to shrink by a further $\sqrt{r}$ in the layer rank (Remark 5); only a downlink compressor contractive in the *spectral* norm would.

A.17 Per-Layer Step Sizes: the Unit-Gain Rule

This appendix derives the heuristic (7) stated in the main text, selects its one free exponent by measurement, and delimits its scope. SGD and Adam have no norm-fixed step, so the rule does not apply to them; both are run at one global rate throughout.

Our counterexamples, like the analysis of Mishra, Trivedi, and Kumar (2026), concern a single matrix, where one scalar step size suffices. A network has layers of very different shapes, and the methods of this paper produce step matrices from *two* families whose norms scale differently with shape, so a single global η cannot be simultaneously correct for both families and across layers. This is not a lacuna we are obliged to tolerate: the layer-wise LMO framework to which our convergence result reduces (Riabinin et al. 2025; Gruntkowska et al. 2025) already carries per-layer norms $\|\cdot\|_{(\ell)}$, smoothness constants L_ℓ and radii η_ℓ; it is only the *experiments* that have hitherto fixed that radius at a constant. What follows instantiates it. The specific rule is this paper's own; the criterion behind it is borrowed (it is the average-case form of the spectral scaling condition of the maximal-update literature (Yang et al. 2021; Yang, Simon, and Bernstein 2023; Large et al. 2024)), and its LMO branch reproduces the aspect factor Muon already uses in practice (Jordan et al. 2024), which is the external check we rely on.

The two families. For a parameter reshaped to $\mathbf{X} \in \mathbb{R}^{m \times n}$ (m the output dimension, n the input dimension, $r = \operatorname{rank} = \min(m, n)$ generically), the step matrix $\mathbf{P}$ that a method applies belongs to one of two families:

$$\mathbf{P} = \mathbf{U}\mathbf{V}^\top \ (\textsc{lmo}), \qquad \mathbf{P} \in \{\pm 1\}^{m \times n} \ (\textsc{sign}). \tag{36}$$

The LMO family comprises Muon, MuonUSign, EF21-MuonUSign, EF21-MuonSign and EF21-SignMuon; the SIGN family comprises SignMuon, MuonSign and SignSGD. Both have *exactly* known Frobenius norms: $\|\mathbf{U}\mathbf{V}^\top\|_F^2 = \operatorname{tr}(\mathbf{V}\mathbf{U}^\top\mathbf{U}\mathbf{V}^\top) = \operatorname{tr}(\mathbf{V}^\top\mathbf{V}) = r$, so $\|\mathbf{P}\|_F = \sqrt{r}$; and a ± 1 matrix has $\|\mathbf{P}\|_F = \sqrt{mn}$. (EF21-SignMuon steps along the error-feedback estimator $\mathbf{d}_t^{\text{est}}$ of $\operatorname{polar}(\tilde{\mathbf{M}}_t)$ rather than along the oracle output itself, so for it $\|\mathbf{P}\|_F = \sqrt{r}$ holds only in the limit; we assign it to the LMO family on that basis.)

The criterion. Define the *RMS gain* of $\mathbf{A} \in \mathbb{R}^{m\times n}$, i.e. how much it amplifies a generic input in root-mean-square terms, with $\mathrm{rms}(\mathbf{v}) := \|\mathbf{v}\|/\sqrt{\dim}$:

$$\gamma(\mathbf{A})^2 := \frac{\mathbb{E}_{\mathbf{u}}\big[\mathrm{rms}(\mathbf{A}\mathbf{u})^2\big]}{\mathbb{E}_{\mathbf{u}}\big[\mathrm{rms}(\mathbf{u})^2\big]}, \qquad \mathbf{u} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_n). \tag{37}$$

Since $\mathbb{E}\|\mathbf{A}\mathbf{u}\|^2 = \mathrm{tr}(\mathbf{A}^\top\mathbf{A}) = \|\mathbf{A}\|_F^2$ and $\mathbb{E}\|\mathbf{u}\|^2 = n$, (37) evaluates in closed form:

$$\gamma(\mathbf{A}) = \frac{\|\mathbf{A}\|_F}{\sqrt{m}}. \tag{38}$$

The single modelling assumption is that $\mathbf{u}$ is isotropic *and independent of* $\mathbf{A}$, to which we return below. Controlling a layer update's RMS-to-RMS effect is precisely the desideratum of the spectral scaling condition (Yang, Simon, and Bernstein 2023) and of the modular norm (Large et al. 2024); (37) is its average-case (isotropic-input) version, and when the independence fails, that is, for aligned inputs, where the operator norm governs the gain instead, one recovers the μP value $a = 1$ discussed below.

Every standard initialization with variance $\propto 1/n$ has *shape-independent* gain. For He normal ($\sigma^2 = 2/n$), $\|\mathbf{X}\|_F = \sigma\sqrt{mn} = \sqrt{2m}$, so $\gamma(\mathbf{X}) = \sqrt{2}$; for PyTorch's default Kaiming-uniform convolution ($\sigma^2 = 1/(3n)$), $\gamma(\mathbf{X}) = 1/\sqrt{3}$. Either way a constant. Requiring the update's gain to be a fixed fraction of the weight's is therefore simply the requirement that *the per-step gain be the same on every layer*, and by (38) that is one formula:

$$\boxed{\eta_\ell = \eta_0\,\lambda_\ell, \qquad \lambda_\ell = \frac{\sqrt{m}}{\|\mathbf{P}\|_F}} \tag{39}$$

which gives $\gamma(\eta_\ell\mathbf{P}) = \eta_0$ exactly, for every shape and both families, so that η_0 *is* the per-step RMS gain. One caveat: the rule is derived from $\|\mathbf{U}\mathbf{V}^\top\|_F = \sqrt{r}$, which holds for the exact oracle. Five Newton–Schulz steps leave the singular values of the returned matrix in a band around 1 rather than at 1, so its Frobenius norm falls below $\sqrt{r}$, by 5–22% on our layer shapes; for the LMO-terminated methods η_0 is therefore the per-step RMS gain of the exact step, realized up to that shape-dependent factor. Substituting the two Frobenius norms of (36),

$$\lambda_\ell^{\text{LMO}} = \sqrt{\max\big(1, \tfrac{m}{n}\big)}, \qquad \lambda_\ell^{\text{SIGN}} = \frac{1}{\sqrt{n}}. \tag{40}$$

Justification of (40)**.** The first expression is exactly the aspect-ratio factor $\sqrt{\max(1, m/n)}$ present in the reference Muon implementation (Jordan et al. 2024), which was introduced as a practical heuristic. It is also the point at which the rule parts company with the geometric alternative: taking the layer norm to be RMS→RMS rather than spectral (Appendix A.1) prescribes $\sqrt{m/n}$, which coincides with unit gain for $m \ge n$ and falls below it for $m < n$, where the two disagree and the implemented factor is the unit-gain one. The disagreement is the informative case: on a wide layer the RMS→RMS ball shrinks the step in proportion to $\sqrt{m/n}$, whereas the gain (38) of $\mathbf{U}\mathbf{V}^\top$ is already $\sqrt{\min(m,n)/m} = 1$ there and needs no correction. Nor does the geometric route reach the SIGN family at all: a ± 1 matrix is the oracle output of no norm (Theorems 1–3), so there is no unit ball whose radius could set its scale, while (39) applies to it unchanged. The unit-gain criterion *derives* the aspect factor, and also explains why Muon's step size is known to transfer across widths: its step has $\gamma = \eta$ independently of n, so no correction in the input dimension is needed. The second expression is the counterpart the sign family has never been given. Only η_0 is tuned, and it is now a shape-free quantity; the shape dependence is determined a priori.

Two consequences follow. First, λ_ℓ is a deterministic function of the layer shape, known to server and clients alike, so per-layer step sizes require *no* communication and leave the one-bit-per-parameter budget intact. Second, on the CIFAR ResNet-18 of our experiments $\lambda_\ell^{\text{SIGN}}$ spans a factor of 13 across layers (from $1/\sqrt{27}$ at the first convolution to $1/\sqrt{4608}$ in the last stage), so a single global rate is necessarily a compromise: roughly correct for the middle of the network, several-fold too large at the first convolution and too small at the last stage. The LMO family is exempt from this, which is one reason full-precision Muon is easier to tune than its sign-compressed variants.

Selecting the exponent. Writing $\lambda_\ell^{\text{SIGN}} = n^{-a}$, the unit-gain rule is $a = \frac{1}{2}$. Identity (38) assumes the input independent of $\mathbf{A}$, the right model for a *single* step; if the *accumulated* update $\sum_t \eta\mathbf{P}_t$ aligns with the activations, its gain is $\Theta(\eta n)$ rather than $\Theta(\eta\sqrt{n})$, giving $a = 1$, the μP rule $\eta \propto 1/n$ for sign-like updates (Yang et al. 2021). A direct measurement decides between the regimes: we track the realized gain $\|\mathbf{X}_t - \mathbf{X}_0\|_F/\sqrt{m}$ at a *constant* step size (under a decaying schedule the accumulation saturates and the fit reports the schedule) and fit its growth exponent h in t, which is $\frac{1}{2}$ for incoherent accumulation and 1 for aligned. Over 20 epochs the fit returns $\hat{h} = 0.513$ for Muon, 0.515 for SignMuon, 0.490 for SignSGD and 0.561 for MuonSign, each at $R^2 \ge 0.999$. Muon is the control: (40) and μP prescribe the LMO family the identical multiplier, so its exponent is the diagnostic's reading when the rule is not in question, and the sign methods match it. All four values lie near $\frac{1}{2}$ and none near 1, so the accumulation is incoherent on this network and we adopt $a = \frac{1}{2}$ for both families in every network experiment. The exponent fixes the shape dependence of λ_ℓ and the transfer of η_0 across widths; it makes no claim about which a maximizes accuracy at one fixed width, where the choice is largely absorbed into η_0.

The placement of weight decay is not arbitrary. The same scale invariance that makes λ_ℓ necessary also dictates *where* an ℓ_2 penalty may be applied: $\mathrm{sign}(c\mathbf{M}) = \mathrm{sign}(\mathbf{M})$ and $\mathrm{polar}(c\mathbf{M}) = \mathrm{polar}(\mathbf{M})$ for all $c > 0$. Folding the decay into the gradient, $\widetilde{\mathbf{G}}_t = \mathbf{G}_t + \lambda_{\mathrm{wd}}\mathbf{X}_t$, the convention of Mishra, Trivedi, and Kumar (2026) and of most sign-method implementations, therefore supplies no contraction: the step length $\eta_t\lambda_\ell\|\mathbf{P}\|_F$ is fixed by (36), and the decay term can only rotate the direction. That rotation is governed by $\rho_t = \lambda_{\mathrm{wd}}\|\mathbf{X}_t\|_F/\|\mathbf{G}_t\|_F$, which drifts from negligible to $\Theta(1)$ as $\|\mathbf{G}_t\|_F$ falls over training and which depends on each method's own momentum scale, so one nominal λ_{wd} is a different perturbation for each method. Decoupled decay, $\mathbf{X}_{t+1} = (1-\eta_t\lambda_{\mathrm{wd}})\mathbf{X}_t - \eta_t\lambda_\ell\mathbf{P}_t$, is by contrast commensurate with the update under the unit-gain rule: its displacement has gain $\eta_t\lambda_{\mathrm{wd}}\gamma(\mathbf{X}_t)$ against the step's η_t, a ratio free of η_t, of the layer shape and of the method. We therefore decouple, and use the coupled form only in an ablation. This explains an observation of Mishra, Trivedi, and Kumar (2026): sweeping $\lambda_{\mathrm{wd}} \in \{0, 0.1, 0.2\}$ coupled over 330 CIFAR-10 ResNet-50 configurations, every Muon and Sign-Muon entry in their top ten (the only entries with a decay sweep) selects $\lambda_{\mathrm{wd}} = 0$; at the two nonzero values the decay term dominates the gradient in $\widetilde{\mathbf{G}}_t$ for most of training, so the transmitted sign approaches $\mathrm{sign}(\mathbf{X}_t)$ and the sweep rejects this placement of the penalty, not regularization as such.

Scope of the analysis. Our theorems are stated for unregularized f, so we report unregularized runs as the primary comparison and weight decay as an ablation; the reference nanoGPT configuration we build on also uses $\lambda_{\mathrm{wd}} = 0$ for every parameter group. Two remarks delimit the gap. First, the *coupled* form is covered verbatim: it is nothing other than running the same method on $f_\lambda = f + \frac{\lambda_{\mathrm{wd}}}{2}\|\mathbf{X}\|_F^2$, so every rate carries over with $L_i \mapsto L_i + \lambda_{\mathrm{wd}} r_i$. The rank factor is not slack in the bound: our smoothness is measured in the nuclear norm against a spectral-norm displacement (Assumption 2), and $\|\lambda_{\mathrm{wd}}\mathbf{Z}\|_* \le \lambda_{\mathrm{wd}} r_i\|\mathbf{Z}\|_{2\to 2}$ is tight at $\mathbf{Z} = \mathbf{I}_{r_i}$; $L \mapsto L + \lambda_{\mathrm{wd}}$ would be the Euclidean statement, and these rates are not Euclidean. The paradox is that this is precisely the variant which does not regularize. Second, the *decoupled* form is not covered by our rates, yet it furnishes something the analysis requires. Since $\|\mathbf{P}_t\|_F$ is a known constant and $\lambda_\ell\|\mathbf{P}_t\|_F = \sqrt{m}$ identically under (39), the triangle inequality gives $\|\mathbf{X}_{t+1}\|_F \le (1-\eta_t\lambda_{\mathrm{wd}})\|\mathbf{X}_t\|_F + \eta_t\sqrt{m}$, and hence, for any $\eta_t\lambda_{\mathrm{wd}} \le 1$,

$$\gamma(\mathbf{X}_t) \;\le\; \max\bigl\{\gamma(\mathbf{X}_0),\ \lambda_{\mathrm{wd}}^{-1}\bigr\} \quad \text{for all } t, \tag{41}$$

a bound on the layer's gain that is uniform in t and *independent of the layer shape*. Norm-constrained updates are what render this possible: for SGD the step length is data-dependent and no such a priori bound exists. Since layer-wise LMO analyses assume smoothness on a bounded region, (41) is the statement that decoupled decay supplies that region. We do not claim a convergence rate for the decoupled variant itself.

Sensitivity to the rule. Equation (7) is a heuristic, so we state its scope precisely. Each candidate rule, one global rate ($\lambda_\ell = 1$), unit gain ($\lambda_\ell = n^{-1/2}$) or μP ($\lambda_\ell = n^{-1}$), shifts the selected η_0 by roughly the multiplier it prescribes; what would matter is a change in the *ordering* of the methods. The exposure is bounded twice over: the LMO family cannot move, unit gain and μP prescribing it the identical multiplier, and the three SIGN methods are tuned and reported under one rule, so a wrong exponent rescales them alike. Neither consideration is a measurement, so Table A8 re-tunes the three SIGN methods from scratch under each rule on federated CNN2, whose three matrix parameters span a factor of 7.8 in $\lambda_\ell^{\text{SIGN}}$, and runs each selected rate at the reporting horizon.

Two things follow. The selected η_0 moves by roughly the multiplier the rule prescribes, which is the rule working and not a defect: measured against one global rate, unit gain raises SignMuon's rate by a factor of 50 and μP by 10^3, against the $\sqrt{n} \in [8.7, 67.9]$ and $n \in [75, 4608]$ that the three layer shapes prescribe. The ordering, meanwhile, does not move: SignMuon, then MuonSign, then SignSGD under every rule, the first ahead of the last by 4.3 points under the global rate, 4.3 under unit gain and 4.0 under μP. Within a method the rules agree to within 0.2 points for SignMuon and 0.1 for SignSGD, at or below the seed spread; MuonSign is the one case where they separate at all, unit gain standing 0.57 above μP and 0.68 above the global rate, about two seed spreads, and in the direction that favours the rule we adopted. The sign-family ordering of Table 2 therefore does not rest on the exponent, which the exposure argument above could only bound rather than establish.

Comparison with concurrent work. Mishra, Trivedi, and Kumar (2026) analyse the normalized update $\mathbf{D}_t = \bar{\mathbf{S}}_t/\sqrt{mn}$, justified by $\|\mathbf{D}_t\|_{\mathrm{op}} \le \|\mathbf{D}_t\|_F = 1$ under their spectral-norm smoothness assumption, and remark that updating with $\bar{\mathbf{S}}_t$ directly is equivalent after absorbing $\sqrt{mn}$ into η_t. That equivalence holds for a single matrix but not across layers of differing shape, and their algorithm applies no shape factor, so their experiments use a single global rate as well. The substitution is also loose in a shape-dependent way: $\bar{\mathbf{S}}_t$ has rank at most $\min(m,n)$, so $\|\bar{\mathbf{S}}_t\|_F \le \sqrt{\min(m,n)}\,\|\bar{\mathbf{S}}_t\|_{\mathrm{op}}$ and the substitution of $\sqrt{mn}$ for the operator norm is loose by up to $\sqrt{\min(m,n)}$. That bound itself grows with depth, from $\sqrt{27} \approx 5.2$ at the first convolution of a ResNet-18 to $\sqrt{512} \approx 22.6$ in the last stage, so the spectral radius the analysis assigns to the step varies across the network instead of remaining uniform. Table A9 sets the rules side by side.

A.18 Algorithms

Federated protocol. At the start of round t each client j holds the global model $\mathbf{X}_{t-1}$ and evaluates one stochastic gradient $\mathbf{G}_t^{(j)} = \nabla f_j(\mathbf{X}_{t-1}; \xi_t^{(j)})$ at it; clients take no local parameter steps, so one round is one server step and no client-drift term arises. (The released runs accumulate three mini-batches of 64 at fixed weights to save activation memory; the BatchNorm statistics

Method	Rule	λ_ℓ	η_0	Test acc (%)
SignMuon	global	1	0.002	85.68 ± 0.17
SignMuon	**unit gain**	$n^{-1/2}$	0.1	85.72 ± 0.24
SignMuon	μP	n^{-1}	2	85.52 ± 0.01
MuonSign	global	1	0.001	82.26 ± 0.40
MuonSign	**unit gain**	$n^{-1/2}$	0.02	82.94 ± 0.19
MuonSign	μP	n^{-1}	0.5	82.37 ± 0.27
SignSGD	global	1	0.0005	81.37 ± 0.21
SignSGD	**unit gain**	$n^{-1/2}$	0.01	81.44 ± 0.15
SignSGD	μP	n^{-1}	0.5	81.47 ± 0.25

Table A8: The per-layer rule ablation, on the federated CNN2 of Table 2. Each (method, rule) pair is re-tuned from scratch on the five-point lattice of Appendix A.11 and then run at 2000 rounds: three seeds under the two alternatives, and under unit gain the five seeds Table 2 reports.

Rule	LMO	SIGN	Equalizes
global ($a=0$)	1	1	nothing
RMS→RMS ball	$\sqrt{\frac{m}{n}}$	—	LMO trust region
Muon default	$\sqrt{\max(1, \frac{m}{n})}$	1	LMO gain
unit gain	$\sqrt{\max(1, \frac{m}{n})}$	$n^{-1/2}$	per-step gain, both
μP ($a=1$)	$\sqrt{\max(1, \frac{m}{n})}$	n^{-1}	aligned accumulation
Mishra et al.	$r^{-1/2}$	$(mn)^{-1/2}$	$\|\mathbf{P}\|_F$

Table A9: Per-layer step-size multipliers λ_ℓ. Only η_0 is tuned; λ_ℓ is fixed a priori by the shape. The last four rules differ only in the SIGN family, and the LMO column of the unit-gain rule coincides with the factor already employed in practice (Jordan et al. 2024), which is our principal evidence that (37) is the correct criterion. The second row is the multiplier implied by taking the layer geometry to be RMS→RMS rather than spectral: it departs from the other four for $m < n$ and, being a property of a unit ball, is undefined for steps that are the oracle of no ball. The last row is the normalization of Mishra, Trivedi, and Kumar (2026), which equalizes $\|\mathbf{P}\|_F$ rather than the gain.

being frozen (Appendix A.11), the loss is separable across samples and the average is a gradient at batch 192, except where a client's shard ends in a shorter mini-batch.) Each client updates its own momentum buffer, applies the LMO (Algorithm A1), and transmits the elementwise sign,

$$\mathbf{M}_t^{(j)} = \mu\mathbf{M}_{t-1}^{(j)} + (1-\mu)\,\mathbf{G}_t^{(j)}, \qquad \mathbf{D}_t^{(j)} = -A\big(\mathbf{M}_t^{(j)}\big), \qquad \mathbf{s}_t^{(j)} = \operatorname{sign}\big(\mathbf{D}_t^{(j)}\big),$$

with the exponential-moving-average momentum of (5), matching Algorithms A8–A9; the uplink is one bit per matrix parameter. The server aggregates by majority vote, $\mathbf{s}_t^{\text{agg}} = \operatorname{sign}\big(\sum_{j=1}^{N} \mathbf{s}_t^{(j)}\big)$, which is ± 1 in each component: client messages are ± 1-valued by the convention of Section 4, so at an odd client count the vote cannot tie, and at an even count a tie is broken by a fair coin. Momentum having been applied at the clients, the server steps directly, $\mathbf{X}_t = \mathbf{X}_{t-1} - \eta_t \mathbf{s}_t^{\text{agg}}$, and the vote rather than the model travels back down the link: every client applies the same ± 1-valued update to its local copy of the model, so client and server models remain identical and the downlink carries one bit per parameter as well (Appendix A.14). The final classification layer is exempt from the rule and trained with AdamW.

Federated error feedback. The repair for the biased sign compressor is EF21 (Richtárik, Sokolov, and Fatkhullin 2021) in the LMO form of Gruntkowska et al. (2025): it changes the uplink message, and for EF21-MuonSign the downlink as well. Client j compresses the residual $\Delta_t^{(j)}$ between its estimator and the quantity it would otherwise send, the polar factor $\mathbf{D}_t^{(j)}$ for EF21-SignMuon, whose oracle runs on the client, and the momentum $\tilde{\mathbf{M}}_t^{(j)}$ for EF21-MuonUSign and EF21-MuonSign, whose oracle runs on the server, and transmits the pair $(\operatorname{sign}(\Delta_t^{(j)}), \alpha_t^{(j)})$ with $\alpha_t^{(j)} = \operatorname{mean}|\Delta_t^{(j)}|$. The server accumulates these into a global estimator ($\mathbf{d}_t$ in Algorithm A8, $\mathbf{g}_t$ in Algorithm A9) and either steps along it or applies one LMO to it. The extra scalar is one full-precision number per matrix layer per round, so the uplink stays at $\approx$1 bit per parameter; the estimator itself is dense, so the downlink carries a full-precision model unless a second error-feedback loop compresses it, as EF21-MuonSign's does.

All six federated methods are instances of just two templates, separated by *where the Muon LMO is evaluated*. When the sign acts *after* the LMO (the SignMuon family), each client must orthogonalize locally, so the LMO runs on the worker and the client transmits a compressed *direction* (Algorithm A8). When the sign acts *before* the LMO (the MuonUSign/MuonSign family), the client transmits a compressed *gradient*, and the server reconstructs it and applies a single LMO (Algorithm A9). Within each

Algorithm A1: MuonLMO

Input: tensor $\mathbf{Y}$; Newton-Schulz coefficients $a = 3.4445$, $b = 4.7750$, $c = 2.0315$; iteration count ns_steps
Output: polar factor $\mathbf{D} \approx \mathbf{U}\mathbf{V}^\top$ of $\mathbf{Y}$

$\mathbf{Y} \leftarrow \mathrm{ReshapeTo2D}(\mathbf{Y})$ ▷ Flatten tensor to matrix $m \times n$
$\mathbf{Y} \leftarrow \mathbf{Y}/\|\mathbf{Y}\|_F$ ▷ Normalize (optional)
for $k = 1$ to ns_steps **do** ▷ 5th-order Newton-Schulz orthogonalization
    $\mathbf{A} \leftarrow \mathbf{Y}\mathbf{Y}^\top$
    $\mathbf{Y} \leftarrow a\,\mathbf{Y} - b\,\mathbf{A}\mathbf{Y} + c\,\mathbf{A}^2\,\mathbf{Y}$
end for
$\mathbf{D} \leftarrow \mathrm{ReshapeToOriginal}(\mathbf{Y})$ ▷ Reshape back to the original tensor shape
return $\mathbf{D}$

Algorithm A2: SignMuon

Input: Initial model $\mathbf{X}_0$, momentum coefficient μ, learning rate η_t
Output: Updated model $\mathbf{X}$

$\mathbf{M}_0 \leftarrow 0$
for $t = 1$ to T **do**
    $\mathbf{G}_t \leftarrow \nabla f(\mathbf{X}_{t-1}; \xi_t)$ ▷ Stochastic gradient
    $\mathbf{M}_t \leftarrow \mu \mathbf{M}_{t-1} + (1-\mu)\,\mathbf{G}_t$ ▷ Momentum accumulation
    $\tilde{\mathbf{M}}_t = \begin{cases} \mathbf{M}_t, & \text{(default)}, \\ (1-\mu)\,\mathbf{G}_t + \mu \mathbf{M}_t, & \text{(Nesterov)} \end{cases}$
    $\mathbf{D}_t \leftarrow \mathrm{MuonLMO}(\tilde{\mathbf{M}}_t)$
    $\mathbf{s}_t^\uparrow \leftarrow \mathrm{sign}(\mathbf{D}_t)$ ▷ Uplink sign compression
    $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t \mathbf{s}_t^\uparrow$ ▷ Update parameters
end for

Algorithm A3: EF21-SignMuon

Input: Initial model $\mathbf{X}_0$, momentum coefficient μ, learning rate η_t
Output: Updated model $\mathbf{X}$

$\mathbf{M}_0 \leftarrow 0, \quad \mathbf{d}_0^{\mathrm{est}} \leftarrow 0$
for $t = 1$ to T **do**
    $\mathbf{G}_t \leftarrow \nabla f(\mathbf{X}_{t-1}; \xi_t)$ ▷ Stochastic gradient
    $\mathbf{M}_t \leftarrow \mu \mathbf{M}_{t-1} + (1-\mu)\,\mathbf{G}_t$ ▷ Momentum accumulation (EMA)
    $\tilde{\mathbf{M}}_t = \begin{cases} \mathbf{M}_t, & \text{(default)}, \\ (1-\mu)\,\mathbf{G}_t + \mu \mathbf{M}_t, & \text{(Nesterov)} \end{cases}$
    $\mathbf{D}_t \leftarrow \mathrm{MuonLMO}(\tilde{\mathbf{M}}_t)$
    $\Delta_t^\uparrow \leftarrow \mathbf{D}_t - \mathbf{d}_{t-1}^{\mathrm{est}}$ ▷ Uplink residual (polar factor)
    $\alpha_t^\uparrow \leftarrow \mathrm{mean}(|\Delta_t^\uparrow|)$
    $\mathbf{d}_t^{\mathrm{est}} \leftarrow \mathbf{d}_{t-1}^{\mathrm{est}} + \alpha_t^\uparrow\,\mathrm{sign}(\Delta_t^\uparrow)$ ▷ Uplink EF21
    $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t \mathbf{d}_t^{\mathrm{est}}$ ▷ Update parameters
end for

template, a method is fixed by its uplink compressor $\mathcal{C}^\uparrow \in \{\mathrm{sign}, \mathrm{EF21}\}$ and downlink compressor $\mathcal{C}^\downarrow \in \{\mathrm{exact}, \mathrm{sign}, \mathrm{EF21\text{-}P}\}$; Table A10 lists the six instantiations.

The two uplinks aggregate differently, and each aggregation is forced. The EF21 uplink *averages* the decompressed messages, $\mathbf{g}_t = \mathbf{g}_{t-1} + \frac{1}{N}\sum_j \alpha_t^{(j)}\,\mathrm{sign}(\Delta_t^{(j)})$, as Gruntkowska et al. (2025, Algorithm 3) prescribe and as the reduction of Appendix A.10 requires; replacing that average by a vote would leave the framework and forfeit Theorem 5. The plain sign uplink instead takes a majority vote, $\mathrm{sign}(\sum_j \mathbf{s}_t^{(j)})$, *before* the server LMO. Voting is what keeps the oracle's argument a ± 1 matrix, so that the server-side method is exactly the centralized MuonUSign, $\mathrm{polar}(\mathrm{sign}(\cdot))$ of (6), evaluated at the aggregated sign; averaging would feed polar an argument valued in $\{-1, -1+2/N, \dots, 1\}$ and define a different method, one that agrees with MuonUSign only at $N = 1$. The choice also matches the sign-compression literature it inherits from (Bernstein et al. 2019). It carries no

Algorithm A4: MuonUSign

Input: Initial model $\mathbf{X}_0$, momentum coefficient μ, learning rate η_t
Output: Updated model $\mathbf{X}$

```
1: M_0 ← 0
2: for t = 1 to T do
3:     G_t ← ∇f(X_{t−1}; ξ_t)                          ▷ Stochastic gradient
4:     M_t ← μM_{t−1} + (1 − μ) G_t                     ▷ Momentum accumulation
5:     M̃_t = { M_t,                    (default),
             { (1 − μ) G_t + μM_t,      (Nesterov)
6:     s_t^↑ ← sign(M̃_t)                               ▷ Uplink sign compression
7:     D_t ← MuonLMO(s_t^↑)
8:     X_t ← X_{t−1} − η_t D_t                          ▷ Update parameters
9: end for
```

Algorithm A5: MuonSign

Input: Initial model $\mathbf{X}_0$, momentum coefficient μ, learning rate η_t
Output: Updated model $\mathbf{X}$

```
1: M_0 ← 0
2: for t = 1 to T do
3:     G_t ← ∇f(X_{t−1}; ξ_t)                          ▷ Stochastic gradient
4:     M_t ← μM_{t−1} + (1 − μ) G_t                     ▷ Momentum accumulation
5:     M̃_t = { M_t,                    (default),
             { (1 − μ) G_t + μM_t,      (Nesterov)
6:     s_t^↑ ← sign(M̃_t)                               ▷ Uplink sign compression
7:     D_t ← MuonLMO(s_t^↑)
8:     s_t^↓ ← sign(D_t)                                ▷ Downlink sign compression
9:     X_t ← X_{t−1} − η_t s_t^↓                        ▷ Update parameters
10: end for
```

Algorithm A6: EF21-MuonUSign

Input: Initial model $\mathbf{X}_0$, momentum coefficient μ, learning rate η_t
Output: Updated model $\mathbf{X}$

```
1: M_0 ← 0,   g_0^est ← 0
2: for t = 1 to T do
3:     G_t ← ∇f(X_{t−1}; ξ_t)                          ▷ Stochastic gradient
4:     M_t ← μM_{t−1} + (1 − μ) G_t                     ▷ Momentum accumulation
5:     M̃_t = { M_t,                    (default),
             { (1 − μ) G_t + μM_t,      (Nesterov)
6:     Δ_t^↑ ← M̃_t − g_{t−1}^est                       ▷ Uplink residual
7:     α_t^↑ ← mean(|Δ_t^↑|)
8:     g_t^est ← g_{t−1}^est + α_t^↑ sign(Δ_t^↑)        ▷ Uplink EF21
9:     D_t ← MuonLMO(g_t^est)
10:    X_t ← X_{t−1} − η_t D_t                          ▷ Update parameters
11: end for
```

consequence for the downlink of this family, the polar factor being dense either way. As in the centralized setting, both templates are applied per matrix parameter, while vector parameters and the final classification layer are optimized with AdamW.

Algorithm A7: EF21-MuonSign

Input: Initial model $\mathbf{X}_0 = \mathbf{W}_0$, momentum coefficient μ, learning rate η_t
Output: Updated model $\mathbf{X}$

1: $\mathbf{M}_0 \leftarrow 0, \quad \mathbf{g}_0^{\text{est}} \leftarrow 0$
2: **for** $t = 1$ to T **do**
3: $\quad \mathbf{G}_t \leftarrow \nabla f(\mathbf{W}_{t-1}; \xi_t)$ ▷ At the broadcast model
4: $\quad \mathbf{M}_t \leftarrow \mu \mathbf{M}_{t-1} + (1-\mu)\,\mathbf{G}_t$ ▷ Momentum accumulation
5: $\quad \tilde{\mathbf{M}}_t = \begin{cases} \mathbf{M}_t, & \text{(default)}, \\ (1-\mu)\,\mathbf{G}_t + \mu \mathbf{M}_t, & \text{(Nesterov)} \end{cases}$
6: $\quad \Delta_t^{\uparrow} \leftarrow \tilde{\mathbf{M}}_t - \mathbf{g}_{t-1}^{\text{est}}$ ▷ Uplink residual
7: $\quad \alpha_t^{\uparrow} \leftarrow \operatorname{mean}(|\Delta_t^{\uparrow}|)$
8: $\quad \mathbf{g}_t^{\text{est}} \leftarrow \mathbf{g}_{t-1}^{\text{est}} + \alpha_t^{\uparrow} \operatorname{sign}(\Delta_t^{\uparrow})$ ▷ Uplink EF21
9: $\quad \mathbf{D}_t \leftarrow \operatorname{MuonLMO}(\mathbf{g}_t^{\text{est}})$
10: $\quad \mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t \mathbf{D}_t$ ▷ Update parameters (server)
11: $\quad \Delta_t^{\downarrow} \leftarrow \mathbf{X}_t - \mathbf{W}_{t-1}$ ▷ Downlink residual
12: $\quad \alpha_t^{\downarrow} \leftarrow \operatorname{mean}(|\Delta_t^{\downarrow}|)$
13: $\quad \mathbf{W}_t \leftarrow \mathbf{W}_{t-1} + \alpha_t^{\downarrow} \operatorname{sign}(\Delta_t^{\downarrow})$ ▷ Downlink EF21-P
14: **end for**

Method	**LMO**	**Uplink** $\mathcal{C}^{\uparrow}$	**Downlink** $\mathcal{C}^{\downarrow}$
SignMuon	worker	sign / MV	exact
EF21-SignMuon	worker	EF21	exact
MuonUSign	server	sign / MV	exact
MuonSign	server	sign / MV	sign
EF21-MuonUSign	server	EF21	exact
EF21-MuonSign	server	EF21	EF21-P

Table A10: The six federated methods as instantiations of the two templates: Algorithm A8 (worker-side LMO; rows 1–2) and Algorithm A9 (server-side LMO; rows 3–6). Each method is fixed by the LMO location and the uplink/downlink compressors. "MV": majority vote; "EF21-P": primal (model-side) error feedback; "exact": the server applies no downlink compressor. That is not the same as a full-precision downlink: SignMuon's server has nothing to compress because the object it distributes, the majority vote, is already ± 1-valued, so its downlink is one bit per parameter all the same (Appendix A.14). The three methods that do broadcast a dense model are MuonUSign, EF21-SignMuon and EF21-MuonUSign.

Algorithm A8: Federated SignMuon / EF21-SignMuon (worker-side LMO)

Input: initial model $\mathbf{X}_0$; clients N; rounds T; learning rate η_t; momentum μ; uplink compressor $\mathcal{C}^{\uparrow} \in \{\text{sign}, \text{EF21}\}$ (Table A10)
Output: global model $\mathbf{X}_T$

$\mathbf{M}_0^{(j)} \leftarrow 0$, $\mathbf{d}_0^{(j)} \leftarrow 0$ for all j; $\quad \mathbf{d}_0 \leftarrow 0$
broadcast $\mathbf{X}_0$ *once*; every client keeps a local copy, refreshed below from the downlink message alone
for $t = 1$ to T **do**
  for $j = 1$ to N **in parallel do** ▷ client j, holding $\mathbf{X}_{t-1}$
    $\mathbf{G}_t^{(j)} \leftarrow \nabla f_j(\mathbf{X}_{t-1}; \xi_t^{(j)})$
    $\mathbf{M}_t^{(j)} \leftarrow \mu \mathbf{M}_{t-1}^{(j)} + (1-\mu)\, \mathbf{G}_t^{(j)}$
    $\tilde{\mathbf{M}}_t^{(j)} \leftarrow \mathbf{M}_t^{(j)}$ ▷ or $(1-\mu)\mathbf{G}_t^{(j)} + \mu \mathbf{M}_t^{(j)}$ (Nesterov)
    $\mathbf{D}_t^{(j)} \leftarrow \mathrm{MuonLMO}\big(\tilde{\mathbf{M}}_t^{(j)}\big)$ ▷ LMO on the client
    if $\mathcal{C}^{\uparrow} = \text{EF21}$ **then** ▷ EF21-SignMuon
      $\Delta_t^{(j)} \leftarrow \mathbf{D}_t^{(j)} - \mathbf{d}_{t-1}^{(j)}; \quad \alpha_t^{(j)} \leftarrow \mathrm{mean}(|\Delta_t^{(j)}|)$
      $\mathbf{d}_t^{(j)} \leftarrow \mathbf{d}_{t-1}^{(j)} + \alpha_t^{(j)}\, \mathrm{sign}(\Delta_t^{(j)})$
      send $\big(\mathrm{sign}(\Delta_t^{(j)}),\, \alpha_t^{(j)}\big)$
    else ▷ SignMuon
      send $\mathbf{s}_t^{(j)} \leftarrow \mathrm{sign}(\mathbf{D}_t^{(j)})$
    end if
  end for
  on the server:
  if $\mathcal{C}^{\uparrow} = \text{EF21}$ **then** ▷ EF21-SignMuon: $\mathbf{d}_t$ is dense, 32 bits down
    $\mathbf{d}_t \leftarrow \mathbf{d}_{t-1} + \frac{1}{N} \sum_j \alpha_t^{(j)}\, \mathrm{sign}(\Delta_t^{(j)})$ ▷ aggregate direction
    $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t\, \mathbf{d}_t$; **broadcast** $\mathbf{X}_t$
  else ▷ SignMuon: the vote is ± 1, 1 bit down
    $\hat{\mathbf{s}}_t \leftarrow \mathrm{sign}\big(\sum_j \mathbf{s}_t^{(j)}\big)$ ▷ majority vote
    $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t\, \hat{\mathbf{s}}_t$; **broadcast** $\hat{\mathbf{s}}_t$ ▷ clients apply the same step
  end if
end for

Algorithm A9: Federated MuonUSign / MuonSign / EF21-MuonUSign / EF21-MuonSign (server-side LMO)

Input: initial model $\mathbf{X}_0$; clients N; rounds T; learning rate η_t; momentum μ; compressors $\mathcal{C}^{\uparrow} \in \{\text{sign}, \text{EF21}\}$, $\mathcal{C}^{\downarrow} \in \{\text{exact}, \text{sign}, \text{EF21-P}\}$ (Table A10)
Output: global model $\mathbf{X}_T$

$\mathbf{M}_0^{(j)} \leftarrow 0$, $\mathbf{g}_0^{(j)} \leftarrow 0$ for all j; $\quad \mathbf{g}_0 \leftarrow 0$
broadcast $\mathbf{X}_0$ *once*; every client holds the broadcast model $\mathbf{W}_0 \leftarrow \mathbf{X}_0$, refreshed below from the downlink message alone ($\mathbf{W}_t = \mathbf{X}_t$ unless $\mathcal{C}^{\downarrow} = \text{EF21-P}$)
for $t = 1$ to T **do**
  for $j = 1$ to N **in parallel do** ▷ client j, holding $\mathbf{W}_{t-1}$
    $\mathbf{G}_t^{(j)} \leftarrow \nabla f_j(\mathbf{W}_{t-1}; \xi_t^{(j)})$
    $\mathbf{M}_t^{(j)} \leftarrow \mu \mathbf{M}_{t-1}^{(j)} + (1-\mu)\, \mathbf{G}_t^{(j)}$
    $\tilde{\mathbf{M}}_t^{(j)} \leftarrow \mathbf{M}_t^{(j)}$ ▷ or $(1-\mu)\mathbf{G}_t^{(j)} + \mu \mathbf{M}_t^{(j)}$ (Nesterov)
    if $\mathcal{C}^{\uparrow} = \text{EF21}$ **then** ▷ EF21-MuonUSign / EF21-MuonSign
      $\Delta_t^{(j)} \leftarrow \tilde{\mathbf{M}}_t^{(j)} - \mathbf{g}_{t-1}^{(j)}$; $\quad \alpha_t^{(j)} \leftarrow \mathrm{mean}(|\Delta_t^{(j)}|)$
      $\mathbf{g}_t^{(j)} \leftarrow \mathbf{g}_{t-1}^{(j)} + \alpha_t^{(j)}\, \mathrm{sign}(\Delta_t^{(j)})$
      send $\big(\mathrm{sign}(\Delta_t^{(j)}),\, \alpha_t^{(j)}\big)$
    else ▷ MuonUSign / MuonSign
      send $\mathbf{s}_t^{(j)} \leftarrow \mathrm{sign}\big(\tilde{\mathbf{M}}_t^{(j)}\big)$
    end if
  end for
  on the server:
  if $\mathcal{C}^{\uparrow} = \text{EF21}$ **then**
    $\mathbf{g}_t \leftarrow \mathbf{g}_{t-1} + \frac{1}{N} \sum_j \alpha_t^{(j)}\, \mathrm{sign}(\Delta_t^{(j)})$ ▷ reconstruct gradient
  else
    $\mathbf{g}_t \leftarrow \mathrm{sign}\big(\sum_j \mathbf{s}_t^{(j)}\big)$ ▷ majority vote
  end if
  $\mathbf{D}_t \leftarrow \mathrm{MuonLMO}(\mathbf{g}_t)$ ▷ single LMO on the server
  if $\mathcal{C}^{\downarrow} = \text{sign}$ **then** ▷ MuonSign: 1 bit/param down
    $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t\, \mathrm{sign}(\mathbf{D}_t)$; **broadcast** $\mathrm{sign}(\mathbf{D}_t)$
    $\mathbf{W}_t \leftarrow \mathbf{W}_{t-1} - \eta_t\, \mathrm{sign}(\mathbf{D}_t)$ ▷ $= \mathbf{X}_t$; one model
  else if $\mathcal{C}^{\downarrow} = \text{EF21-P}$ **then** ▷ EF21-MuonSign: 1 bit/param down
    $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t \mathbf{D}_t$ ▷ server model
    $\Delta_t^{\downarrow} \leftarrow \mathbf{X}_t - \mathbf{W}_{t-1}$; $\quad \alpha_t^{\downarrow} \leftarrow \mathrm{mean}(|\Delta_t^{\downarrow}|)$
    $\mathbf{W}_t \leftarrow \mathbf{W}_{t-1} + \alpha_t^{\downarrow}\, \mathrm{sign}(\Delta_t^{\downarrow})$; **broadcast** $\big(\mathrm{sign}(\Delta_t^{\downarrow}), \alpha_t^{\downarrow}\big)$ ▷ clients apply the same refresh
  else ▷ exact downlink: MuonUSign / EF21-MuonUSign
    $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t \mathbf{D}_t$; **broadcast** $\mathbf{X}_t$; $\quad \mathbf{W}_t \leftarrow \mathbf{X}_t$
  end if
end for

References

Amsel, N.; Persson, D.; Musco, C.; and Gower, R. M. 2025. The Polar Express: Optimal Matrix Sign Methods and Their Application to the Muon Algorithm. arXiv:2505.16932.

Bernstein, J. 2025. Deriving Muon. https://jeremybernste.in/writing/deriving-muon. Blog post.

Bernstein, J.; and Newhouse, L. 2024. Old Optimizer, New Norm: An Anthology. arXiv:2409.20325.

Bernstein, J.; Wang, Y.-X.; Azizzadenesheli, K.; and Anandkumar, A. 2018. signSGD: Compressed Optimisation for Non-Convex Problems. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, 560–569. PMLR.

Bernstein, J.; Zhao, J.; Azizzadenesheli, K.; and Anandkumar, A. 2019. signSGD with Majority Vote is Communication Efficient and Fault Tolerant. In *International Conference on Learning Representations (ICLR)*. arXiv:1810.05291.

Cesista, F. L. 2025. Steepest Descent Under Schatten-p Norms. https://leloykun.github.io/ponder/steepest-descent-schatten-p/. Blog post.

Grishina, E.; Smirnov, M.; and Rakhuba, M. 2025. Accelerating Newton–Schulz Iteration for Orthogonalization via Chebyshev-type Polynomials. arXiv:2506.10935.

Gruntkowska, K.; Gaponov, A.; Tovmasyan, Z.; and Richtárik, P. 2025. Error Feedback for Muon and Friends. arXiv:2510.00643.

Jordan, K.; Jin, Y.; Boza, V.; You, J.; Cesista, F.; Newhouse, L.; and Bernstein, J. 2024. Muon: An optimizer for hidden layers in neural networks. https://kellerjordan.github.io/posts/muon/.

Karimireddy, S. P.; Rebjock, Q.; Stich, S.; and Jaggi, M. 2019. Error Feedback Fixes SignSGD and other Gradient Compression Schemes. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, 3252–3261. PMLR.

Kovalev, D. 2025. Understanding Gradient Orthogonalization for Deep Learning via Non-Euclidean Trust-Region Optimization. arXiv:2503.12645.

Kravatskiy, A.; Kozyrev, I.; Kozlov, N.; Vinogradov, A.; Merkulov, D.; and Oseledets, I. 2025. Ky Fan Norms and Beyond: Dual Norms and Combinations for Matrix Optimization. arXiv:2512.09678.

Krizhevsky, A. 2009. Learning Multiple Layers of Features from Tiny Images. Technical report, University of Toronto. https://www.cs.toronto.edu/~kriz/learning-features-2009-TR.pdf.

Large, T.; Liu, Y.; Huh, M.; Bahng, H.; Isola, P.; and Bernstein, J. 2024. Scalable Optimization in the Modular Norm. In *Advances in Neural Information Processing Systems (NeurIPS)*. arXiv:2405.14813.

Li, J.; and Hong, M. 2025. A Note on the Convergence of Muon. arXiv:2502.02900.

Liu, J.; Su, J.; Yao, X.; Jiang, Z.; Lai, G.; Du, Y.; Qin, Y.; Xu, W.; Lu, E.; Yan, J.; Chen, Y.; Zheng, H.; Liu, Y.; Liu, S.; Yin, B.; He, W.; Zhu, H.; Wang, Y.; Wang, J.; Dong, M.; Zhang, Z.; Kang, Y.; Zhang, H.; Xu, X.; Zhang, Y.; Wu, Y.; Zhou, X.; and Yang, Z. 2025. Muon is Scalable for LLM Training. arXiv:2502.16982.

Mishra, N.; Trivedi, K.; and Kumar, P. 2026. SignMuon: Communication-Efficient Distributed Muon Optimization. arXiv:2605.16311.

Penedo, G.; Kydlíček, H.; Ben Allal, L.; Lozhkov, A.; Mitchell, M.; Raffel, C.; Von Werra, L.; and Wolf, T. 2024. The FineWeb Datasets: Decanting the Web for the Finest Text Data at Scale. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*. arXiv:2406.17557.

Pethick, T.; Xie, W.; Antonakopoulos, K.; Zhu, Z.; Silveti-Falls, A.; and Cevher, V. 2025a. Training Deep Learning Models with Norm-Constrained LMOs. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, 49069–49104. PMLR.

Pethick, T.; Xie, W.; Erdogan, M.; Antonakopoulos, K.; Silveti-Falls, A.; and Cevher, V. 2025b. Generalized Gradient Norm Clipping & Non-Euclidean (L_0, L_1)-Smoothness. In *Advances in Neural Information Processing Systems (NeurIPS)*. arXiv:2506.01913.

Riabinin, A.; Shulgin, E.; Gruntkowska, K.; and Richtárik, P. 2025. Gluon: Making Muon & Scion Great Again! (Bridging Theory and Practice of LMO-based Optimizers for LLMs). arXiv:2505.13416.

Richtárik, P.; Sokolov, I.; and Fatkhullin, I. 2021. EF21: A New, Simpler, Theoretically Better, and Practically Faster Error Feedback. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34, 4384–4396.

Shah, I.; Polloreno, A. M.; Stratos, K.; Monk, P.; Chaluvaraju, A.; Hojel, A.; Ma, A.; Thomas, A.; Tanwer, A.; Shah, D. J.; Nguyen, K.; Smith, K.; Callahan, M.; Pust, M.; Parmar, M.; Rushton, P.; Mazarakis, P.; Kapila, R.; Srivastava, S.; Singla, S.; Romanski, T.; Vanjani, Y.; and Vaswani, A. 2025. Practical Efficiency of Muon for Pretraining. arXiv:2505.02222.

Shulgin, E.; AlRashed, S.; Orabona, F.; and Richtárik, P. 2026. Beyond the Ideal: Analyzing the Inexact Muon Update. In *Proceedings of the 29th International Conference on Artificial Intelligence and Statistics (AISTATS)*. arXiv:2510.19933.

Yang, G.; Hu, E. J.; Babuschkin, I.; Sidor, S.; Liu, X.; Farhi, D.; Ryder, N.; Pachocki, J.; Chen, W.; and Gao, J. 2021. Tuning Large Neural Networks via Zero-Shot Hyperparameter Transfer. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34, 17084–17097. arXiv:2203.03466.

Yang, G.; Simon, J. B.; and Bernstein, J. 2023. A Spectral Condition for Feature Learning. arXiv:2310.17813.